# THE VOLUME GAP AT FIVE-VALENT VERTICES

*Every five-valent Ashtekar–Lewandowski vertex whose edge tangents are in general position is zero or one of three operators. The gap closes when the two triple products through the distinguished edge carry unequal weights, and survives when they do not compete; at equal weights it survives on every class we can reach.*

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Abstract. Whether the volume operator of loop quantum gravity has a spectral gap had been addressed at valence four semiclassically, numerically and at leading order in the spins, and at valence five and higher only numerically: Brunnemann and Rideout observed that some embeddings produce eigenvalues accumulating at zero and others do not, and that the decaying ones are exactly those whose reduced sign factors contain $\pm 4$. We explain this rule for the Ashtekar–Lewandowski operator and prove most of it in the convention in which spins are assigned to the edges of a fixed embedding. Up to relabelling, every five-valent vertex operator whose edge tangents are in general position (no three coplanar) is zero or $-4i[\alpha\, J_1 \cdot (J_2 \times J_3) + \beta\, J_1 \cdot (J_4 \times J_5)]$ with $(\alpha, \beta) \in \{(2,0),(2,2),(4,2)\}$, and "contains $\pm 4$" means $t = \beta/\alpha = 1/2$. One direction of the rule is proved in full: for every $|t| \notin \{0,1\}$ the gap closes, through a dimerised chain with two edge modes, exponentially in the spin; at the Ashtekar–Lewandowski value $t = 1/2$ the rate is $q_{\min} \asymp N^{3/2} 2^{-N}$. The other direction is proved for $t = 0$, where the gap is exactly $2\sqrt{3}$, and in part for $t = 1$: there the gap is proved when the distinguished spin is $1/2$, $1$, $3/2$ or $2$, on every tuple in which three of the other four spins are at most $3$ (where it grows linearly in the distinguished spin with the sharp slope $4\sqrt{10 - 6\sqrt{2}}$), and on several further infinite classes, and it is certified for all spins up to $19$. The rest of $t = 1$ is stated as a conjecture: a uniform gap, growing with that slope in the distinguished spin and like $M/\ln M$ in the largest spin, for which we prove the upper bound. In Brunnemann and Rideout's own convention, with the spins sorted, the decay is proved for $6$ of the $60$ sign vectors (up to sign) that contain $\pm 4$. Along the way we prove the sharp four-valent law $\lambda^2 \geq j_{\max}(j_{\max}+1)/4$ for the nonzero eigenvalues $\lambda$ of $J_1 \cdot (J_2 \times J_3)$, whose constant and minimisers Brunnemann and Thiemann found numerically, through a factorisation into a hydrogen Stark matrix; a factorial gap closing at a six-valent vertex in general position; and a uniform gap for the Rovelli–Smolin operator on every five-valent vertex with an edge of spin $1/2$, which includes the vertices on which we prove that the Ashtekar–Lewandowski gap closes. What is new is the proofs, the mechanisms and the sharp constants. The computer-assisted steps are exact rational, interval, or rigorously bounded floating-point certificates, and all ship with replayable scripts.

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# 1 INTRODUCTION

In loop quantum gravity the volume of a region is an operator whose spectrum is discrete: the volume of each vertex of a spin network acts within the finite-dimensional space of intertwiners at that vertex and can be diagonalised there, and it depends on the spins of the edges meeting there and on how those edges are embedded. Discreteness alone does not say that volume is *granular*. A pure point spectrum does not by itself give a positive lower bound: its eigenvalues can still accumulate at zero, and then there are quanta of space that are arbitrarily small. The question whether the smallest nonzero eigenvalue of the volume of a vertex is bounded below uniformly in the spins is the *volume gap* question. Brunnemann and Thiemann asked it explicitly [16]; Bianchi and Haggard asked it for the four-valent vertex "for all spins" [10]; and the recent review of Haggard, Lewandowski and Sahlmann records that "for vertices of valence higher than 4, there are only numerical results for the volume spectrum" [26].

This paper answers it for the Ashtekar–Lewandowski volume operator [4] at five-valent vertices, and explains the answer.

## 1.1 WHAT WAS KNOWN

At valence four the answer was expected to be yes. Brunnemann and Thiemann derived the matrix of the operator in closed form, computed all spectra to spin 50, and found that the smallest eigenvalues fall into series, the lowest of which is $2\sqrt{j(j+1)}$ at the spins $(1/2, 1/2, j, j)$ [16, table on p. 38]. Brunnemann and Rideout state in their abstract that for four-valent vertices "the presence of a volume gap is shown" [13]. What they prove is a lower bound "in leading order of the maximal spin", derived under assumptions on how the three smaller spins scale with the largest [13, §5.4]; it says nothing about the tuples outside those assumptions. Bianchi and Haggard reproduced the spectrum from Bohr–Sommerfeld quantisation of the classical tetrahedron and found it gapped [10, 11]. None of these is a bound valid for every spin assignment.

At valence five and higher Brunnemann and Rideout found, numerically, that the answer depends on the embedding [13, 14, 15]. The operator at a vertex depends on the edge tangents only through the signs $\epsilon(IJK)$ of the triple products of tangent vectors, which after gauge reduction enter through reduced sign factors $\sigma$. For some $\sigma$ the smallest nonzero eigenvalue fell exponentially with the largest spin, for others it rose or stayed constant, and the falling ones were exactly those whose $\sigma$ contains an entry $\pm 4$. Their later paper regrouped the embeddings into four relabelling classes [15]. Nothing beyond valence four had been proved.

## 1.2 THE ANSWER

The first result organises the problem. Write $J_1, \dots, J_5$ for the angular momentum operators on the five edges, acting on the space $\mathrm{Inv}(j_1 \otimes \cdots \otimes j_5)$ of gauge-invariant states.

*Theorem 1.1 — Normal form*

Up to relabelling the edges and an overall sign, the Ashtekar–Lewandowski operator of every five-valent vertex whose edge tangents are in general position (no three of them coplanar) is either zero or

$$A_{\alpha,\beta} = -4i\big[\alpha\, J_1 \cdot (J_2 \times J_3) + \beta\, J_1 \cdot (J_4 \times J_5)\big], \qquad (\alpha, \beta) \in \{(2,0), (2,2), (4,2)\}. \tag{1}$$

The 384 sign configurations of tangents in general position fall into orbits of size 24 (on which the operator vanishes), 120, 120 and 120, one for each normal form. A reduced sign vector contains $\pm 4$ exactly when $t := \beta/\alpha = 1/2$.

So a five-valent vertex in general position carries at most two triple products, and both pass through one edge, which we call the *distinguished edge*. When some three tangents are coplanar the picture is less tidy, and we describe it only briefly (remark 4.1). The whole dichotomy is a statement about the ratio $t$ of their weights. Write $q_{\min}$ for the smallest nonzero eigenvalue modulus of $A$, a function of the spins; Brunnemann and Rideout's $\lambda_V$ is $\sqrt{q_{\min}}$.

*Theorem 1.2 — The gap closes for unequal weights*

If $\alpha\beta \neq 0$ and $|t| < 1$, the spins $(1/2;\ j-1/2, j;\ j, j)$ give $q_{\min} \le |\alpha|\, N\sqrt{2N+1}\,|t|^N \to 0$, where $N = 2j$; if $|t| > 1$, the same holds with $(\alpha, t)$ replaced by $(\beta, 1/t)$ on the spins $(1/2;\ j, j;\ j-1/2, j)$, in which the two pairs are exchanged. For the Ashtekar–Lewandowski value $t = 1/2$, $\tfrac{3}{8}P_N \le q_{\min} \le P_N$ with $P_N = 4N\sqrt{2N+1}\,2^{-N}$, and $q_{\min}/P_N \to (\tfrac{16}{3}\ln\tfrac{4}{3})^{-1/2}$.

The two remaining ratios behave differently: a single product keeps a sharp gap, and equal weights keep a gap on every class we can reach.

*Theorem 1.3 — The gap survives when the products do not compete*

For $t = 0$ every nonzero eigenvalue satisfies $q \ge 2\sqrt{3}$, with equality exactly when the three edges of the triple product carry spin $1/2$ and the other two spins differ by $1/2$.

*Theorem 1.4 — The gap survives when they compete equally, on these classes*

For $t = 1$ the smallest nonzero eigenvalue is bounded below by a positive constant, uniformly in the other four spins, when the distinguished spin $j_1$ is $1/2$, 1, $3/2$ or 2; and, for every $j_1$, on the tuples in which three of the other four spins are at most 3, on the tuples in which one of them is $1/2$, and on the tuples whose recoupling region has a "closed" colour class and either an odd dimension sum or a pair containing spin $1/2$ (theorem 6.5). Every tuple with all spins at most 19, and every tuple with $j_1 = 5/2$ and spins at most 35, satisfies $q \geq 4.5579607$.

What is left at $t = 1$ is a precise conjecture (section 11): the infimum is $2\sqrt{18 - 2\sqrt{41}} = 4.55796\ldots$, attained at $(1/2; 1/2, 1; 1, 1)$, and the gap grows at least like the larger of $c_1 j_1$ and $c_2 M/\ln M$, where $M = 2j_{\max}$. We prove that $c_1$ cannot exceed $4\sqrt{10 - 6\sqrt{2}} = 4.92296\ldots$, that a linear bound with exactly this slope, $q \geq 4\sqrt{10 - 6\sqrt{2}}\, j_1 - C$, holds on the tuples with three bounded legs, and that $M/\ln M$ cannot be improved to $M$. The conjectured growth law for all tuples remains open.

The rule concerns a fixed embedding with every assignment of spins to its edges, and the theorems above are stated in that convention. Brunnemann and Rideout's 2007 tables attach each sign vector to sorted spins instead. In that convention theorem 1.2 gives the decay for 6 of the 60 sign vectors (up to sign) that contain $\pm 4$; for the other 54 we have exhaustive numerics to $2j_{\max} = 34$ and no proof (section 5).

## 1.3 THE PICTURE

Why should the ratio of two weights decide whether a spectrum reaches zero? The answer is visible already when the distinguished edge carries spin $1/2$. Then the gauge-invariant states form a single chain: each of the two triple products moves along it by one step, they alternate, and the operator is a zero-diagonal Jacobi matrix whose links are the classical four-valent matrix elements. A zero-diagonal chain whose links alternate weak, strong, weak, strong is the Su–Schrieffer–Heeger chain of polymer physics [36]: if it starts and ends with a weak link it carries two states localised at its ends, and their energies split only by the product of the weak-to-strong ratios along the chain, which is exponentially small. If it starts and ends with a strong link it carries nothing of the kind.

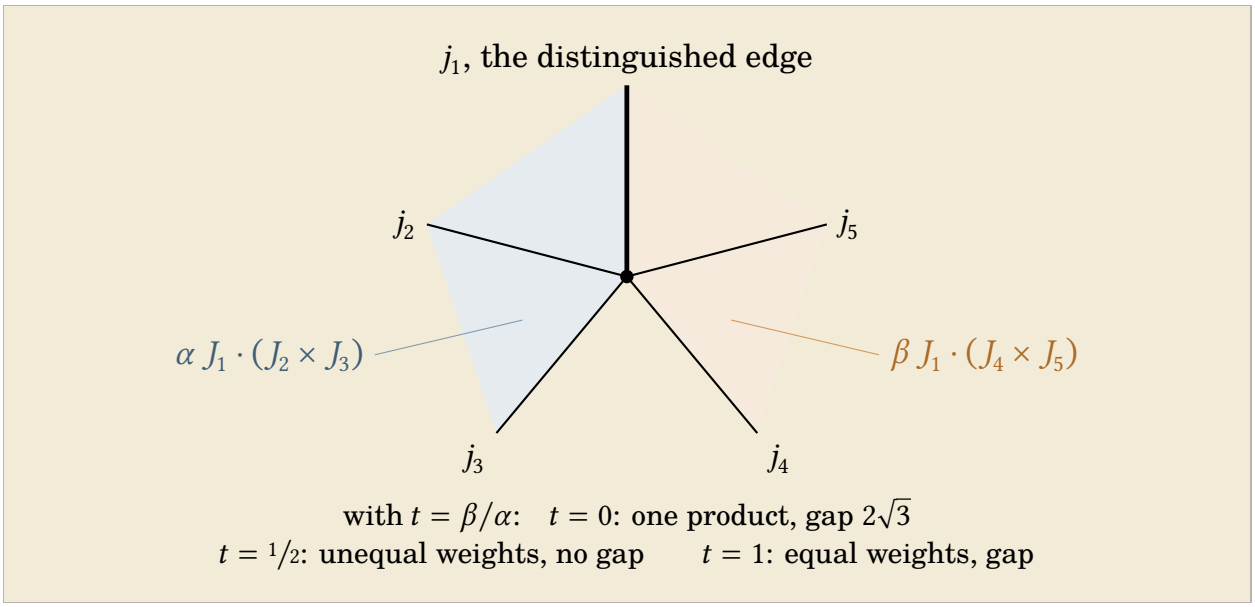



FIG. 1. *After relabelling, a five-valent vertex whose tangents are in general position carries at most two triple products, and both pass through the distinguished edge. The sign factors of the embedding only fix their weights, and the only ratios that occur are* $t = 0, 1/2, 1$. *The sign factors enter only through* $t$.

Which links are weak is decided by the two pairs of edges. One pair has the narrower range of recoupling values and sits inside the other; its links are the inner ones. With equal weights the inner links are only polynomially weaker than their neighbours, the dimerisation is trivial, and the gap grows. With weights 4 and 2, the heavier pair can be put inside, the inner links become twice as strong as the outer ones, the chain is dimerised in the topologically nontrivial phase, and the gap closes like $2^{-N}$. With a single triple product there is nothing to compete, and the operator is a four-valent operator in disguise, whose gap we know sharply. Figure 1 shows the vertex, figure 3 the chain, and figure 4 what happens to $q_{\min}$ as $t$ moves from $1/4$ to 1.

For larger distinguished spins the states no longer form a chain but a strip of a two-dimensional lattice, and the mechanism that protects the gap at $t = 1$ is more interesting. The operator is congruent, through an explicit positive diagonal matrix, to the compression of a sum of two hydrogen Stark matrices to the strip (theorem 6.2). A Stark matrix has the spectrum $N - 1, N - 3, \ldots$, so the uncompressed sum has a gap; the triangle inequalities of the recoupling cut it, and the question becomes how much the cut can cost. On the parts of the strip that the cut does not reach it costs nothing (theorem 6.5), and the rest of the proof is about the cut edges, one slice of $j_1$ at a time.

## 1.4 SUPPORTING RESULTS

Three results support the main one.

The first is the four-valent law, sharp for every spin: $\lambda^2 \geq j_{\max}(j_{\max}+1)/4$ for every nonzero eigenvalue $\lambda$ of $J_1 \cdot (J_2 \times J_3)$, with equality exactly on $(1/2, 1/2, j, j)$ (theorem 3.1). The constant and the minimising family are Brunnemann and Thiemann's numerical observation [16, p. 38]; what is new is the proof. It is used for $t = 0$, and its proof, a factorisation of the four-valent matrix into a hydrogen Stark matrix and a smooth multiplier, is the first appearance of the Stark structure that carries $t = 1$.

The second is the six-valent vertex. At Brunnemann and Rideout's minimising embedding, whose tangents are in general position, the operator is a two-row ladder, its end modes tunnel through $n$ dimers each costing a factor of order $1/h$, and the gap closes at least as fast as $\exp(-n \ln n + O(n))$, faster than the exponential their data suggested (theorem 8.1).

The third is a contrast. The Rovelli–Smolin operator takes the absolute value of each triple product before summing. On every five-valent vertex with an edge of spin $1/2$, which includes every vertex used in theorem 1.2, it has a uniform gap (theorem 9.1). This is not an argument for that operator: it too has zero-volume states, and it has been disfavoured for dynamics since the consistency check of Giesel and Thiemann [24].

## 1.5 WHAT IS NEW

Most of these statements confirm what the numerics of Brunnemann, Rideout and Thiemann suggested. The four-valent law, with its constant and its minimisers, was observed numerically [16] and derived at leading order [13]; the decay at valence five and six was observed; the $\pm 4$ rule was observed; and the constant $2\sqrt{3}$ of theorem 1.3 was observed, as $\lambda_V = \sqrt[4]{12}$, in [15, p. 35]. What is new is the proofs, the normal form that turns the $\pm 4$ rule into a statement about $t$, the mechanisms (dimerised chains, Stark matrices, ladders), the constants $(16/3 \ln 4/3)^{-1/2}$ for the decay and $4\sqrt{10 - 6\sqrt{2}}$ for the slope, and the factorial rate at valence six. That rate corrects the exponential fit of [14]; it is the one quantitative statement here that goes against an earlier expectation.

We screened the literature through the citing records of the papers above and found no earlier proof of any of these statements, and no earlier statement of the normal form or of the identification of the four-valent matrix with a Stark matrix (section 10.4 says what we read). Two sources that we know only in abstract are close enough to matter. Lévy-Leblond and Lévy-Nahas [29] studied the symmetric coupling scheme generated by the mixed product $J_1 \cdot (J_2 \times J_3)$, which Chakrabarti [18] had proposed as a label for the states of three angular momenta. Aquilanti, Caligiana and Cavalli [2] relate the alternative bases of the hydrogen atom, the parabolic (Stark) basis among them, through recoupling coefficients. Our priority statement for the Stark identification does not cover these two papers.

### 1.6 COMPUTER ASSISTANCE AND CONVENTIONS

Several proofs end in a finite computation. Each is one of three kinds: an exact rational computation (pivots of an $LDL^T$ factorisation, ranks, polynomial coefficient signs), an interval-arithmetic computation, or a floating-point computation with a rigorous a priori error bound. Section 10 lists them with their scope, and every one ships with a script and a checksum. We also reproduce, with our own code, the published numbers that fix the normalisation: Brunnemann and Rideout's seven-valent value 0.000597922, their minimising spins at valence five and six, and Bianchi and Haggard's 0.310.

The normalisation is fixed once in section 2: $q$ is an eigenvalue modulus of Brunnemann and Rideout's matrix with $Z = 1$, so their $\lambda_V$ is $\sqrt{q}$, and at $t = 1$ the operator is $-8iT$ with $T = J_1 \cdot (J_2 \times J_3 + J_4 \times J_5)$, so that $q$ is 8 times an eigenvalue modulus of $T$. Every edge carries a positive spin, $j_I \geq 1/2$; an edge of spin 0 is not an edge of the vertex. Spins are *labelled*: a statement about an embedding quantifies over all assignments of positive spins to its edges. Brunnemann and Rideout's 2007 tables use *sorted* spins; we say so whenever the difference matters.

**Organisation.** Section 2 fixes the operator. Section 3 proves the four-valent law. Section 4 proves the normal form. Section 5 proves that the gap closes for $|t| \neq 0, 1$ and that it survives for $t = 0$. Section 6 treats $t = 1$, section 7 its large-spin behaviour, section 8 valence six and section 9 the Rovelli–Smolin operator. Section 10 describes the computer-assisted steps and section 11 what remains open. Routine estimates and the bookkeeping of the certificates are in the appendices. Section 12 says how the work began and how AI tools were used, and appendix J how the results were reached.

## 2 THE OPERATOR AND ITS NORMALISATION

This section fixes the operator once, from its definition by Ashtekar and Lewandowski, in the normalisation in which Brunnemann and Rideout published their numbers, and checks the normalisation against those numbers.

### 2.1 FROM THE DEFINITION TO A MATRIX

Throughout, $J_I = (J_I^1, J_I^2, J_I^3)$ is the Hermitian angular momentum operator of edge $I$, with $[J_I^a, J_I^b] = i\epsilon^{abc} J_I^c$, acting in the spin-$j_I$ representation; operators on different edges commute. Every edge carries a positive spin, $j_I \geq 1/2$, and all statements below assume this: an edge of spin 0 carries $J_I = 0$, so it is not an edge of the vertex, and a five-valent tuple with a spin-0 entry is really a vertex of lower valence. (Allowing such entries would change the constants: at $t = 1$ the tuple $(1/2; 1/2, 1/2; 0, 1/2)$ is the four-valent vertex with four spins $1/2$ and has $q_{\min} = 2\sqrt{3}$, below the equal-weight constant of section 6.) A *half-integer* is an odd multiple of $1/2$; when we contrast integer and half-integer spins, "half-integer" excludes the integers. The volume of a region acts on a spin network through its vertices. At a vertex with edges $e_1, \ldots, e_N$, Ashtekar and Lewandowski [4, eq. (4.13)] define

$$\hat{V} = \kappa_0 \sqrt{|\hat{q}_v|}, \qquad \hat{q}_v = \frac{1}{48} \epsilon_{ijk} \sum_{e,e',e''} \epsilon(e,e',e'')\, J_e^i J_{e'}^j J_{e''}^k,$$

with the sum over ordered triples of edges at the vertex, $\epsilon(e,e',e'')$ the sign of the determinant of their tangent vectors (0 if they are linearly dependent), and $\kappa_0$ an undetermined constant. Brunnemann and Rideout [14, eqs. (2.24)–(2.26)] write the same operator with the building block

$$\hat{q}_{IJK} := \left[(J_I + J_J)^2, (J_J + J_K)^2\right] = -4i\, J_I \cdot (J_J \times J_K) \tag{2}$$

and the matrix $Q := \sum_{I<J<K\leq N} \epsilon(IJK)\, \hat{q}_{IJK}$. Since the summand of $\hat{q}_v$ is symmetric under permutations of the three edges,

$$\hat{q}_v = \frac{1}{8} \sum_{I<J<K} \epsilon(IJK)\, J_I \cdot (J_J \times J_K) = \frac{1}{8} \sum_{I<J<K} \epsilon(IJK)\, \frac{i}{4}\, \hat{q}_{IJK} = \frac{i}{32}\, Q,$$

and $\hat{V} = \ell^3 \sqrt{|Z|\,|Q|}$ with all constants (including $\kappa_0$, the Immirzi parameter and $1/32$) collected in $Z$. Brunnemann and Rideout set $Z = 1$ and publish $\lambda_V := \sqrt{|\lambda_Q|}$, where $\lambda_Q$ runs over the eigenvalues of $Q$.

**Normalisation used in this paper.** We write $q$ for an eigenvalue modulus of $Q$ (equivalently of the Hermitian matrix $iQ$) and $q_{\min}$ for the smallest nonzero one. So $\lambda_V = \sqrt{q}$, and the volume eigenvalue is $\ell^3\sqrt{|Z|q}$. A volume gap means $\inf q_{\min} > 0$, the infimum taken, with the embedding fixed, over the assignments of positive spins for which the operator is not zero on the invariant space.

### 2.2 GAUGE INVARIANCE AND THE REDUCED SIGNS

On gauge-invariant states $\sum_I J_I = 0$, so the last edge can be eliminated: $J_N = -\sum_{L<N} J_L$ acting on $\mathrm{Inv}(j_1 \otimes \cdots \otimes j_N)$, and since $J_N$ commutes with the other edges, $J_I \cdot (J_J \times J_N) = -\sum_{L \neq I,J,N} J_I \cdot (J_J \times J_L)$. Collecting terms gives Brunnemann and Rideout's formula [14, eq. (4.12)],

$$Q = \sum_{I<J<K<N} \sigma(IJK)\, \hat{q}_{IJK}, \qquad \sigma(IJK) = \epsilon(IJK) - \epsilon(IJN) + \epsilon(IKN) - \epsilon(JKN). \tag{3}$$

When no three tangents are coplanar, each $\sigma$ lies in $\{0, \pm 2, \pm 4\}$. At valence five we list $\sigma = (\sigma_{123}, \sigma_{124}, \sigma_{134}, \sigma_{234})$ in their order. In a real recoupling basis each $\hat{q}_{IJK}$ is a real antisymmetric matrix, so the spectrum of $iQ$ is symmetric under $q \mapsto -q$ and only moduli matter.

**Labelled and sorted spins.** An embedding fixes $\epsilon$, and we may assign any positive spins to its edges; this is the *labelled* convention, and statements "for all spins" are statements over all assignments. It is the convention in which a statement about an embedded vertex is a statement about the spin-network states on that one embedded graph, whatever spins they carry. Brunnemann and Rideout's 2007 papers instead sort the spins, $j_1 \leq \ldots \leq j_5$, label the edges accordingly, and attach each $\sigma$ to the sorted tuple [14]. A relabelling that sorts the spins permutes the signs, so one embedding's spectrum is spread over several sorted $\sigma$; their 2010 paper regroups the data by relabelling classes [15]. The infimum over sorted spins with $\sigma$ fixed is never smaller than the labelled infimum of the same embedding, and we note below the two places where they differ.

### 2.3 THE FOUR-VALENT MATRIX

At valence four only $\sigma(123) \in \{0, \pm 2, \pm 4\}$ survives and $Q = -4i\sigma(123)\, J_1 \cdot (J_2 \times J_3)$. In the basis $|k\rangle$ in which edges 1 and 2 couple to spin $k$, the operator $J_1 \cdot (J_2 \times J_3)$ is a zero-diagonal Jacobi matrix with $|\langle k-1|\, J_1 \cdot (J_2 \times J_3)\, |k\rangle| = a_k$,

$$a_k^2 = \frac{\Delta(j_1, j_2; k)\, \Delta(j_3, j_4; k)}{16\,(4k^2 - 1)}, \qquad \Delta(a, b; k) = \big((a+b+1)^2 - k^2\big)\big(k^2 - (a-b)^2\big), \tag{4}$$

for $\max(|j_1 - j_2|, |j_3 - j_4|) < k \leq \min(j_1 + j_2, j_3 + j_4)$. The operator $J_1 \cdot (J_2 \times J_3)$ was proposed by Chakrabarti [18] as an operator whose eigenvalues label the states of three coupled angular momenta, noting that its classical analogue is the volume generated by the three angular momentum vectors [18, p. 302]; he wrote it as $i/4$ times a commutator of two pair Casimirs, as in (2) [18, eq. (2.4)], and, following a remark of Racah, gave its matrix elements in the coupled basis as a product of a $6j$ and a $9j$ symbol [18, p. 327]. Lévy-Leblond and Lévy-Nahas [29] studied it as the generator of a symmetric coupling

scheme. The closed product form (4) of the matrix elements is in Brunnemann and Thiemann [16, eq. (67)], and, in the form of a product of two Heron areas ($\Delta = 16\,\mathrm{area}^2$ of the triangle with sides $a + 1/2, b + 1/2, k$), Aquilanti, Marinelli and Marzuoli [1, eq. (2)]. We recomputed it twice: by brute force from the generators on the invariant subspace of the tensor product, and from Racah's formula for the $6j$ symbols in exact arithmetic (section 10).

The same four-valent blocks reappear at every valence, because a triple product $J_I \cdot (J_J \times J_K)$ commutes with the total spin of the three edges and with every coupling of the remaining ones; on each block it is the four-valent matrix (4) with the fourth spin replaced by the total.

### 2.4 REPRODUCTION OF PUBLISHED NUMBERS

With our own code (two independent constructions, section 10) we reproduce the numbers that pin down the normalisation, listed in the table below. For Bianchi and Haggard's column the eigenvalues are those of $J_1 \cdot (J_2 \times J_3)$, written $\lambda$ as in section 3; elsewhere they are the $q$ just defined.

| SOURCE | PUBLISHED | REPRODUCED |
|---|---|---|
| Bianchi–Haggard [10], "loop gravity" | $\frac{\sqrt{2}}{3}\sqrt{\lvert\lambda\rvert}$, four-valent | all printed digits: 0.310 at $(1/2, 1/2, 1/2, 1/2)$, 0.396 at $(1/2, 1/2, 1, 1)$, 0.464 at $(1/2, 1/2, 3/2, 3/2)$, 0.498 at $(1/2, 1, 1, 3/2)$, 0 and 0.620 at $(1, 1, 1, 1)$, 0.522 at $(1/2, 1/2, 2, 2)$, 0.577 at $(1/2, 1, 3/2, 2)$, 0.620 at $(1, 1, 1, 2)$, and the six values $1.828, \ldots, 6.817$ at $(6, 6, 6, 7)$ |
| Brunnemann–Rideout [14, §6.5] | seven-valent $\lambda_V = 0.000597922$, $\sigma$ of their Table 10, spins $(1, 1, 1, 1, 1, 3/2, 5/2)$ | $\sqrt{q_{\min}} = 0.000597922$ ($q_{\min} = 3.5751089 \times 10^{-7}$, $\dim \mathrm{Inv} = 44$) |
| Brunnemann–Rideout [14, Table 5] | minimising sorted spins of $\sigma = (4, 2, 2, 0)$ at each $2j_{\max}$ | all 29 rows, $2 \le 2j_{\max} \le 30$, by exhaustive sorted search |
| Brunnemann–Rideout [14, Table 9] | six-valent minimisers for $\sigma = (2, 0, 0, 0, 0, 0, 0, 0, -2, 0)$ | every level $2j_{\max} \le 9$; e.g. $\lambda_V = 0.0261984$ at $(1/2, 1/2, 5/2, 5/2, 3, 3)$, where their fit gives 0.0282 |
| Kisielowski [28, Table 2], Rovelli–Smolin | e.g. 1.201406 six times at valence five, spin one; a zero at spin two | all printed digits, with $V = \sqrt{Q_{RS}/32}$ ($\kappa_0 = 1$ in [4]) |

Four remarks on the table. At $2j_{\max} = 30$ the Table 5 minimum is already $2.5 \times 10^{-12}$ of the norm of the matrix, where double precision can no longer rank the candidates, so we stopped there; some text extractions of the published PDF misalign the rows of that table from $2j_{\max} = 10$ on, so we compare against the page image. Brunnemann and Rideout later reported a code bug that affected all their seven-valent results [15, §5]; we use their seven-valent value only as a check of the normalisation at the stated $\sigma$ and spins, where our own code reproduces it. Brunnemann and Rideout print no eigenvalues at valence six, only minimising spins (their Table 9 and Figs. 29–32), and their fit $\lambda_V \approx e^{3.72} e^{-1.215 \cdot 2j_{\max}}$ cannot be extrapolated (section 8). At odd $2j_{\max}$ their six-valent minimisers differ from the family of section 8.

## 3 THE FOUR-VALENT LAW

This section proves the sharp lower bound for the four-valent operator. It is needed for $t = 0$ in section 5, and its proof introduces the object that carries the rest of the paper: the hydrogen Stark matrix.

*Theorem 3.1 — Sharp four-valent law*

Let $j_1, \dots, j_4 \geq 1/2$ with $\dim \mathrm{Inv}(j_1 \otimes \cdots \otimes j_4) \geq 2$. Every nonzero eigenvalue $\lambda$ of $J_1 \cdot (J_2 \times J_3)$ on the invariant space satisfies

$$\lambda^2 \;\geq\; \tfrac{1}{4}\, j_{\max}(j_{\max} + 1),$$

with equality exactly when the sorted spins are $(1/2, 1/2, j, j)$.

The constant and the minimisers are not new: Brunnemann and Thiemann found numerically that the lowest eigenvalue series of $\hat{q}_{123}$ is $2\sqrt{j(j+1)}$, at the spins $(1/2, 1/2, j, j)$, for all spins up to 50 [16, table on p. 38], which is theorem 3.1 in their normalisation. What is new is the proof, and the Stark structure it exposes. In particular $|\lambda| \geq \sqrt{3}/4$, with equality only at four spins $1/2$. The eigenvalues of $\hat{q}_{123} = -4i J_1 \cdot (J_2 \times J_3)$ have modulus at least $2\sqrt{j_{\max}(j_{\max}+1)} > 2j_{\max}$, twice the bound that Brunnemann and Rideout derived at leading order in $j_{\max}$ under assumptions on the scaling of the spins [14, (7.56)]; and $\frac{\sqrt{2}}{3}\sqrt{\sqrt{3}/4} = 0.3102$ is Bianchi and Haggard's smallest volume [10].

The matrix entries (4) are a product of two triangle factors, one for each pair of edges. Each triangle factor, divided by $4k^2-1$, is a matrix element that every student of the Stark effect has met. In the shell of principal quantum number $n$ of hydrogen, the component $A_z$ of the Runge–Lenz vector connects the states $|L,m\rangle$ and $|L-1,m\rangle$ with squared modulus $(n^2-L^2)(L^2-m^2)/(4L^2-1)$, and its eigenvalues are Pauli's parabolic quantum numbers, the integers $n_1-n_2$, equally spaced by two. So one pair of edges contributes a matrix whose spectrum we know exactly and whose smallest positive eigenvalue is 1 or 2, and the other pair contributes a slowly varying multiplier. The four-valent operator is a Stark matrix with stretched links. The question becomes how much the stretching can lower the bottom of the spectrum, and the answer is: by no more than the stretch at the first link.

**Definition 3.1** (Stark matrix). For real $m \geq 0$ and an integer $N \geq 1$ put $n = m+N$. The *Stark matrix* $H(n,m)$ is the $N \times N$ zero-diagonal Jacobi matrix on the labels $L = m, m+1, \dots, n-1$ whose link between $L-1$ and $L$ has

$$h_L^2 = \frac{(n^2-L^2)(L^2-m^2)}{4L^2-1}. \tag{5}$$

*Lemma 3.2 — The Stark spectrum*

The eigenvalues of $H(n,m)$ are $N-1, N-3, \dots, -(N-1)$, for every real $m \geq 0$. Its smallest positive eigenvalue is 1 if $N$ is even and 2 if $N \geq 3$ is odd.

*Proof.* For integer $m$, take two spins $j \otimes j$ with $2j+1 = n$, and the sector of total $J_z = m$. In the coupled basis $|L,m\rangle$ ($L = m, \dots, 2j$) the operator $S_z := J_z^{(1)} - J_z^{(2)}$ is tridiagonal; its diagonal vanishes because $S_z$ is odd under exchanging the two equal spins while $|L,m\rangle$ has parity $(-1)^{2j-L}$; and its off-diagonal elements are the standard ones, $|\langle L,m|S_z|L-1,m\rangle|^2 = (L^2-m^2)(n^2-L^2)/(4L^2-1)$ [20]. In the product basis $|m_1, m-m_1\rangle$ the same operator is diagonal with eigenvalues $2m_1 - m$, which are the $N = n-m$ integers $N-1, N-3, \dots$. For real $m$, fix $N$: the coefficients of the characteristic polynomial are polynomials in the $h_L^2$, hence rational functions of $m$ with no pole on $m \geq 0$, and they agree with those of $\prod_k (x-(N-1-2k))$ at every integer $m \geq 0$. So they agree identically. □

Sort the spins, $j_1 \le j_2 \le j_3 \le j_4 = j_{\max}$, and couple edges $1, 2$. Write $n = j_1 + j_2 + 1$, $m_> = \max(|j_1 - j_2|, |j_3 - j_4|)$, $m_< = \min(|j_1 - j_2|, |j_3 - j_4|)$ and $\varpi = 2j_4 + 1$, so that $\bar{\mu} := {}^1\!/_4 j_4(j_4+1) = (\varpi^2 - 1)/16$.

**Lemma 3.3** (Factorisation). The links of the four-valent chain are $a_k = h_k w_k$ for $m_> < k \le n-1$, where $h$ is the Stark matrix $H(n, m_>)$ on exactly the range of the chain, and

$$w_k^2 = \tfrac{1}{16} f(k), \qquad f(k) = (k^2 - m_<^2)\big((j_3 + j_4 + 1)^2 - k^2\big).$$

*Proof.* $\{|j_1 - j_2|, |j_3 - j_4|\} = \{m_>, m_<\}$, so the four factors of (4) regroup; sorting gives $k_{\max} = j_1 + j_2 = n - 1$ and $k_{\min} = m_>$. □

The regrouping is the point: putting the larger of the two differences into the Stark factor makes the Stark matrix live on exactly the range of the four-valent chain. With the natural grouping by pairs it would not.

**Lemma 3.4** (Site scaling). Let $c_1, \dots, c_L > 0$ be the links of a zero-diagonal Jacobi matrix $J(c)$ and $g_1, \dots, g_L > 0$. Choose $\gamma_1 > 0$ and $\gamma_{s+1} = g_s^2/\gamma_s$. Then $J(cg) = SJ(c)S$ with $S = \operatorname{diag}(\sqrt{\gamma_i})$, and the smallest positive eigenvalue satisfies $\lambda(cg) \ge \sqrt{\min_{\text{odd}} \gamma \cdot \min_{\text{even}} \gamma}\ \lambda(c)$.

*Proof.* $\sqrt{\gamma_s \gamma_{s+1}} = g_s$, so $SJ(c)S$ has links $g_s c_s$. In the bipartite form $J = \left(\begin{smallmatrix} 0 & B \\ B^T & 0 \end{smallmatrix}\right)$ (odd sites against even sites) the nonzero eigenvalues are $\pm$ the singular values of $B$, $B$ has full column rank, and $B(cg) = S_{\text{odd}} B(c) S_{\text{even}}$, so $\sigma_{\min}(B(cg)) \ge \sigma_{\min}(S_{\text{odd}})\sigma_{\min}(B(c))\sigma_{\min}(S_{\text{even}})$. The free constant $\gamma_1$ cancels in the product. □

Unwinding the recursion, $\min_{odd} \gamma \cdot \min_{even} \gamma$ is a minimum over alternating products of ratios $g_{s+1}/g_s$, so the lemma needs no monotonicity of $g$. It needs two facts about the multiplier $f$: that it is large enough at the first link, and that it does not dip later in a way the alternating products can see. Both are elementary and are proved in appendix A.

**Lemma 3.5** (The multiplier). Write $f_s = f(m_> + s)$, $s = 1, \dots, L = N-1$, where $N = n - m_>$ is the dimension. (F) $f_1 \ge \varpi^2 - 1$, that is $w_1^2 \ge \bar{\mu}$, with equality iff $j_1 = j_2$ and $j_3 = j_4$. (C) $f$ is log-concave and unimodal along the chain, and (C0) $f_2 > f_1$ if $N \ge 3$; (C2) $f_L > f_2$ if $N \ge 4$; (C3) $f_L^2 \ge f_{L-1} f_2$ if $N \ge 4$.

*Proof of theorem 3.1.* Apply lemma 3.4 to $c = h$ and $g = w$. By lemma 3.2,

$$\lambda \ \ge\ \sqrt{\min_{odd} \gamma \cdot \min_{even} \gamma}\ \lambda_H,$$

with $\lambda_H = 1$ for even $N$ and $2$ for odd $N$. For $N = 2$ the Stark link is $h = 1$ and $\lambda^2 = f_1/16 \ge \bar{\mu}$, with equality iff $j_1 = j_2 = {}^1\!/_2$ and $j_3 = j_4$.

For $N \geq 3$ write $r_s = \log(f_{s+1}/f_s)$, which is nonincreasing by log-concavity. With $\gamma_1 = 1$ the weights are $\gamma_{2i+1} = \omega_i := \prod_{l \leq i} f_{2l}/f_{2l-1}$ at the odd sites and $\gamma_{2i} = \eta_i := \frac{f_1}{16} \prod_{l<i} f_{2l+1}/f_{2l}$ at the even sites. Pairing each odd step with the following even one gives $\log \omega_i \geq \tfrac{1}{2} \log(f_{2i+1}/f_1)$, or $\tfrac{1}{2} \log(f_L^2/(f_{L-1} f_1))$ at the last site; pairing each even step with the following odd one gives $\log(\eta_i/\eta_1) \geq \tfrac{1}{2} \log(f_{2i}/f_2)$, or $\tfrac{1}{2} \log(f_L^2/(f_{L-1} f_2))$ at the last site. By unimodality, (C0) and (C2), $f_t \geq f_1$ for all $t$ and $f_t \geq f_2$ for $t \geq 2$, and the end terms are at least 1 by (C3) (for $L = 2$ by (C0)). So $\min \omega \geq 1$ and $\min \eta \geq f_1/16$, and $\lambda^2 \geq (f_1/16)\lambda_H^2 \geq \bar{\mu}\lambda_H^2$. For odd $N$ this is at least $4\bar{\mu}$.

It remains to show strictness for even $N \geq 4$. Suppose $\lambda^2 = \bar{\mu}$. Then every inequality in the chain is an equality. In particular, for a minimising vector $x$, equality in $\|S_{odd} B S_{even} x\| \geq \sigma_{\min}(S_{odd}) \sigma_{\min}(B) \sigma_{\min}(S_{even}) \|x\|$ forces $S_{even} x$ to be supported on the even sites where $\gamma$ is minimal, and $B S_{even} x$ on the odd sites where $\gamma$ is minimal. The eigenvector $u$ of the Stark matrix at its smallest positive eigenvalue has nonzero components at sites 1, 2, $N-1$ and $N$, because $u_1 \neq 0$ for any eigenvector of an irreducible Jacobi matrix and then $u_2 = \lambda u_1/h_1 \neq 0$, and likewise at the other end. So $\gamma$ is minimal at sites 1 and $N-1$ and at 2 and $N$, which gives $\omega_{(L-1)/2} = 1$ and $\eta_{(L+1)/2} = \eta_1$, whose product is $f_L/f_1 = 1$. This contradicts $f_L > f_2 > f_1$. □

*Remark 3.1* (A cruder route, not used). For even dimension the inverse of the bipartite block is the lower-triangular matrix of lemma 5.1, whose entries factor as $u_i v_j$ for $i \geq j$: a discrete Hardy operator. Muckenhoupt's weighted Hardy inequality [33] then places its norm between A and 2A, with $\mathrm{A} = \max_k (\sum_{i \geq k} u_i^2)^{1/2} (\sum_{j \leq k} v_j^2)^{1/2}$, for every chain with positive links. Evaluating A for the four-valent links relates the gap to the harmonic sum $\sum_k 1/a_k$, the discrete form of the zero-volume period of the classical tetrahedron, which is the Bohr–Sommerfeld scaling of Bianchi and Haggard [11]. We do not carry this out; it would give only an unspecified constant where the Stark factorisation gives the sharp one.

## 4 THE NORMAL FORM

This section proves theorem 1.1 and sets up the lattice on which all later proofs live. The argument is short: relabelling acts on the sign configurations, there are four orbits, and one operator identity on the invariant space collapses each orbit to two triple products.

## 4.1 ORBITS OF SIGN CONFIGURATIONS

Five tangent directions in general position, that is with no three of them coplanar, define ten nonzero signs $\epsilon(IJK)$, $I < J < K \le 5$; such a sign vector is a uniform chirotope of rank three, so it satisfies the three-term Grassmann–Plücker relations. This is the hypothesis of theorem 1.1; remark 4.1 says what changes without it.

**Lemma 4.1** (Relabelling). Let $\pi$ permute the edges. The unitary that permutes the tensor factors maps the operator of $(\epsilon; j_1, \dots, j_5)$ to the operator of $(\epsilon^\pi; j_{\pi(1)}, \dots, j_{\pi(5)})$, where $\epsilon^\pi$ is the relabelled chirotope. Moreover $A_{-\epsilon} = -A_\epsilon$.

*Proof.* $A_\epsilon = \tfrac{1}{6} \sum \chi_\epsilon(I,J,K)\, \hat{q}_{IJK}$ over ordered triples of distinct edges, where $\chi_\epsilon$ is the alternating extension of $\epsilon$; the operator $\hat{q}_{IJK} = -4i J_I \cdot (J_J \times J_K)$ is alternating because the three vector operators commute. This expression is covariant under relabelling. □

So the set of spectra over all spin assignments is an invariant of the relabelling orbit. The orbits are found by an exact finite computation (section 10).

*Proposition 4.2 — Orbits*

Exactly 384 of the $2^{10}$ sign vectors satisfy the three-term Grassmann–Plücker relations, and each is realised by integer tangent vectors with entries in $[-4, 4]$. Under relabelling they form four orbits, of sizes 24, 120, 120 and 120, each closed under $\epsilon \mapsto -\epsilon$. Through (3) they produce 1, 20, 30 and 120 distinct reduced vectors $\sigma$, 171 in all (85 nonzero up to sign), and no $\sigma$ occurs in two orbits. Every $\sigma$ of the last orbit contains $\pm 4$, and no other $\sigma$ does.

These counts are Brunnemann and Rideout's: 384 configurations and 171 reduced vectors at valence five [14, Table 1], and four classes under permutation [15, Table 3], whose representatives $(2,0,0,0)$, $(2,2,2,0)$ and $(2,2,4,0)$ lie in the second, third and fourth orbit.

## 4.2 TWO TRIPLE PRODUCTS

**Lemma 4.3.** On $\mathrm{Inv}(j_1 \otimes \cdots \otimes j_5)$, $J_1 \cdot \big((J_2 + J_3) \times J_4\big) = J_1 \cdot (J_4 \times J_5)$.

*Proof.* Write $B = J_2 + J_3$. On invariant vectors $J_5 = -(J_1 + B + J_4)$, acting first. The terms $J_1 \cdot (J_4 \times J_1) = -i J_4 \cdot J_1$ and $J_1 \cdot (J_4 \times J_4) = i J_1 \cdot J_4$ cancel, so $J_1 \cdot (J_4 \times J_5) = -J_1 \cdot (J_4 \times B) = J_1 \cdot (B \times J_4)$. □

*Proof of theorem 1.1.* For each orbit we exhibit a labelling with $\sigma = (\alpha, \beta, \beta, 0)$: the orbits give $(\alpha, \beta) = (2,0)$, $(2,2)$ and $(4,2)$ up to signs (the computation of proposition 4.2). Then by (3)

$$A = \alpha \hat{q}_{123} + \beta(\hat{q}_{124} + \hat{q}_{134}) = -4i\big[\alpha J_1 \cdot (J_2 \times J_3) + \beta J_1 \cdot \big((J_2 + J_3) \times J_4\big)\big],$$

which is (1) by lemma 4.3. The signs of $\alpha$ and $\beta$ do not affect the spectrum: in the basis of the next subsection the first product preserves $c = |J_4 + J_5|$ and the second shifts it by $\pm 1$, so the diagonal unitary $(-1)^c$ reverses the sign of $\beta$ alone, and $(-1)^b$ that of $\alpha$. Finally, $|\beta/\alpha|$ takes the value $1/2$ only on the last orbit, which is the orbit of the $\sigma$ containing $\pm 4$. □

For each $\epsilon$ we also solved the linear equation $\sum \epsilon(IJK)\, J_I \cdot (J_J \times J_K) = \lambda J_a \cdot (J_b \times J_c) + \mu J_a \cdot (J_d \times J_e)$ on the invariant space over all choices of the leg $a$ and of the split of the other four. In the last two orbits exactly one leg $a$ admits a solution with $\lambda\mu \neq 0$; this is the *distinguished edge*. In the third orbit the split into pairs is not unique, because on the invariant space

$$J_1 \cdot (J_2 \times J_3 + J_4 \times J_5) = J_1 \cdot (J_3 \times J_4 - J_2 \times J_5), \tag{6}$$

by the computation of lemma 4.3 applied to $J_1 \cdot ((J_2 + J_4) \times (J_3 + J_5))$. This is one face of a larger symmetry of the $t = 1$ operator, proposition 4.5 below.

*Remark 4.1* (Tangents not in general position). If some three tangents are coplanar but not all five, some signs $\epsilon(IJK)$ vanish and theorem 1.1 does not apply. The same exact enumeration, run over $\{0, \pm 1\}^{10}$, finds 1 680 such sign vectors with pairwise non-parallel tangents (each satisfies the chirotope axioms and is realised by integer tangents with entries in $[-4, 4]$), in 16 relabelling orbits, and their reduced signs take every value from $-4$ to 4. In 9 orbits the operator still has the form (1), now with $|t| \in \{0, 1/3, 1/2, 2/3, 1\}$ and $|\alpha| \in \{1, 2, 3\}$, so the results of sections 5 and 6 apply to it after rescaling $q$ by $|\alpha|/2$; in particular theorem 5.3(i) closes the gap on the four orbits with $|t| \in \{1/3, 1/2, 2/3\}$. In the other 7 orbits (780 sign vectors) the operator is not of the form (1): in one of them it is a combination of two triple products that share two edges, in the other six no two triple products suffice, and we do not treat them. Edges with parallel tangents are the degenerate case of the remark at the end of section 8.

### 4.3 THE RECOUPLING GRID

Fix the normal form (1) and write $s = j_1$ for the distinguished spin. The basis adapted to it couples the two pairs,

$$|b, c\rangle, \qquad b = |J_2 + J_3|, \quad c = |J_4 + J_5|, \qquad |b - c| \le s \le b + c.$$

For a pair $P = (j_a, j_b)$ put $n_P = j_a + j_b + 1$ and $m_P = |j_a - j_b|$, so that the pair spin ranges over $[m_P, n_P - 1]$, and write $h_P(L)$ for the Stark links (5) of $H(n_P, m_P)$. The sites $(b, c)$ fill a rectangle cut by the strip $|d| \le s$ and the corner $u \ge s$, where

$$d = b - c, \qquad u = b + c, \qquad F(x) = (s + \tfrac{1}{2})^2 - x^2, \qquad G(u) = (u + s + 1)(u - s).$$

The product $X = J_1 \cdot (J_2 \times J_3)$ commutes with $c$ and moves $b$ by one; on fixed $c$ it is the four-valent operator of $\mathrm{Inv}(j_2, j_3, s, c)$. The product $D = J_1 \cdot (J_4 \times J_5)$ commutes with $b$ and moves $c$.

*Proposition 4.4 — The grid*

A diagonal unitary gauge carries $iA_{\alpha,\beta}$, up to an overall sign, to the real symmetric, entrywise nonnegative matrix on the sites with links

$$(b-1,c)\text{–}(b,c) \;:\quad |\alpha|\,h_1(b)\sqrt{F(d-\tfrac{1}{2})\,G(u)},$$
$$(b,c-1)\text{–}(b,c) \;:\quad |\beta|\,h_2(c)\sqrt{F(d+\tfrac{1}{2})\,G(u)},$$

where $(b,c)$ is the upper end of the link and $d,u$ are taken there. In particular the eigenvalue moduli of $A_{\alpha,\beta}$ are those of this matrix.

*Proof.* By (4) applied to $\mathrm{Inv}(j_2,j_3,s,c)$, the $X$-link has squared modulus ${}^1\!/_{16}h_1(b)^2P(b;s,c)$ with $P(b;s,c)=((s+c+1)^2-b^2)(b^2-(s-c)^2)$, and the polynomial identity

$$P(b;s,c)=F(d-\tfrac{1}{2})\,G(u),\qquad P(c;s,b)=F(d+\tfrac{1}{2})\,G(u) \tag{7}$$

gives the stated form after multiplying by $4|\alpha|$; the $D$-links are the same with the pairs exchanged. It remains to remove the signs. On invariant vectors $X=-L\cdot(J_2\times J_3)$ with $L=J_4+J_5$, and by the Wigner–Eckart theorem for a scalar product of tensor operators on two subsystems [20, (7.1.6)],

$$\langle b-1,c|X|b,c\rangle=(-1)^{b+c+s}\begin{Bmatrix}s & c & b-1\\ 1 & b & c\end{Bmatrix}\langle b-1\|J_2\times J_3\|b\rangle\,\langle c\|L\|c\rangle.$$

The $6j$ symbol equals $\{s\ c\ b;\ 1\ b-1\ c\}$, whose sign is $(-1)^{s+b+c}$ [20, Table 5]; the two $c$-dependent signs cancel, $\langle c\|L\|c\rangle>0$, and the remaining factor does not involve $c$. So the phase of an $X$-link depends on $b$ alone, and likewise the phase of a $D$-link depends on $c$ alone. A diagonal unitary depending on $(b,c)$ then makes every link positive. □

The identity (7) is where the Heron factors of the $6j$ coefficient split: the $s$-dependence of every link is a function of the diagonal $d$ times a function of the antidiagonal $u$. It is the source of the congruence in theorem 6.2. The sites carry a natural two-colouring, the parity of $u-s$, and every link joins the two colours, so the matrix is bipartite and its nonzero spectrum is $\pm$ the nonzero singular values of the block $B$ from one colour to the other. Figure 2 draws a small example.

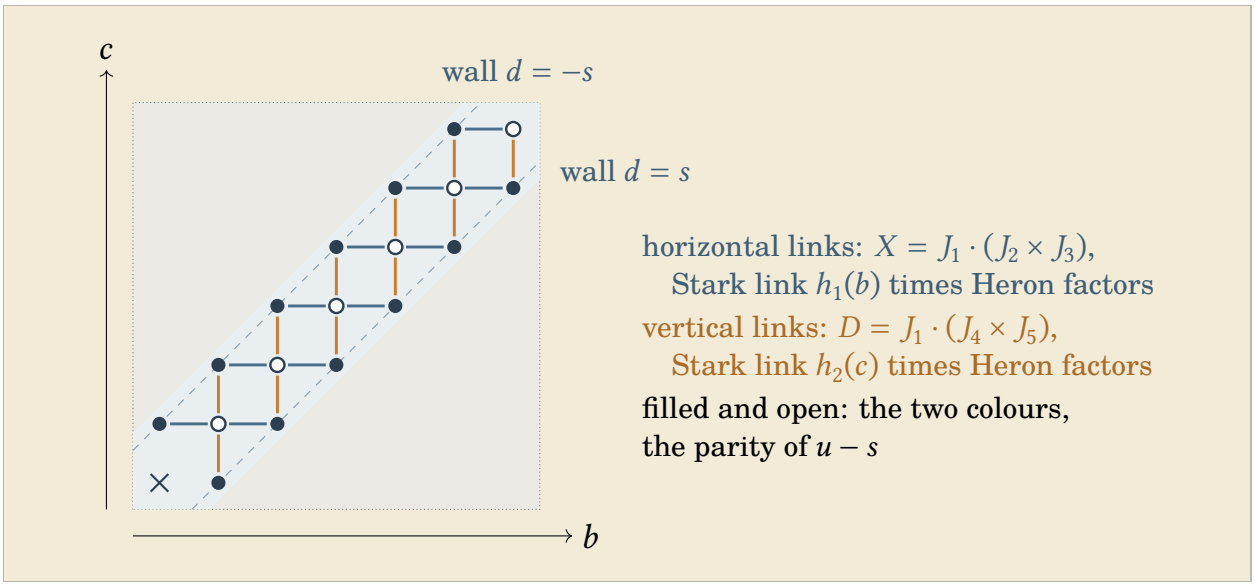



FIG. 2. *The recoupling grid for $s = j_1 = 1$ and two pairs whose spins range over $0, \dots, 6$; the cross marks the corner site $(0,0)$, excluded because $b + c < s$. The states fill the rectangle of the two pair spins $(b, c)$, cut to the strip $|b - c| \le s$ and above the corner $b + c \ge s$. The first triple product moves horizontally, the second vertically, and every link joins the two colours. For $s = 1/2$ the strip is a single zigzag path: the chain of section 5.*

*Proposition 4.5 — Symmetry at equal weights*

At $t = 1$, with $T = J_1 \cdot (J_2 \times J_3 + J_4 \times J_5)$, the moduli of the spectrum of $T$ are invariant under every permutation of $j_2, j_3, j_4, j_5$. On the invariant space $T = -\sum_{2 \le a < b < c \le 5} J_a \cdot (J_b \times J_c)$ and $T = -i\,[J_2 \cdot J_3 - J_4 \cdot J_5,\ J_3 \cdot J_4 - J_2 \cdot J_5]$.

*Proof.* On invariant vectors $T = -V \cdot L$ with $V = J_2 \times J_3 + J_4 \times J_5$ and $L = J_2 + \cdots + J_5$, because $J_1$ commutes with $V$; and $(J_2 \times J_3) \cdot (J_2 + J_3) = 0$ since $J_2 \cdot (J_2 \times J_3) = i J_2 \cdot J_3 = -J_3 \cdot (J_2 \times J_3)$. What remains are the four triple products of the other legs, a cyclically symmetric expression; the commutator form follows from $J_a \cdot (J_b \times J_c) = i/4[(J_a + J_b)^2, (J_b + J_c)^2]$. The transposition of legs $2, 3$ maps $T$ to $-X + D$, which the gauge $(-1)^b$ maps back to $X + D$. A transposition and the four-cycle generate the symmetric group. □

# 5 UNEQUAL WEIGHTS CLOSE THE GAP; ONE PRODUCT KEEPS IT

This section proves theorems 1.2 and 1.3. The first is about a chain, the second about a four-valent operator in disguise. We start with the chain, because the same chain also shows why equal weights behave differently.

## 5.1 THE GUIDING PICTURE: A DIMERISED CHAIN

When the distinguished edge carries spin $1/2$ the strip $|b - c| \le 1/2$ of figure 2 is a single zigzag path: from each site one link changes $b$ and one changes $c$. So $iA$ is a zero-diagonal Jacobi matrix whose links alternate between the two triple products. For such a chain with an even number of sites there is an exact formula for the inverse, and the smallest eigenvalue is the reciprocal of the norm of a lower-triangular matrix whose entries are products of ratios of consecutive links.

*Lemma 5.1 — Even chains*

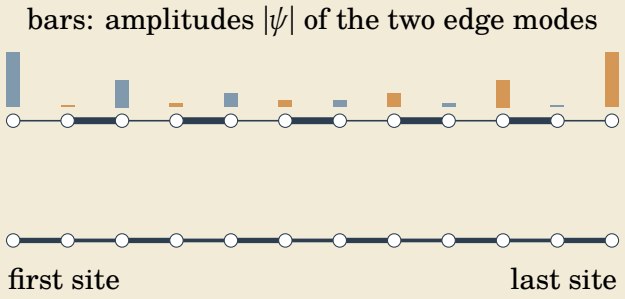
Let $J$ be a zero-diagonal Jacobi matrix of size $2K$ with links $c_1, \dots, c_{2K-1} > 0$. Then $J$ is invertible, its eigenvalues are $\pm$ the singular values of the $K \times K$ lower bidiagonal matrix $B$ with diagonal $c_1, c_3, \dots$ and subdiagonal $c_2, c_4, \dots$, and $(B^{-1})_{ij} = (-1)^{i-j} E_{ij}$ with

$$E_{ij} = \frac{1}{c_{2j-1}} \prod_{l=j}^{i-1} \frac{c_{2l}}{c_{2l+1}} \quad (i \ge j), \qquad E_{ij} = 0 \quad (i < j).$$

Hence $1/\|E\|_F \le q_{\min} \le 1/\max_{ij} E_{ij} \le 1/E_{K1}$.

*Proof.* Reorder the sites odd-then-even; then $J = \left(\begin{smallmatrix} 0 & B \\ B^T & 0 \end{smallmatrix}\right)$, $\det J = \pm \prod c_{odd}^2 \neq 0$, and forward substitution in $Bx = e_j$ gives $E$. □

So everything depends on the ratios $c_{2l}/c_{2l+1}$ of an even link to the following odd one. If they are all larger than one by a fixed factor, the corner entry $E_{K1}$ is exponentially large and the chain has an exponentially small eigenvalue: this is the Su–Schrieffer–Heeger mechanism, and the eigenvector is a pair of edge modes. If they are all smaller than one, the entries of $E$ are bounded by the diagonal ones and the gap is controlled by the links themselves. Figure 3 shows the two situations.




$t = 1/2$, heavier pair inside: inner links twice the outer; two edge modes, $q_{\min} \asymp N^{3/2} 2^{-N}$

$t = 1$: inner links weaker by only $\tau/(\tau+1)$; no edge modes; the gap grows

FIG. 3. *The chain of the spin-1/2 sector, drawn schematically: thick lines are strong links and thin lines weak ones. Odd links belong to the outer pair of edges, even links to the inner pair. With weights 4 and 2 and the heavier pair inside, the chain starts and ends weak and carries two edge modes whose amplitudes fall by about half at every second site (bars); their splitting is the product of the ratios, about $2^{-N}$. With equal weights the inner links are weaker, the chain starts and ends strong, and no mode is localised.*

Which pair is inner is a property of the spins. Write $n_P = j_a + j_b + 1$ and $m_P = |j_a - j_b|$ for the two pairs of the normal form. The chain runs over the pair spins between $\max(m_1, m_2)$ and $\min(n_1, n_2)$, and the pair that attains the larger lower end and the smaller upper end is *inner*: its range sits inside the other. Its links vanish at both ends of the chain, because its Stark factor $(n^2 - L^2)(L^2 - m^2)$ does; the outer pair's do not. With equal weights this makes the inner links the weak ones, by a factor $\tau/(\tau+1)$ at distance $\tau$ from the end (lemma 6.9). Multiplying the inner pair by 2 against the outer one turns weak into strong. Figure 4 follows $q_{\min}$ along the family $(1/2; j - 1/2, j; j, j)$, in which the pair $(j - 1/2, j)$ is inner, as $t$ moves towards 1.

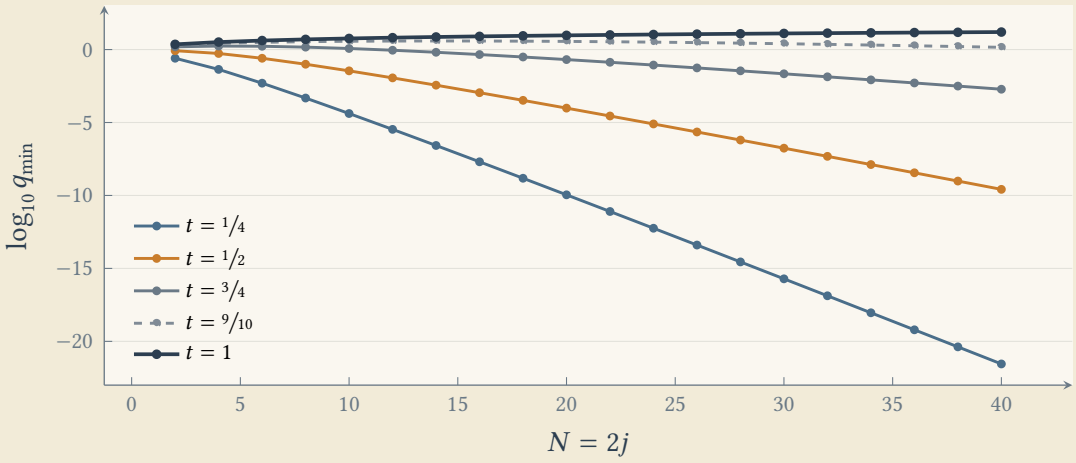



FIG. 4. $q_{\min}$ *on the spin-1/2 chain of* $-4i[J_1 \cdot (J_2 \times J_3) + t\, J_1 \cdot (J_4 \times J_5)]$ *along* $(1/2; j - 1/2, j; j, j)$ *(60-digit arithmetic). Every* $0 < t < 1$ *eventually decays like* $t^N$*, the closer to* 1 *the later; at* $t = 1$ *the decay never starts.*

*Proposition 5.2 — The chain*

Let $j_1 = 1/2$, $\alpha\beta \neq 0$ and $N = 2j$. For the spins $(1/2;\ j-1/2, j;\ j, j)$ the operator $iA_{\alpha,\beta}$ is a zero-diagonal Jacobi matrix on $2N$ sites whose links alternate $D_1, X_1, D_2, \dots, X_{N-1}, D_N$ with

$$D_l = |\beta|\, l\sqrt{(N+1)^2 - l^2}, \qquad X_l = |\alpha|\sqrt{(N-l)(N+l+1)\, l(l+1)}.$$

*Proof.* With $s = 1/2$ the sites of proposition 4.4 satisfy $|b-c| = 1/2$, and ordering them by $u = b + c$ gives a path. Here $b \in [1/2, N-1/2]$ and $c \in [0, N]$, so the path has the $2N$ sites $u = 1/2, 3/2, \dots, 2N-1/2$ and starts and ends with a $D$-link. Since $F(0) = 1$ and, at an $X$-link, $G(u) = 4b^2 - 1$, the $X$-link is $|\alpha|\sqrt{(n_1^2 - b^2)(b^2 - m_1^2)}$ with $n_1 = N + 1/2$, $m_1 = 1/2$, which is $X_l$ at $b = l + 1/2$; the $D$-link is the same with $n_2 = N+1$, $m_2 = 0$ and $c = l$. □

*Theorem 5.3 — Gap closing*

In the setting of proposition 5.2, with $t = \beta/\alpha$:

(i) if $0 < |t| < 1$, then $A$ is invertible and $q_{\min} \le |\alpha|\, N\sqrt{2N+1}\, |t|^N$; if $|t| > 1$, then on the spins $(1/2;\ j, j;\ j-1/2, j)$, obtained by exchanging the pairs $(j_2, j_3)$ and $(j_4, j_5)$, $A$ is invertible and $q_{\min} \le |\beta|\, N\sqrt{2N+1}\, |t|^{-N}$;

(ii) if $(\alpha, \beta) = (4, 2)$, then $3/8 P_N \le q_{\min} \le P_N$ with $P_N = 4N\sqrt{2N+1}\, 2^{-N}$, and $q_{\min}/P_N \to \kappa = (16/3 \ln 4/3)^{-1/2} = 0.80732$.

For every embedding in general position whose $\sigma$ contains $\pm 4$, therefore, $\inf q_{\min} = 0$ over spin assignments, and the smallest nonzero volume is $\ell^3\sqrt{|Z| q_{\min}} \asymp N^{3/4} 2^{-N/2}$.

*Proof.* (i) By lemma 5.1, $q_{\min} \le 1/E_{N1} = \prod_l D_l / \prod_l X_l$, and the products telescope:

$$\frac{\prod_{l=1}^{N} D_l}{\prod_{l=1}^{N-1} X_l} = \frac{|\beta|^N}{|\alpha|^{N-1}} \cdot \frac{N!\sqrt{N!\,(2N+1)!/(N+1)!}}{\sqrt{(N-1)!^2\, N!\,(2N)!/(N+1)!}} = |\alpha|\, |t|^N\, N\sqrt{2N+1}.$$

For $|t| > 1$ exchange the two pairs (lemma 4.1): $A_{\alpha,\beta}$ becomes $A_{\beta,\alpha}$ on the exchanged spins, so the same family with the pairs exchanged works.

(ii) The upper bound is (i). For the lower bound write $E_{ij}/E_{N1}$ as a product of ratios over the $a = j-1$ first and $b = N-i$ last links. With $(\alpha, \beta) = (4, 2)$, $(X_l/D_l)^2 = 4(N-l)(l+1)/(l(N+1-l)) \ge 2$ because $2(N-l)(l+1) - l(N+1-l) = l(N-l) + 2N - 3l > 0$, and $(X_l/D_{l+1})^2 = 4l(N+l+1)/((l+1)(N+l+2)) \ge 8/5$. Hence $E_{ij} \le E_{N1}\, (5/8)^{(a+b)/2}$, so $\|E\|_F \le E_{N1}/(1 - 5/8)$ and $q_{\min} \ge 3/8 P_N$. For the limit, $E/E_{N1}$ has entries $f_N(a) g_N(b)$ that are dominated by the square-summable $(5/8)^{(a+b)/2}$ and converge to $2^{-a}(a+1)^{-1/2}$ and $2^{-b}$; so $E/E_{N1}$ converges in Hilbert–Schmidt norm to a rank-one matrix of norm $\sqrt{4 \ln 4/3}\sqrt{4/3} = \kappa^{-1}$. The last statement follows from theorem 1.1 and lemma 4.1: every

embedding in general position whose $\sigma$ contains $\pm 4$ is a relabelling of one with $(\alpha, \beta) = (4, 2)$ and the pair $(2, 3)$ inner. □

The family of (ii) is Brunnemann and Rideout's numerical minimiser for $2j_{\max} \le 9$ [14, Table 5]; beyond that the minimisers have $j_1 = 3/2, 5/2, \ldots$ and decay somewhat faster (our reproduction of their Table 5 in section 2). In their variables the family gives $\lambda_V = \sqrt{q_{\min}} \asymp j^{3/4} e^{-(\ln 2) j}$, with $\ln 2 = 0.693$, against their fit $\lambda_V \approx 30\, e^{-0.79\, j_{\max}}$ to the smallest eigenvalue over all five-valent sign vectors [13, Table 3]. The larger fitted rate is consistent with the faster decay of the later minimisers; we have not fitted it ourselves. Over $2 \le N \le 400$ the ratio $q_{\min}/P_N$ lies in $[0.712, 0.807]$ ($N = 1$ would put spin 0 on a leg), so the constant $3/8$ is loose and the limit $\kappa$ is approached at rate $O(1/N)$.

**Sorted spins.** In Brunnemann and Rideout's 2007 convention each $\sigma$ is attached to sorted spins. Theorem 5.3 then gives $\inf q_{\min} = 0$ for the six sorted $\sigma$ containing $\pm 4$ whose distinguished edge is the smallest spin and whose heavy pair contains the second smallest, namely $(4, 2, 2, 0)$, $(4, -2, -2, 0)$, $(2, \pm 4, -2, 0)$ and $(4, 4, \pm 2, 0)$ up to sign, because the sorted tuple $(1/2, j - 1/2, j, j, j)$ realises the family. For the other 54 sorted $\sigma$ with $\pm 4$ we have only numerics: an exhaustive sorted search to $2j_{\max} = 34$ finds every one of them decreasing, by factors between $7 \times 10^{-4}$ and 0.16 between $2j_{\max} = 22$ and 34, including the fourteen that a shorter window had suggested were constant.

## 5.3 ONE PRODUCT: THE FOUR-VALENT LAW IN DISGUISE

*Theorem 5.4 — One triple product*

For $t = 0$, $A = -8i\, J_1 \cdot (J_2 \times J_3)$, and every nonzero eigenvalue satisfies $q \ge 2\sqrt{3}$, with equality exactly when $j_1 = j_2 = j_3 = 1/2$ and $|j_4 - j_5| = 1/2$.

*Proof.* The product $J_1 \cdot (J_2 \times J_3)$ commutes with $C = J_1 + J_2 + J_3$, so the invariant space splits over the value $c$ of $|C|$ into $\mathrm{Inv}(j_1, j_2, j_3, c) \otimes \mathrm{Inv}(c, j_4, j_5)$, with the second factor of dimension at most one. On a block of dimension at least two, theorem 3.1 gives $|\lambda| \ge 1/2\sqrt{m(m+1)}$ with $m = \max(j_1, j_2, j_3, c) \ge 1/2$, so $q = 8|\lambda| \ge 4\sqrt{m(m+1)} \ge 2\sqrt{3}$. Equality needs $m = 1/2$ and the equality case of theorem 3.1, that is $j_1 = j_2 = j_3 = c = 1/2$; and $c = 1/2$ occurs in $j_4 \otimes j_5$ exactly when $|j_4 - j_5| = 1/2$. □

Brunnemann and Rideout observed numerically that this configuration "leads to a smallest non-zero eigenvalue of $\sqrt[4]{12}$ for every $j_{\max}$" at valences five to seven [15, p. 35]; that is $\lambda_V = \sqrt{2\sqrt{3}}$, and theorem 5.4 proves it at valence five, with the equality set. In the sorted convention the constant depends on $\sigma$. If the triple product involves the largest sorted spin, then $m \ge j_{\max}$, and $m = 1/2$ would force all five spins to be $1/2$, which admit no invariant vector; so $m \ge 1$ and $q \ge 4\sqrt{2}$. This happens for six of the ten sorted

$\sigma$ of this orbit, one for each triple through the largest spin, and for the other four the sorted infimum is $2\sqrt{3}$. An exhaustive sorted search over doubled spins up to 6 finds exactly these two values, $4\sqrt{2}$ attained at $(1/2, 1/2, 1/2, 1/2, 1)$ or $(1/2, 1/2, 1, 1, 1)$ and $2\sqrt{3}$ at $(1/2, 1/2, 1/2, 1/2, 1)$.

## 6 EQUAL WEIGHTS: THE GAP SURVIVES

This section is about the one operator that section 5 leaves open,

$$T = J_1 \cdot (J_2 \times J_3 + J_4 \times J_5), \qquad A = -8iT, \qquad q = 8|\text{eigenvalue of } T|,$$

on $\mathrm{Inv}(j_1 \otimes \cdots \otimes j_5)$, with $s = j_1$ the distinguished spin.

The chain of figure 3 already says why equal weights should keep the gap: the inner pair's links are the weak ones, so the dimerisation is trivial and nothing localises at the ends. For $j_1 = 1/2$ that picture is the whole story. For larger $j_1$ the chain widens into the strip of figure 2, and the question is whether anything in two dimensions can do what the topological dimerisation did in one.

The answer we can prove is no on every class listed below, and the plan is the same each time. Up to a positive rescaling, the strip carries two Stark matrices, whose spectra are integers (section 6.1). Where the triangle inequalities do not cut the strip, that settles the matter (section 6.2). Where they do, we look, for a suitable pairing of the legs, for a one-dimensional object (a chain, a two-point pencil, a pair of coupled chains, a tridiagonal stage) whose explicit inverse is a product of link ratios, and show that the ratios do not compound. That is the quantitative form of trivial dimerisation. What changes from slice to slice is only the shape of the one-dimensional object: for $j_1 = 1/2$ the strip is a chain; for $j_1 = 1$ one colour class sits on a single diagonal and the problem becomes a pencil; for half-integer $j_1$ the diagonals pair up into zigzag chains coupled through diagonal matrices; for integer $j_1 \geq 2$ one diagonal is left over and is shared. The difficulty grows with $j_1$ because the number of coupled chains does, and the losses of each coupling compound: at $j_1 = 3/2$ the bound left over is close enough to the threshold that a finite range of parameters has to be checked by computer, and at $j_1 = 9/2$ the simple chained inequality no longer certifies every tuple.

*Theorem 6.1 — The gap at equal weights*

Every nonzero eigenvalue of $8T$ satisfies $q \geq q_*$ in each of the following cases.

(a) $j_1 = 1/2$, with $q_* = 6\sqrt{2}/\pi = 2.7009$ (theorem 6.10).

(b) $j_1 = 1$, with $q_* = 4.558$ (theorem 6.12).

(c) $j_1 = 3/2$, with $q_* = 3.7$ (theorem 6.14), with a finite parameter box certified in outward-rounded interval arithmetic.

*Theorem 6.1 — The gap at equal weights (continued)*

(d) $j_1 = 2$, with $q_* = 4.5579607$ (theorem 6.15).

(e) Any $j_1$, if for some pairing of the legs 2–5 the recoupling region has a closed colour class and either the two pair dimensions have odd sum or one pair contains a spin $1/2$; then $q_* = 2\sqrt{\mu} \geq 4\sqrt{2s(s+1)/\pi}$, with $\mu$ the rescaling constant of lemma 6.4 (theorem 6.5). For integer $j_1$ this covers every tuple except those whose legs 2–5 are all integers or all half-integers at least $3/2$.

(f) Any $j_1$, if one of the legs 2–5 has spin $1/2$; then $q_* = 2.7009$, and $q_* = 4\sqrt{2}$ if $j_1 \geq 1$ (corollary 6.8).

(g) Any $j_1$, if three of the legs 2–5 have spin at most 3; then $q_* = 4.5579607$, and moreover $q \geq 4\sqrt{10 - 6\sqrt{2}}\, j_1 - 1370$, with a slope that cannot be improved (theorem 7.3).

(h) Every tuple with all spins at most 19, and every tuple with $j_1 = 5/2$ and all spins at most 35, with $q_* = 4.5579607$ (finite certificates, section 10).

| CASE | PROVED $q_*$ | MIN. FOUND | METHOD |
|---|---|---|---|
| (a) $j_1 = 1/2$ | 2.7009 | 4.5579607 | analytic; two constants in interval arithmetic |
| (b) $j_1 = 1$ | 4.558 | 7.3534 | analytic; finite part exact |
| (c) $j_1 = 3/2$ | 3.7 | 7.4763 | analytic; finite box in interval arithmetic |
| (d) $j_1 = 2$ | 4.5579607 | 10.5410 | analytic; finite box in interval arithmetic |
| (e) closed colour class | $4\sqrt{2s(s+1)/\pi}$ | — | analytic |
| (f) a leg of spin $1/2$ | 2.7009; $4\sqrt{2}$ for $j_1 \geq 1$ | — | analytic; windows $5 \leq N \leq 149$ in interval arithmetic |
| (g) three legs at most 3 | 4.5579607 and $4\sqrt{10 - 6\sqrt{2}}\, j_1 - 1370$ | — | analytic; exact and interval certificates |
| (h) spins at most 19; $j_1 = 5/2$, spins at most 35 | 4.5579607 | 4.5579607 | rigorous floating point; exact |

The column "min. found" gives the smallest value over all tuples of the slice with spins at most 19; it is not a proved infimum. The smallest value we have found anywhere is $2\sqrt{18 - 2\sqrt{41}} = 4.5579607392$, at $(1/2; 1/2, 1; 1, 1)$, where the operator is a four-site chain with squared links $32, 32, 80$ and characteristic polynomial $x^4 - 144x^2 + 2560$; the minimum at $j_1 = 5/2$ over the same range is 12.8129. So none of the proved constants is

the minimum of its slice; 4.5579607 is, up to $4 \times 10^{-8}$, the conjectured global minimum of section 11, used as a common threshold.

### 6.1 THE GUIDING PICTURE: TWO STARK MATRICES AND A CUT

Proposition 4.4 writes every link of $T$ as a Stark link of one pair times $\frac{1}{4}\sqrt{F\,G}$, where $F$ depends only on the diagonal $d = b - c$ and $G$ only on the antidiagonal $u = b + c$. A function of $d$ times a function of $u$ is exactly what a diagonal congruence (a rescaling of the sites) can produce, so the Heron factors can be scaled away. Look at figure 2 with this in mind: every horizontal link carries $h_1(b)$, every vertical link $h_2(c)$, and the factors $\sqrt{F(d \mp \frac{1}{2})}$ and $\sqrt{G(u)}$ are the same along each diagonal and each antidiagonal. Rescale each site by a number that depends on its two diagonals and these factors disappear; what remains is $h_1 \otimes 1 + 1 \otimes h_2$, the Kronecker sum of the two Stark matrices, with the links that cross the walls or the corner removed. For $s = \frac{1}{2}$ the strip is the zigzag chain of section 5, and the rescaled chain has the Stark links of the two pairs, alternately.

*Theorem 6.2 — Heron–Stark congruence*

For every spin tuple and the pairing (23|45), the entrywise nonnegative matrix $W$ of $T$ satisfies $W = \frac{1}{4}SW_0S$, where $W_0 = P_R K P_R$ is the compression to the region $R$ of the Kronecker sum $K = H(n_1, m_1) \otimes 1 + 1 \otimes H(n_2, m_2)$ on the rectangle, and $S = \operatorname{diag}(\vartheta)$ with $\vartheta(b,c)^2 = \phi(d)\,\omega(u) > 0$, where $\phi(d)\phi(d+1) = F(d + \frac{1}{2})$ and $\omega(u-1)\omega(u) = G(u)$.

*Proof.* An $X$-link joins $(b-1, c)$ and $(b, c)$, so it changes $d$ from $d-1$ to $d$ and $u$ from $u-1$ to $u$, and $\vartheta\vartheta' = \sqrt{\phi(d-1)\phi(d)\omega(u-1)\omega(u)} = \sqrt{F(d-\frac{1}{2})G(u)}$; a $D$-link gives $\sqrt{F(d+\frac{1}{2})G(u)}$ in the same way. By proposition 4.4 what remains of each link is $\frac{1}{4}h_1(b)$ or $\frac{1}{4}h_2(c)$, the link of $K$. Links of $K$ that leave $R$ are absent from $W_0$ by definition, and $F$, $G$ are positive inside $R$. □

So $T$ and the compressed Kronecker sum have the same inertia and the same nullity; their kernels correspond through $\ker W = S^{-1} \ker W_0$. By lemma 3.2 the uncompressed sum $K$ has spectrum $\{\lambda + \lambda'\}$ with $\lambda, \lambda'$ running over two Stark spectra: odd integers if the pair dimensions $N_P = n_P - m_P = 2\min(j_a, j_b) + 1$ have odd sum, even integers (with a kernel) if they have even sum. All the difficulty of the two-dimensional problem is in the cut $P_R$, the triangle inequalities of the recoupling, and in how much the rescaling $S$ costs.

**Lemma 6.3** (Scaling with kernels). For any real matrix $B$ and positive diagonal $S_E$, $S_O$, every nonzero singular value of $S_E B S_O$ is at least $\min S_E \cdot \min S_O$ times the smallest nonzero singular value of $B$.

*Proof.* Put $M = B^T S_E^2 B$, which is positive semidefinite. The nonzero eigenvalues of $S_O M S_O$ are those of $M^{1/2} S_O^2 M^{1/2}$. This matrix dominates $(\min S_O)^2 M$ and has the kernel of $M$, so Courant–Fischer on the range compares its nonzero eigenvalues with those of $M$. Apply this twice. □

**Lemma 6.4** (The cost of the rescaling). Let $\mu = \min_{\text{one colour}} \vartheta^2 \cdot \min_{\text{other colour}} \vartheta^2$. Then $\mu \geq (2s+2)\Phi_s \geq 8s(s+1)/\pi$, where $\Phi_s = 2s$ for integer $s$ and $\Phi_s = 2s/\mathscr{W}_{s-1/2}$ for half-integer $s$, with Wallis's product $\mathscr{W}_K = \prod_{l \leq K} 4l^2/(4l^2-1) < \pi/2$. Hence

$$q \;\geq\; 2\sqrt{\mu}\,\lambda(W_0),$$

with $\lambda(W_0)$ the smallest nonzero singular value of the bipartite block of $W_0$.

*Proof.* The free constants of $\phi$ and $\omega$ cancel in $\mu$. Since $G$ increases for $u > -1/2$, $\omega(u+2)/\omega(u) = G(u+2)/G(u+1) \geq 1$, so the minima over the two parities of $u$ are $\omega(u_{\min})$ and $\omega(u_{\min}+1)$, with product $G(u_{\min}+1) \geq 2s+2$ because $u_{\min} \geq s$. Along each parity class of $d$, $\phi(d+2)/\phi(d) = F(d+3/2)/F(d+1/2)$ is nonincreasing because $F$ is log-concave, so $\log\phi$ is concave there and its minimum is at an end; the end values over $[-s, s]$ are equal for integer $s$ (by $F(x) = F(-x)$), giving $\Phi_s = F(s-1/2) = 2s$, and differ by Wallis's product $\mathscr{W}_{s-1/2}$ for half-integer $s$. A site's colour, the parity of $u-s$, fixes the parity of $u$; and since $u - d = 2c$ and $c$ runs over $m_2 + \mathbb{Z}$, the parity of $d$ is then fixed too. So $\mu$ is at least the product. The last statement is theorem 6.2 and lemma 6.3. □

## 6.2 CLOSED REGIONS AND THE LADDER

The cut can cost something only where it removes a link of $K$, and it removes links only at the sites next to the walls $d = \pm s$ and the corner $u = s$; the outer edges of the rectangle are not cuts, since $K$ itself lives on the rectangle. A colour class $C$ of the region is *closed* if every neighbour in the rectangle of every site of $C \cap R$ lies in $R$. The cut sites of the region lie on the wall $d = s$, which has colour $2(j_4 + j_5) \bmod 2$; on the wall $d = -s$, of colour $2(j_2 + j_3) \bmod 2$; and at the corner $u = s$, of colour 0 (the colour being the parity of $u - s$). The class of colour $k$ is closed exactly when no cut site has colour $k$.

*Theorem 6.5 — Closed regions*

Suppose a colour class $C$ is closed, and either $N_1 + N_2$ is odd or one of the two pairs contains a spin $1/2$. Then every nonzero singular value of the bipartite block of $W_0$ is at least 1, and every nonzero eigenvalue satisfies $q \geq 2\sqrt{\mu} \geq 4\sqrt{2s(s+1)/\pi}$.

*Proof.* For $\psi$ supported on $C \cap R$, $K\psi$ is supported on neighbours of $C \cap R$, all in $R$ by closedness, so the block of $W_0$ from $C$ is $K$ restricted to $\ell^2(C \cap R)$. If $N_1 + N_2$ is odd the spectrum of $K$ consists of odd integers, so $\|K\psi\| \geq \|\psi\|$. If the first pair is $(1/2, j)$, its Stark

matrix is $2\times 2$ with link 1, the rectangle is two rails $b = j \pm 1/2$, the two sites of a position $c$ have opposite colours, and $K$ restricted to $C$ acts as $\hat{\psi} \mapsto (H' \pm 1)\hat{\psi}$ on the positions $J$ of $C$, where $H'$ is the Stark matrix of the other pair. If $N'$ is odd, $H' \pm 1$ has odd-integer spectrum. If $N'$ is even and $J$ is the whole range of $H'$, the nonzero singular values of $H' \pm 1$ are at least 2. If $J$ misses an end, lemma 6.6 gives $\|(H' \pm 1)\hat{\psi}\| \geq \|\hat{\psi}\|$. In every case lemma 6.4 finishes. □

The one step in this proof that is not spectral bookkeeping is the case of a chain with an end removed. Removing an end site from a Stark matrix cannot create a near-zero mode, because the end components of its eigenvectors are Clebsch–Gordan weights, largest in the middle of the spectrum and decreasing away from it; so a vector that avoids the end cannot concentrate on the eigenvalue closest to the shift.

**Lemma 6.6** (One-sided Stark bound). Let $H = H(n,m)$ have even dimension $N$. If $\psi$ vanishes at one end site of the chain, then $\|(H \pm 1)\psi\| \geq \|\psi\|$.

The proof is in appendix B: $Q = (H-1)^2 - 1 = H(H-2)$ has exactly one negative eigenvalue, at the eigenvalue 1 of $H$, and a form with one negative eigenvalue is nonnegative on the hyperplane $e^\perp$ if and only if $e^T Q^{-1} e \leq 0$; for the end vector $e$ this is a sum over the eigenvalues weighted by the squared end components, and their monotonicity settles the sign.

When no colour class is closed but a leg carries spin $1/2$, the block is not a restriction of $H' \pm 1$ but a shifted window of it: rows $[a+1, b+1]$ against columns $[a,b]$. These windows have a uniform bound too.

*Theorem 6.7 — Shifted Stark windows*

For every Stark matrix $H(n,m)$ and every window $0 \leq a \leq b \leq N-2$, the smallest singular value of rows $[a+1, b+1]$ times columns $[a,b]$ of $H - 1$ is at least $c_* = \sqrt{(15 - \sqrt{97})/6} = 0.92657$, with equality at $N = 3$, $m = 0$.

The proof (appendix B) reduces every window to the full window $[0, N-2]$, turns the bound into an explicit inequality between two end-weighted resolvent sums, proves it for even $N$ and for odd $N \geq 151$ by hand, and certifies odd $5 \leq N \leq 149$ in interval arithmetic. A fully analytic constant $2/\pi \arcsin 3/4 = 0.5399$ holds for all windows.

**Corollary 6.8** (Ladders). If one of the legs 2–5 has spin $1/2$, then $q \geq 6\sqrt{2}/\pi$ if $j_1 = 1/2$, $q \geq 4\sqrt{2}$ if $j_1$ is an integer, and $q \geq 2\sqrt{\mu}\, c_* \geq 6.215$ if $j_1 \geq 3/2$ is a half-integer.

*Proof.* For $j_1 = 1/2$ this is theorem 6.10. For integer $j_1$ the number of half-integer spins among the legs 2–5 is even; pairing them together puts both walls in colour 0, the colour of the corner, so the other class is closed, and theorem 6.5 applies with the spin-$1/2$ pair. For half-integer $j_1 \geq 3/2$ either some class is closed, or (lemma B.2 in appendix B) the

block is a shifted window of $H' + 1$, and theorem 6.7 and lemma 6.4 give $q \geq 2\sqrt{\mu}\, c_*$ with $\mu \geq 45/4$. □

### 6.3 THE FIRST SLICE: $j_1 = 1/2$

For $j_1 = 1/2$ the region is the chain of section 5, and the question is the dimerisation of figure 3 for general spins. We use doubled labels and work with the links of $8T$ itself, so that the eigenvalues are $q$ directly, with no factor 8: let $m_X = 2|j_2 - j_3|$, $n_X = 2(j_2 + j_3 + 1)$, likewise $m_D, n_D$, and $m = \max(m_X, m_D)$, $n = \min(n_X, n_D)$. The links carry the labels $l = m + 1, \dots, n - 1$ in order and alternate in type, and a link of type $P$ of $8T$ has squared weight $c^2 = \tfrac{1}{4}(n_P^2 - l^2)(l^2 - m_P^2)$ (proposition 4.4 with $s = 1/2$).

**Lemma 6.9** (Inner and outer pairs). If the chain has an even number $N = n - m$ of sites, its odd-numbered links all belong to one pair $O$ with $m_O \leq m - 1$ and $n_O \geq n + 1$, and the other pair owns both $m$ and $n$. For an even link at label $L = m + 2\tau$ followed by an odd link,

$$\frac{c_{2\tau}^2}{c_{2\tau+1}^2} \leq \frac{n + L}{n + L + 2} \cdot \frac{L - m}{L - m + 2} \leq \frac{\tau}{\tau + 1}.$$

If $N$ is odd, the odd links belong to the pair owning $n$ and the even links to the pair owning $m$.

*Theorem 6.10 — The spin-1/2 slice*

For $j_1 = 1/2$ and all other spins, every nonzero eigenvalue satisfies $q \geq 6\sqrt{2}/\pi > 2.70$.

The proofs are in appendix C. For even $N$ the ratio bound telescopes, so $E_{ij}^2 \leq (j/i)/c_{2j-1}^2$, and a lower bound on the odd links turns $\|E\|_F^2$ into a harmonic double sum. For odd $N$ the ratios increase along the chain; deleting the row where they cross 1 splits the chain into two even pieces, one read from each end. The smallest value found on this slice, over spins at most 19, is 4.5579607, so the constant is not sharp.

### 6.4 THE SECOND SLICE: $j_1 = 1$

For $s = 1$ the strip has three diagonals, $d = -1, 0, 1$, and the two colour classes are the middle diagonal and the two walls. So the bipartite block $B$ goes from the diagonal to the walls, and $B^T B$ lives on the diagonal alone. A diagonal site $(\ell, \ell)$ touches only the four wall sites $(\ell \pm 1, \ell)$, $(\ell, \ell \pm 1)$, and two neighbouring diagonal sites share exactly two wall sites; so $B^T B$ is tridiagonal: the two-dimensional problem at $j_1 = 1$ is a one-dimensional birth–death problem, which proposition 6.11 writes down. Each pair of neighbouring diagonal sites sees its two shared wall sites through two rows, one weighted by $h_1$ and one by $h_2$. In the nonnegative gauge of proposition 4.4 every link is positive, so every plaquette (four sites around a unit square) has a positive product of links; we call such

a gauge flux-free. With the sign $(-1)^\ell$ on the diagonal, the two rows then combine into the square of a difference, $A^+(\Phi_\ell - \Phi_{\ell-1})^2$, plus a nonnegative term. With a plaquette of negative product the cross terms would cancel instead. The difference form is a weighted Hardy problem, and Hardy's inequality bounds it.

*Proposition 6.11 — The pencil at $j_1 = 1$*

Fix a pairing with pairs $(n_1, m_1)$, $(n_2, m_2)$. Let $\mathrm{lo} = \max(m_1, m_2)$, raised to 1 if both are 0, and $\mathrm{hi} = \min(n_1, n_2) - 1$. Put $P_i(e) = \max(0, (n_i^2 - e^2)(e^2 - m_i^2))$,

$$A_e^+ = \frac{(\sqrt{P_1(e)} + \sqrt{P_2(e)})^2}{8e}, \qquad A_e^- = \frac{(\sqrt{P_1(e)} - \sqrt{P_2(e)})^2}{8e}, \qquad \mu_\ell = \frac{2\ell + 1}{\ell(\ell + 1)}.$$

Then the nonzero values of $(q/8)^2$ are the nonzero eigenvalues of the pencil

$$\mathscr{E}(\Phi) = \sum_{e=\mathrm{lo}}^{\mathrm{hi}+1} \Big[ A_e^+ (\Phi_e - \Phi_{e-1})^2 + A_e^- (\Phi_e + \Phi_{e-1})^2 \Big] \quad \text{against} \quad \sum_\ell \mu_\ell \Phi_\ell^2,$$

with $\Phi_{\mathrm{lo}-1} = \Phi_{\mathrm{hi}+1} = 0$.

*Proof.* For $s = 1$ every wall link has $F(\pm 1/2) = 2$. In the gauge $(-1)^\ell$ the two rows that see $\phi_{\ell-1}$ and $\phi_\ell$ give $\frac{\sqrt{2}}{4}(h_1\xi - h_2 v)$ and $\frac{\sqrt{2}}{4}(h_2\xi - h_1 v)$ with $\xi = \sqrt{G(2\ell - 1)}\phi_{\ell-1}$, $v = \sqrt{G(2\ell)}\phi_\ell$, and $(h_1\xi - h_2 v)^2 + (h_2\xi - h_1 v)^2 = 1/2[(h_1 + h_2)^2(\xi - v)^2 + (h_1 - h_2)^2(\xi + v)^2]$. Substituting $\phi_\ell = z_\ell \Phi_\ell$ with $z_\ell^2 = \mu_\ell$, which solves $\sqrt{G(2\ell - 1)}z_{\ell-1} = \sqrt{G(2\ell)}z_\ell$, and $(4\ell^2 - 1)h_i^2 = P_i$ gives the pencil. Missing sites and links are exactly the zeros of $P_i$ and of $G(1)$. □

*Theorem 6.12 — The slice $j_1 = 1$*

For $j_1 = 1$ and all other spins every nonzero eigenvalue satisfies $q \geq 4.558$.

The proof (appendix D) uses the Hardy inequality with constant 4, a nodal-domain argument for identical pairs (whose kernel is the constants), and explicit estimates of the Hardy product when $\max(n_1, n_2) \geq 33$; the remaining finitely many pairs of pairs are certified by exact $LDL^T$ factorisation.

### 6.5 HALF-INTEGER $j_1$: ZIGZAGS

For half-integer $s$ the strip $|d| \le s$ has an even number $2s+1$ of diagonals, and they pair up: the sites on two neighbouring diagonals $d_k$, $d_k+1$, with $d_k = -s+2k$, form a single zigzag path, which we call a *zigzag*. Rows of the bipartite block are the sites with $d+s$ even, columns those with $d+s$ odd. Each zigzag is then a chain of the kind solved in section 5, and neighbouring zigzags are coupled through the links between their diagonals. The picture to keep is a stack of such chains, glued by couplings whose Gram matrices are a constant multiple of the neighbouring chain's, up to a diagonal correction.

*Proposition 6.13 — Zigzag structure*

For half-integer $s$ the bipartite block $B$ is block lower-bidiagonal in the zigzag index: each diagonal block $D_k$ is a single path alternating between rows and columns, with imbalance #rows − #columns $\in \{-1,0,1\}$, and the off-diagonal block $N_k$ couples zigzag $k$ to zigzag $k-1$. Moreover, for adjacent zigzags,

$$N_k^T N_k = \mathsf{r}_k\, D_{k-1}^T D_{k-1} + \Lambda_k, \qquad \mathsf{r}_k = \frac{F(d_k - 1/2)}{F(d_k - 3/2)},$$

with $\Lambda_k$ diagonal.

*Proof.* A row on $d_k$ has its neighbours on $d_k \pm 1$; those on $d_k+1$ are the columns of zigzag $k$ and those on $d_k-1$ the columns of zigzag $k-1$. On each diagonal the sites form an interval, which gives the path structure and the imbalance. For the identity, an off-diagonal entry of $D_{k-1}^T D_{k-1}$ (respectively $N_k^T N_k$) between two adjacent columns is the product of the two links at their unique shared row of zigzag $k-1$ (respectively $k$). These two rows, $(b,c)$ on $d$ and $(b+1,c-1)$ on $d+2$, form a plaquette with the two columns, and by proposition 4.4 the two link products contain the same Stark and $G$ factors and differ only in $F(d+1/2)$ against $F(d+3/2)$. □

For $j_1 = 3/2$ there are two zigzags, and in a suitable ordered pairing neither has surplus columns, so $B$ is injective and $\lambda(T) = \sigma_{\min}(B)$. Splitting $\|B(x,y)\|^2 = \|D_0 x\|^2 + \|Nx + D_1 y\|^2$ and giving up a fraction of the first term to absorb the coupling gives, for every $\tau > 0$,

$$\sigma_{\min}(B)^2 \;\ge\; \min\Big(\lambda_{\min}\big(D_0^T D_0 - \tau N^T N\big),\; \tfrac{\tau}{1+\tau}\lambda_{\min}(D_1^T D_1)\Big), \tag{8}$$

and by proposition 6.13, $D_0^T D_0 - \tau N^T N = (1 - 4/3\tau) D_0^T D_0 - \tau\Lambda$. So everything reduces to two single chains, whose inverses are explicit products of link ratios, and one diagonal correction. The price is the factor $\tau/(1+\tau)$, paid once at $j_1 = 3/2$. What is left is a bound of 3.777 at the worst tuple, against a computed value of 8.87. The closed-form estimates that make this bound uniform hold only for large spins (beyond $n_1 = 4000$

or $10^5$, depending on the regime), so below that the bound is evaluated parameter by parameter, by computer.

*Theorem 6.14 — The slice $j_1 = 3/2$*

For $j_1 = 3/2$ and all other spins every nonzero eigenvalue satisfies $q \geq 3.7$. This is proved with a finite parameter box certified in outward-rounded interval arithmetic. Independently of that box, every tuple with $j_1 = 3/2$ and spins at most 20 satisfies $q \geq 7.18$, by exact $LDL^T$ certificates.

The proof (appendix E) sorts the tuples into eight regimes by the gaps between the sorted spins of legs 2–5, chooses in each regime a pairing in which both zigzags have no surplus column, bounds the two chains and the coupling by explicit telescoping products, evaluates the resulting bound on a finite box of parameters in outward-rounded interval arithmetic, and controls everything beyond the box by written tail lemmas. The binding regime is the family $(3/2; j, j+1/2, j+1/2, j+1/2)$ at $j = 1$.

## 6.6 THE SLICE $j_1 = 2$: A TRIANGULAR REDUCTION

For integer $j_1$ the strip has an odd number of diagonals, so pairing them into zigzags leaves one over. Instead, take the colour class whose diagonals are $d = \pm 1, \pm 3, \ldots$ as columns and let each row diagonal be *owned* by one of its two neighbouring column diagonals: the owner keeps most of that row's contribution to $\|Bx\|^2$ and the other neighbour pays a small penalty. Each column diagonal then carries one tridiagonal problem of the kind solved above. For $j_1 = 2$ there are only the two column diagonals $d = \pm 1$, the walls $d = \pm 2$ are owned by their single neighbours, and only the middle row diagonal $d = 0$ is shared.

Since $T$ links only adjacent diagonals,

$$\|Bx\|^2 = \sum_{\rho} \left\| Z(\rho - 1, \rho) x_{\rho-1} + Z(\rho + 1, \rho) x_{\rho+1} \right\|^2,$$

where $Z(\delta, \rho)$ is the zigzag block from column diagonal $\delta$ to row diagonal $\rho$, a path matrix; and the Gram matrices of the two zigzags at one column diagonal are proportional off the diagonal, $Z(\delta, \delta+1)^T Z(\delta, \delta+1) - \mathrm{r}_\delta Z(\delta, \delta-1)^T Z(\delta, \delta-1) = \mathrm{diag}(\Lambda)$, by the same plaquette argument as proposition 6.13. Applying $\|a+b\|^2 \geq (1-\theta)\|a\|^2 - (\theta^{-1}-1)\|b\|^2$, $0 < \theta < 1$, row diagonal by row diagonal reduces positivity of $B^T B - \lambda$ to positivity of one tridiagonal matrix per column diagonal.

*Theorem 6.15 — The slice $j_1 = 2$*

For $j_1 = 2$ and all other spins every nonzero eigenvalue satisfies $q \geq 4.5579607$.

The proof (appendix F) routes each tuple to a pairing in which the wall chain on the wall $d = -2$ has no surplus columns, and in one of the three cases neither wall chain does, bounds the chains and the one coupling in closed form, and closes with an interval-arithmetic box and an explicit tail. Tuples outside the even-sum class are already covered by theorem 6.5, and tuples with all spins at most 19 by the finite certificate.

A proof for all $j_1$ along these lines would need either a uniform version of lemma 6.4 for the compressed Kronecker sum or the large-$j_1$ structure of the next section; section 11 describes how far both go.

## 7 LARGE SPINS AT EQUAL WEIGHTS

The slices of section 6 fix $j_1$. This section asks how the gap at $t = 1$ grows, and in which spin. The answer is that it grows linearly in the distinguished spin $j_1$, with a sharp slope that we compute exactly, but only like $M/\ln M$ in the largest spin $M = 2j_{\max}$ when $j_1$ is fixed. The first statement is proved on the class where three of the other legs are bounded; the second is proved as an upper bound.

### 7.1 THE GUIDING PICTURE

Figure 5 shows the two growth laws. Along $(1/2; j - 1/2, j, j, j)$, the per-level minimisers of the exhaustive surveys, $q_{\min} \ln M/M$ rises slowly, from 2.97 at $M = 60$ to 3.13 at $M = 4096$. Along $(s; 1/2, 1, 1, s - 1/2)$, the per-$j_1$ minimisers, $q_{\min}$ is a line of slope $4\sqrt{10 - 6\sqrt{2}}$ in $s$ up to a bounded correction.

Both laws have a structural explanation. When $j_1$ is large and three of the other legs are small, the fourth leg is forced to within a bounded distance of $j_1$, and $T/j_1$ converges spectrally to an operator of the three small spins alone: minus the $z$-component of $U = J_2 \times J_3 + J_2 \times J_4 + J_3 \times J_4$ on one weight space. Its smallest nonzero eigenvalue is the slope. When $j_1$ is fixed and the other spins are large, the low modes sit in the corner $u \approx s$ of the grid, where the links $\frac{1}{4}h\sqrt{F}\sqrt{G}$ look like $n/8$ times the matrix elements of $J_x$ in spin $s$ (the $\sqrt{F}$) times those of a non-compact $\mathfrak{su}(1,1)$ generator $K_1$ (the $\sqrt{G}$, with Bargmann index $s + 1$). The generalised eigenfunctions of $K_1$ are $u^{-1/2+i\varkappa}$, and cutting $u$ to $[s, n]$ quantises $\varkappa \approx \pi/(2 \ln n)$; with the smallest $|J_x|$-eigenvalue $1/2$ (half-integer $s$) or 1 (integer $s$) this predicts growth like $\pi M/\ln M$ or $2\pi M/\ln M$. The data confirm the order $M/\ln M$, and at the sizes we can compute the constants are close to $\pi$ and $2\pi$, but they do not settle them (see the end of this section). This is a heuristic, but it is also where the upper bound of theorem 7.4 comes from: on the chain of $j_1 = 1/2$ the site index $i$ plays the role of $u$, and the test vector there is $i^{-1/2} \sin(\varkappa \ln i)$ with $\varkappa = \pi/(2 \ln K)$, the real form of the generalised eigenfunction, up to a slowly varying weight.

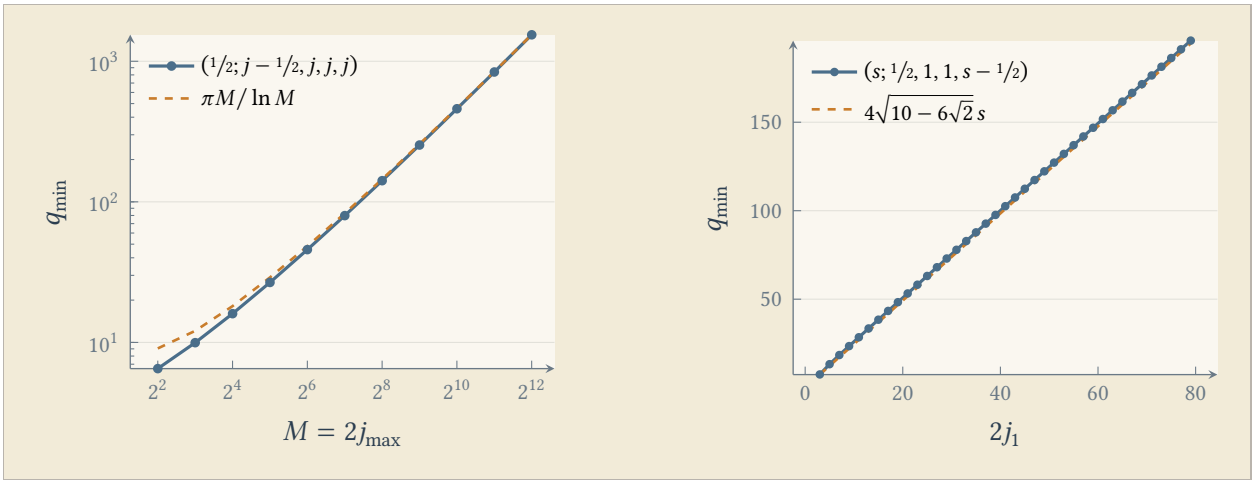



FIG. 5. *The two growth laws at $t = 1$. Left: at fixed $j_1 = 1/2$ the gap grows only like $M/\ln M$ in the largest spin ($M$ up to 4096); the dashed line $\pi M/\ln M$ is the heuristic of the text, which the data cross near $M = 2^{14}$, beyond the plot. Right: along the per-$j_1$ minimisers it grows linearly in $j_1$, with the slope $4\sqrt{10-6\sqrt{2}} = 4.92296$ of theorem 7.2.*

## 7.2 THE THREE-SPIN PENCIL

The idea is to remove the distinguished leg altogether. Removing it turns $T$ into an operator on the other four legs, restricted to the highest-weight vectors of weight $s = j_1$; removing a fifth leg as well leaves an operator on three spins. The price is the projector onto highest-weight vectors, the extremal projector of $\mathfrak{su}(2)$, which is an explicit series in the lowering and raising operators; its matrix elements are the alternating Pochhammer sums below. The result is a generalised eigenvalue problem on a space whose size does not depend on $j_1$.

Single out one of the legs 2–5 and call it leg 5. Write $V = j_2 \otimes j_3 \otimes j_4$, $\mathrm{G} = J_2 + J_3 + J_4$, $U = J_2 \times J_3 + J_2 \times J_4 + J_3 \times J_4$, $M_0 = s - j_5$, and $V[M_0]$ for the weight space $\mathrm{G}_z = M_0$. Put $\chi_k = (-1)^k/(k!(2s+2)_k)$, $\chi'_k = (-1)^k/(k!(2s+3)_k)$, $\mathrm{X} = sU_z + 1/2(U_+\mathrm{G}_- + U_-\mathrm{G}_+)$ and

$$\mathrm{B} = \sum_k \chi_k \mathrm{G}_-^k \mathrm{G}_+^k, \qquad \mathrm{B}' = \sum_k \chi'_k \mathrm{G}_-^k \mathrm{G}_+^k, \qquad \mathrm{A} = \mathrm{X}^*\mathrm{B} - \frac{j_5}{2s+2} U_- \mathrm{B}' \mathrm{G}_+.$$

*Theorem 7.1 — Three-spin pencil*

A is Hermitian and $\mathrm{B} \succeq 0$ on $V[M_0]$, and there $\mathrm{A} = (sU_z + 1/2 U_+\mathrm{G}_-)\mathrm{B}$. On the part of $V[M_0]$ with $\mathrm{G}^2 = g(g+1)$, B is the scalar $\beta_g = (s + j_5 + 1 - g)_{g-M_0}/(2s+2)_{g-M_0} \in [0,1]$, positive exactly when $|s - j_5| \le g \le s + j_5$. The spectrum of $T$ on $\mathrm{Inv}(j_1, \dots, j_5)$, with multiplicities, is the spectrum of $-\mathrm{B}^{-1/2}\mathrm{A}\mathrm{B}^{-1/2}$ on the range of B.

The proof is in appendix G; the theorem was also checked against the full tensor-product construction on 15 tuples (section 10).

As $s \to \infty$ with the three spins and $M_0$ fixed, $\chi_k = O(s^{-k})$ for $k \ge 1$, so $\mathrm{B} \to 1$ and, by the second form of A, $\mathrm{A}/s \to U_z$. So the spectrum of $T/j_1$ converges to that of $-U_z$,

where $U_z$ is the *three-spin chirality* on $(j_2 \otimes j_3 \otimes j_4)_{M_0}$; the sign does not matter for the moduli, because the spectrum of $U_z$ is symmetric.

*Theorem 7.2 — The sharp slope*

On $(1/2 \otimes 1 \otimes 1)_{M_0=1/2}$, $U_z$ has characteristic polynomial $x(4x^4 - 20x^2 + 7)/4$ up to sign, so its smallest nonzero eigenvalue modulus is $\nu_0 = \sqrt{(5 - 3\sqrt{2})/2} = 0.61537$. For every $s \geq 3/2$ the nonzero spectrum of $T$ at $(s; 1/2, 1, 1, s - 1/2)$ is $\pm x_-, \pm x_+$ with

$$x_\pm^2 = \tfrac{5}{2}s^2 + \tfrac{3}{2}s - \tfrac{9}{4} \pm \tfrac{1}{4}(3s + 4)\sqrt{8s^2 - 12s + 5},$$

so $q_{\min} = 8x_- = 8\nu_0\, s + O(1)$. Consequently no bound $q \geq c\, j_1 - C$ valid for all spins can have $c > 8\nu_0 = 4\sqrt{10 - 6\sqrt{2}} = 4.92296$.

*Proof.* For $s \geq 3/2$ the grid has five sites and a single plaquette whose product is a perfect square of a polynomial in $s$; its characteristic polynomial is $x(16x^4 - (80s^2 + 48s - 72)x^2 + 28s^4 + 36s^3 - 29s^2 - 36s + 1)/16$, identically in $s$, and the stated roots follow. The value at $s = 3/2$ is 7.4763 and at $s = 2$ is 10.5410, the minima of these two slices over all spins at most 19. □

## 7.3 THREE BOUNDED LEGS

Let $\mathscr{C}_K$ be the set of tuples in which three of the legs 2–5 have $2j \leq K$, with $j_1$ and the fourth leg arbitrary. By proposition 4.5 the bounded legs may be taken to be $2, 3, 4$.

*Theorem 7.3 — Linear law on three bounded legs*

On $\mathscr{C}_6$, every nonzero eigenvalue of $8T$ satisfies $q \geq 4.5579607$ and

$$q \;\geq\; 8\nu_0\, j_1 - 1370, \qquad 8\nu_0 = 4\sqrt{10 - 6\sqrt{2}} = 4.9229562\ldots$$

The same holds on $\mathscr{C}_2$ and $\mathscr{C}_4$ with 1370 replaced by 97 and 453. By theorem 7.2 the slope cannot be larger.

So on these classes the gap grows linearly in $j_1$ with the sharp slope; the additive constants are crude. The proof (appendix G) has a picture behind it: for $s$ beyond an explicit threshold the grid of a tuple in $\mathscr{C}_6$ has a fixed shape, its links are $s$ times a limit matrix plus a correction bounded in Frobenius norm, and the limit matrix is injective with smallest singular value at least $\nu_0$. For all but two limit matrices this is certified in interval arithmetic with room to spare; the two exceptions are the extremal triple $(1, 1, 1/2)$ of theorem 7.2, where the smallest singular value is exactly $\nu_0$. Weyl's inequality does the rest. Below the threshold there are finitely many tuples, certified individually, and there the linear bound is negative anyway.

*Theorem 7.4 — Sublinear growth in the largest spin*

Along $(1/2; j-1/2, j, j, j)$, with $K = 2j \geq 2$,

$$q_{\min} \;\leq\; 2\pi\,\frac{\sqrt{2j(4j+1)(\ln K + \pi^2/6)}}{(\ln K)^{3/2}},$$

so $q_{\min} \ln M/M$ is bounded and $q_{\min}/j_{\max} \to 0$. Along $(1; j, j, j, j)$, with $n = 2j+1$, $q_{\min} \leq 8\pi n\sqrt{(\ln n + \pi^2/6)/(2\ln^3(n-1))}$.

*Proof.* By lemma 6.9 and proposition 4.4 the chain has $4j$ sites with $a_{2i-1}^2 = 4(2j+1-i)(2j+1+i)i^2$ and $a_{2i-2}^2 = 4(2j+1-i)(2j+i)(i-1)i$, and $B$ is $K \times K$ lower bidiagonal and invertible, so $q_{\min} = \min \|Bx\|/\|x\|$. Take $x_i = (-1)^i z_i \varphi_i$ with $z_i^2 = \prod_{2\leq l\leq i} a_{2l-2}^2/a_{2l-1}^2 = (2j+2)/(i(2j+1+i))$. Then

$$\|Bx\|^2 = a_1^2\varphi_1^2 + \sum_{i\geq 2} 4(2j+2)(2j+1-i)\,i\,(\varphi_i - \varphi_{i-1})^2, \qquad \|x\|^2 = \sum_i \frac{(2j+2)\varphi_i^2}{i(2j+1+i)}.$$

Choose $\varphi_i = \sin(\pi \ln i/(2\ln K))$, so $\varphi_1 = 0$ and $|\Delta\varphi_i| \leq \frac{\pi}{2\ln K}\ln\frac{i}{i-1}$. Since $\ln\frac{i}{i-1} \leq \frac{1}{i-1}$, $i\ln^2\frac{i}{i-1} \leq \ln\frac{i}{i-1} + \frac{1}{(i-1)^2}$, and the sum over $2 \leq i \leq K$ is at most $\ln K + \pi^2/6$. This bounds $\|Bx\|^2 \leq 2\pi^2 j(2j+2)(\ln K + \pi^2/6)/\ln^2 K$, while $\varphi_i^2/i \geq 1/2\int_{i-1}^{i} \varphi^2/x\,dx$ gives $\|x\|^2 \geq (2j+2)\ln K/(4(4j+1))$. The second family is the same computation on the pencil of proposition 6.11 with identical pairs, where $A_e^- = 0$ and the corner end is Dirichlet. □

Numerically both families grow like $M/\ln M$, but the constants are not settled by the sizes we can reach. On the first family $q_{\min}\ln M/M$ is 2.97 at $M = 60$ and 3.13 at $M = 4096$, the end of figure 5; it passes $\pi$ near $M = 2^{14}$ and is 3.151 at $M = 2^{16}$ and 3.154 at $M = 2^{18}$, still rising. On the second family it is 5.52 at $M = 60$, 5.87 at $M = 4096$, 5.97 at $M = 2^{16}$ and 6.00 at $M = 2^{18}$, below $2\pi$ and still rising. So the heuristic constants $\pi$ and $2\pi$ are the right size, and we do not know whether they are the limits. The test vector of the proof overestimates the first family's $q_{\min}$ by a factor 1.31 at $M = 60$, 1.15 at $M = 4096$ and 1.11 at $M = 2^{16}$. For half-integer $j_1 \geq 5/2$ the same corner mechanism is visible in a single one-dimensional chain at the wall: if some ordered pairing has an injective block, then $\lambda(T)$ is at most the smallest singular value of the chain on the rows $d = j_1 - 1$ and the columns $d = j_1$, because the wall columns meet no other rows. We know the decay rate of that chain only numerically.

## 8 VALENCE SIX: A LADDER AND A FACTORIAL

At valence six the sign configurations are too many for a normal form (there are 23 808 of them [14, Table 1]), and we treat only the configuration that Brunnemann and Rideout single out as producing the smallest eigenvalues,

$$\sigma = (2,0,0,0,0,0,0,0,-2,0) \quad \text{in the order } 123,124,125,134,135,145,234,235,245,345,$$

so that $A = 2\hat{q}_{123} - 2\hat{q}_{245} = -8i(X - Y)$ with $X = J_1 \cdot (J_2 \times J_3)$ and $Y = J_2 \cdot (J_4 \times J_5)$. It is realised by the integer tangents $\dot{e}_1 = (-1,2,-3)$, $\dot{e}_2 = (-2,4,1)$, $\dot{e}_3 = (2,-1,-4)$, $\dot{e}_4 = (-3,1,4)$, $\dot{e}_5 = (-4,3,3)$, $\dot{e}_6 = (-2,-3,2)$, all of whose twenty triple determinants are nonzero. The spins are $2j = (1,1,n,n,n+1,n+1)$, Brunnemann and Rideout's minimising spins at even $2j_{\max} = n+1$ [14, Table 9]; at odd $2j_{\max}$ their minimisers are different. Their Table 9 labels the configuration both with $-2$ and with $+2$ in the ninth place; both labels are realisable in general position, and they give the same ladder and identical spectra.

### 8.1 THE PICTURE: TWO ROWS OF DIMERS

The two triple products share leg 2, so the useful labels are the spin $c = |J_1 + J_3| \in \{c_-, c_+\} = \{\frac{n-1}{2}, \frac{n+1}{2}\}$ of legs 1, 3 and the spin $e = |J_4 + J_5|$ of legs 4, 5. In the basis that also fixes $h = |J_2 + J_4 + J_5|$ the product $Y$ is a set of dimers: it pairs $e = h - 1/2$ with $e = h + 1/2$ on each row $c$, with weight $y_h = \tfrac{1}{4}\sqrt{(n+1-h)(n+2+h)h(h+1)}$, and it has exactly two zero rows, at the two ends of the row $c_+$. The product $X$ is a rung: in each block of fixed $e$ it is a rank-two antisymmetric map of fixed size $x = \tfrac{1}{4}\sqrt{n(n+2)}$ between the two rows. So the matrix is a two-row ladder (figure 6). The two zero rows of $Y$ are end modes, and to couple them through the ladder costs a factor $x/y_h$ per dimer, about $1/h$ for $h$ small against $n$. After $n$ dimers the cost is the product, and it can be written down exactly: $\prod_{h=1}^{n}(x/y_h)^2$ is $(n(n+2))^n$ divided by $n!\,(n+1)!\,n!\,(2n+2)!/(n+2)!$, which by Stirling is $\exp(-2n\ln n + O(n))$. So the cost is about $1/n!$, up to a factor $e^{O(n)}$.

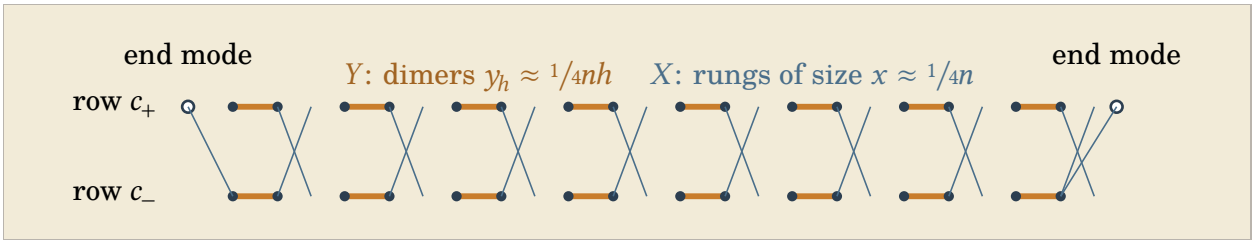



FIG. 6. *The six-valent operator as a ladder. Each row is a chain of dimers of Y; the rungs of X couple the rows only through a fixed direction in each block. The two sites that Y leaves unpaired, at the ends of the upper row, carry the nearly-zero mode; it reaches across through n dimers, each costing a factor of order $x/y_h \approx 1/h$, so the splitting is factorial, not exponential.*

*Theorem 8.1 — Factorial gap closing at valence six*

For the configuration and spins above:

(i) for odd $n$ the operator has trivial kernel, and for even $n$ its kernel is exactly two-dimensional;

(ii) for odd $n \geq 7$,

$$0 < q_{\min}(n) \;\leq\; 22700\Big(\frac{(n+1)\sqrt{2n}}{4} + \frac{\sqrt{n}}{4}\Big)\sqrt{12}\,\frac{(n+2)^{(n-3)/2}}{((n-2)!)^{3/2}} \;=\; \exp\big(-n\ln n + O(n)\big).$$

So the smallest nonzero volume tends to zero along a six-valent embedding in general position, faster than any exponential.

The proof is in appendix H. It runs the zero-energy recursion along the ladder: the equations of $A\psi = 0$ split into two independent chains of $2 \times 2$ transfer steps, one killed by the left end, the other carrying the end modes, which reaches the right end only for odd $n$ (this is (i)). For odd $n$ a quasimode built from the transfer matrix $\mathsf{T}$ gives $\sigma_{\min} \leq \text{const} \cdot |\det \mathsf{T}|/\|\mathsf{T}\|$; the determinant telescopes to a bounded quantity, and $\|\mathsf{T}\|$ grows like the product $\prod(y_j/x)$ over the dimers, whose square is $(n-1)!\,n!\,(n-2)!\,\prod_{k=n+5}^{2n+1} k/(12(n(n+2))^{n-3})$. The proof uses the moduli of the matrix elements and the sign structure of proposition H.1, in which each rung block has rank two; it does not cover arbitrary independent signs of the entries.

**The rate, and Brunnemann and Rideout's fit.** By Stirling, $\ln\prod_{j=3}^{n-1}(y_j/x) = n\ln n + O(n)$, so the bound is $\exp(-n\ln n + O(n))$. Exact characteristic polynomials of the operator for odd $n \leq 51$ (computed from (3), and identical to those of the full twenty-triple operator for $n \leq 6$) give $\ln q_{\min} = -1.0016\,n\ln n + 0.316\,n + \ldots$ for a four-parameter fit over $15 \leq n \leq 51$, consistent with the bound having the right leading term. (The factor $((n-2)!)^{3/2}$ in the bound alone would suggest a coefficient $3/2$; the numerator $(n+2)^{(n-3)/2}$ contributes $+1/2 n\ln n$, and the coefficient is 1.) In Brunnemann and Rideout's variables this is $\lambda_V = \sqrt{q_{\min}} \approx \exp(-1/2 n\ln n)$, a factorial law; their fit $\lambda_V \approx e^{3.72}e^{-1.215\cdot 2j_{\max}}$ over $2j_{\max} \leq 13$ is an exponential fit to it over a short window, where $n\ln n$ is close to linear, and it should not be extrapolated. Our own values on this family are $q_{\min} = 0.840877$, $0.0459024$, $6.86354 \times 10^{-4}$ and $4.65514 \times 10^{-6}$ for $n = 1, 3, 5, 7$.

*Remark 8.1* (Every valence, degenerately). If several edges share one tangent direction, every triple containing two of them has $\epsilon = 0$, and the operator of an $N$-valent vertex is that of a five-valent vertex with one leg replaced by the total spin of the bundle, because the triple products of the bundle's edges with any two other legs carry the same sign and add up to the triple product of the bundle's total spin, while those with two edges inside the bundle have $\epsilon = 0$. So bundling edges onto the fifth edge of the vertex of

theorem 5.3 closes the gap at every valence $N \geq 5$. These vertices are five-valent in disguise, and we do not count them as evidence about generic higher valence.

## 9 A CONTRAST: THE ROVELLI–SMOLIN OPERATOR

The vertices on which theorem 5.3 closes the gap all have an edge of spin $1/2$. The Rovelli–Smolin operator, in the form of De Pietri and Rovelli [19, eqs. (7.15), (7.42)], takes the absolute value of each triple before summing and has no orientation signs:

$$Q_{RS} = \sum_{I<J<K\leq N} |\hat{q}_{IJK}|, \qquad \hat{V}_{RS} = c_{RS}\sqrt{Q_{RS}},$$

which is also the form Ashtekar and Lewandowski re-derive in their appendix [4] and Kisielowski studies [28]. On exactly the spin labels where the Ashtekar–Lewandowski gap closes, it has a uniform gap.

*Theorem 9.1 — Rovelli–Smolin gap with a spin-$1/2$ edge*

Let a five-valent vertex have nonzero spins, one of them $1/2$, and $\dim \mathrm{Inv} \geq 2$. Then $Q_{RS} \geq 3\sqrt{3}$ on the whole invariant space; in particular $Q_{RS}$ has trivial kernel. The variant $\sum_T \sqrt{|\hat{q}_T|}$ of Giesel and Thiemann [24] is at least $3 \cdot 3^{1/4}$.

*Proof.* Label the spin-$1/2$ edge 1. The six triples containing it split into three complementary pairs $\{1JK\}, \{1LM\}$ with $\{J, K, L, M\} = \{2, 3, 4, 5\}$.

The triple $\hat{q}_{1JK}$ commutes with $S = |J_L + J_M|$, and on each value of $S$ it is the four-valent operator on $\mathrm{Inv}(1/2, j_J, j_K, S)$, of dimension at most two because $x = |J_J + J_K| \in \{S \pm 1/2\}$. On a two-dimensional block it is $\left(\begin{smallmatrix} 0 & -4a \\ 4a & 0 \end{smallmatrix}\right)$, so $|\hat{q}_{1JK}| = 4a$ there, and by (4) with the factor $4k^2 - 1$ cancelled ($k = j_J + 1/2$), $16a^2 = ((j_K + S + 1)^2 - k^2)(k^2 - (j_K - S)^2) \geq k(2k + 1) \geq 3$. So $|\hat{q}_{1JK}| \geq \sqrt{3}\, P_{JK}$, with $P_{JK}$ its range projector.

The kernel of $\hat{q}_{1JK}$ is spanned by the one-dimensional blocks, each a joint eigenvector of $x = |J_J + J_K|$ and $y = |J_L + J_M|$ with $(x, y)$ equal to $(j_J + j_K,\ j_J + j_K + 1/2)$ or $(|j_J - j_K|,\ |j_J - j_K| - 1/2)$. Joint eigenspaces of $(x, y)$ are at most one-dimensional (the spin-$1/2$ leg must couple $x$ and $y$ to zero), so these are lines of one orthonormal basis. The kernel lines of $\hat{q}_{1LM}$ lie in the same basis with the roles of $x$ and $y$ exchanged. A common line would need $j_J + j_K = j_L + j_M \pm 1/2$ simultaneously in both directions, which is impossible, or one spin equal to the sum of the other four, which forces $\dim \mathrm{Inv} = 1$. So the two kernel projectors are orthogonal, $P_{JK} + P_{LM} \geq 1$, and

$$Q_{RS} \;\geq\; \sum_{T \ni 1} |\hat{q}_T| \;\geq\; \sqrt{3} \sum_{pairs} (P_{JK} + P_{LM}) \;\geq\; 3\sqrt{3}.$$

For the variant, $|\hat{q}_T| \geq \sqrt{3} P_T$ gives $\sqrt{|\hat{q}_T|} \geq 3^{1/4} P_T$. □

The bound is not sharp: the minimum over all spin-$1/2$ tuples with spins up to 20 is 15.93, at $(1/2, 1/2, 1/2, 1/2, 1)$ (and we reproduce it with our own code on small tuples). Along the Ashtekar–Lewandowski family $(1/2; j - 1/2, j, j, j)$ the smallest eigenvalue of $Q_{RS}$ grows, from 21.4 at $2j = 2$ to 2709 at $2j = 40$, while theorem 5.3 has the Ashtekar–Lewandowski value falling like $2^{-2j}$.

**What the contrast does and does not say.** The mechanism of theorem 5.3 is a cancellation between two triple products that enter with signs; taking the absolute value of each triple first removes it. It is the order of the absolute value and the sum that matters, not the presence of orientation signs: the signless sum $|\sum_T \hat{q}_T|$ has exact zero eigenvalues on spin-$1/2$ tuples too. The Rovelli–Smolin operator is not therefore gapped in general. It has zero-volume states at valence five, for instance at five spins 2 (section 2); its general gap at valence five is open, with numerical support; and since the consistency check of Giesel and Thiemann [24], which found it incompatible with the quantisation of the fluxes, it has been disfavoured for dynamics, which is why Brunnemann and Rideout study the Ashtekar–Lewandowski operator. The original paper of Rovelli and Smolin [34] (in its arXiv version, which was never revised; we could not obtain the two-page erratum) states the higher-valence operator in a form that, read literally, takes the absolute value after summing over triples; theorem 9.1 does not apply to that form, and the later works cited here [19, 4, 24, 28] use the per-triple form above.

## 10 THE COMPUTER-ASSISTED STEPS

Several proofs end in a finite computation. This section says what each computation decides, in which arithmetic, and where its script is. The scripts and outputs are at `https://github.com/johnzfitch/five-valent-volume-gap` (release `v1.0`, which also carries the outputs too large for the repository): `code/` holds the code written for this paper, `certificates/` the certificates on which the proofs rest, each subfolder with the checksums of its source, and `SHA256SUMS.txt` the checksums of everything.

### 10.1 THREE KINDS OF CERTIFICATE

An *exact* certificate decides a statement in rational arithmetic: the signs of the pivots of an $LDL^T$ factorisation (Sylvester's law of inertia), the rank of a rational matrix, or the signs of the coefficients of a polynomial after substituting nonnegative shift variables that encode the constraints. An *interval* certificate evaluates an inequality with outward rounding (mpmath's interval type or Arb balls, or binary64 with every operation followed by one step of `nextafter` in the safe direction) and checks it strictly. A *rigorous-float* certificate computes in binary64 and bounds the error a priori: the

matrix is perturbed by the rounding of square roots (Weyl's inequality), a congruence $X^T(\widehat{A}-tI)X$ is formed and its entrywise error bounded with Higham's $\gamma_k$ [27, Lemma 3.5], and strict Gershgorin separation of the computed matrix then fixes the inertia of $A-tI$ by Sylvester's law; the kernel dimension it is compared with is computed exactly.

The error model of the rigorous-float certificates needs each floating-point entry $y$ of the matrix to approximate the exact square root of its rational squared link $r$ with relative error at most $2.0001u$, where $u = 2^{-53}$ is the unit roundoff. The certificates of theorem 6.1(h) verify this rather than assume it: each entry is computed as `math.sqrt` of the rational, and is then checked in exact rational arithmetic, $r(1-E)^2 \le y^2 \le r(1+E)^2$ with $E = 2.0001 \cdot 2^{-53}$, before it is used; the run stops if any check fails. All 1 843 571 tuples with spins at most 19 and the 430 481 tuples with $j_1 = 5/2$ and $39 \le 2j_{\max} \le 70$ pass, with the same kernel dimensions and level minima as an earlier run that computed the square roots differently.

## 10.2 THE LIST

The first table lists the computations on which a proof rests; the last column names an independent computation of the same statement, which is a check and not part of the proof. All of them ran on a CPU except the interval box for $j_1 = 3/2$, which ran on a GPU with the same outward rounding.

| STATEMENT | KIND | WHAT THE PROOF USES | INDEPENDENT CHECK |
|---|---|---|---|
| Proposition 4.2 (orbits) | exact | all $2^{10}$ sign vectors, relabellings, $\sigma$ | a separate implementation |
| Lemma 3.5 (four-valent) | exact | coefficient signs of the polynomials (F), (C0), (C2), (C3) | a second encoding; exact $LDL^T$ of the law on $3.78 \times 10^6$ tuples with spins $\le 60$ |
| Theorem 6.10 ($j_1 = 1/2$) | interval | two explicit constants for $n \le 2 \times 10^4$, analytic beyond | exact rational checks of the link inequalities on all chains with spins $\le 13$, with floating-point spectral cross-checks |
| Theorem 6.7 (windows) | interval, exact | odd $5 \le N \le 149$ (interval, all $m \ge 0$); inequality (9) for even $N \le 400$ (exact; the analytic argument covers $N \ge 200$) | exact $LDL^T$ for $N \le 151$ at 15 values of $m$, and at four values for $N = 51$, 75, 101, 149, 151, 201 |

| STATEMENT | KIND | WHAT THE PROOF USES | INDEPENDENT CHECK |
|---|---|---|---|
| Theorem 6.12 ($j_1 = 1$) | exact | $LDL^T$ of the pencil for all 164 176 pairs of pairs with $n_1, n_2 \le 32$, and the 10 416 with max = 32.5 | exhaustive exact certificate of $j_1 = 1$, spins $\le 19$ |
| Theorem 6.14 ($j_1 = 3/2$) | exact; interval | 30 polynomial certificates; the box of $6.4 \times 10^7$ parameter sets; 50 tail certificates | exact $LDL^T$ of the 54 019 tuples with spins $\le 20$ (legs 2–5 sorted, $\dim \mathrm{Inv} \ge 2$) |
| Theorem 6.15 ($j_1 = 2$) | exact; interval | 24 polynomial certificates; the box of 1 489 059 configurations | exhaustive exact certificate, spins $\le 19$; stage certificates for $39 \le 2j_{\max} \le 44$ |
| Theorem 7.3 ($\mathscr{C}_6$) | exact; interval | exact error bounds; interval Cholesky of 532 limits and 6 137 finite tuples; exact certificates of the 17 376 finite tuples with spins $\le 19$; exact comparison of each certified limit bound with $v_0$; exact characteristic polynomials of the two limits below it | — |
| Theorem 6.1(h), spins $\le 19$ | rigorous float | 1 843 571 tuples, kernels and every square root checked exactly | exact $LDL^T$ on 7 860 tuples at four thresholds (31 440 cases) and on 369 sampled tuples |
| Theorem 6.1(h), $j_1 = 5/2$ | exact | stage certificates of 430 481 tuples | rigorous-float certificate of the same tuples |

Both finite boxes are certified in outward-rounded interval arithmetic. For $j_1 = 2$ the smallest margin is 3.2356 (truncated), at $(n_1, m_1, n_2, m_2) = (21, 0, 22, 1)$; for $j_1 = 3/2$ the smallest certified cancellation margin $1 - \tau(4/3 + C_\Lambda)$ is $3.02 \times 10^{-5}$ (truncated), at the regime Tm, and the smallest certified value of the bound is 3.7774. Here and below, certified lower bounds are printed truncated and certified upper bounds rounded up. Tuples are counted in two conventions: theorem 6.14's 54 019 are leg multisets (legs 2–5 sorted, $\dim \mathrm{Inv} \ge 2$), while the exhaustive certificates of `certificates/refB` enumerate pairings, with $j_2 \le j_3$, $j_4 \le j_5$ and $(j_2, j_3)$ not after $(j_4, j_5)$, which gives 147 480 tuples with nonzero spectrum for the same range.

The second table lists computations that only check statements proved by hand.

| STATEMENT | KIND | WHAT IS CHECKED |
|---|---|---|
| Theorem 7.1 (pencil) | float | the pencil against the full tensor product, 15 tuples |
| Theorem 8.1(i) (valence six) | exact | characteristic polynomials of the full twenty-triple operator for $n \leq 6$, and of the exact $\sigma$-reduced operator for $n \leq 51$ |
| Lemmas H.2 to H.4 | exact | the identities of their proofs (symbolic in $n$, $i$); the bound against the full operator for odd $n \leq 81$ |
| Remark 4.1 | exact | sign vectors off general position, their orbits and two-product presentations |

### 10.3 OUR OWN CODE

The constructions of section 2 were written for this paper and import nothing from the certificate folders: `code/su2.py` builds the operators from the Hermitian generators on the tensor product, with the invariant space realised as the highest-weight vectors of weight $j_N$ in the first $N-1$ legs, and evaluates Racah's formula for $6j$ symbols in exact rationals; `code/grid.py` builds the grid of proposition 4.4. `code/validate.py` checks the four-valent closed form against brute force and against Racah recoupling, the identity $\hat{q}_{IJK} = -4i J_I \cdot (J_J \times J_K)$, the reduction (3), and the grid against the full ten-triple operator built from integer tangents of each orbit (1 332 cases, relative agreement $3 \times 10^{-15}$). `code/checks.py` checks the identities and constants quoted in the text, `code/reproduce.py` the published numbers of section 2, and `code/nongeneric.py` the counts of remark 4.1.

### 10.4 RELATED WORK AND PRIORITY

Besides the papers discussed in section 1, the volume spectrum has been studied through closed formulas and recoupling theory [37, 19, 32, 31, 17, 1, 30, 35, 12], through quantum and semiclassical tetrahedra and polyhedra [6, 5, 9, 25, 7, 8], and for the Rovelli–Smolin form [34, 28]. The operator $J_1 \cdot (J_2 \times J_3)$ is older than loop quantum gravity: Chakrabarti proposed its eigenvalues, in place of a pair Casimir, as labels for the states of three coupled angular momenta [18, p. 301], and Lévy-Leblond and Lévy-Nahas [29] studied the symmetric coupling scheme it generates.

We read in full the papers on the volume operator and on the mixed product cited in this paragraph and in section 1 (for Rovelli and Smolin, the arXiv version), with these exceptions: of Lévy-Leblond and Lévy-Nahas [29], Aquilanti, Caligiana and Cavalli [2] and Bendjoudi and Mebarki [7, 8] we have read only the abstracts, and the two-page erratum to Rovelli and Smolin [34] we could not obtain. We also screened by title and abstract the records on INSPIRE that cite Brunnemann and Rideout, Brunnemann

and Thiemann, and Bianchi and Haggard. In none of this did we find a proof of a volume gap or of its absence beyond numerics, a statement of the normal form, or the identification of the four-valent matrix with a hydrogen Stark matrix. The priority statements of section 1 are limited accordingly: the two papers we know only in abstract, [29] and [2], are where an earlier appearance of the Stark structure would be most likely, and our statements do not cover them.

## 11 WHAT REMAINS OPEN

The classification of theorem 1.1 reduces the five-valent problem in general position to one operator, $T = J_1 \cdot (J_2 \times J_3 + J_4 \times J_5)$, and what is not proved about it can be stated precisely.

**Conjecture 11.1** (The equal-weight gap). Every nonzero eigenvalue of $8T$ satisfies

$$q \;\geq\; 2\sqrt{18 - 2\sqrt{41}} = 4.5579607\ldots,$$

with equality only at $(1/2; 1/2, 1; 1, 1)$ up to the symmetries of proposition 4.5.

For $j_1 = 1$ the conjecture is proved, since theorem 6.12 gives $q \geq 4.558 > 2\sqrt{18 - 2\sqrt{41}}$, which also excludes equality on that slice. For $j_1 = 2$, for all spins at most 19 and for $j_1 = 5/2$ with spins at most 35 the proofs give $q \geq 4.5579607$, short of the conjectured constant by less than $4 \times 10^{-8}$; a second certified pass at a threshold between 4.5579608 and 5.56 that excludes the minimiser would give the sharp form on the finite ranges. For $j_1 = 1/2$ and $3/2$ only the weaker constants 2.70 and 3.7 are proved. With theorems 5.3 and 5.4 the conjecture would complete the labelled dichotomy for tangents in general position: such an embedding has a gap exactly when its $\sigma$ does not contain $\pm 4$.

**Conjecture 11.2** (The growth law). There are constants $c_2, C > 0$ such that every nonzero eigenvalue of $8T$ satisfies

$$q \;\geq\; \max\Big(4\sqrt{10 - 6\sqrt{2}}\; j_1,\; c_2\,\frac{M}{\ln M}\Big) - C, \qquad M = 2j_{\max}.$$

Both terms are sharp in form: the slope in $j_1$ cannot be larger (theorem 7.2) and the growth in $M$ cannot be faster than $M/\ln M$ (theorem 7.4). Numerically the lower envelope in $M$ is about $1.5\,M/\ln M$. Along the two families of theorem 7.4, $q_{\min} \ln M/M$ is 3.151 and 5.97 at $M = 2^{16}$, close to the heuristic values $\pi$ and $2\pi$ of section 7 but still rising, so its limits are not established. The linear part, with exactly this slope, is proved on $\mathscr{C}_6$ (theorem 7.3); for unrestricted tuples it is open. Behind it is a statement about three spins alone: the smallest nonzero eigenvalue modulus $\nu$ of $U_z$ on any weight space of any $j_2 \otimes j_3 \otimes j_4$ is at least $\nu_0 = \sqrt{(5 - 3\sqrt{2})/2}$, attained only at $(1/2, 1, 1)$ with

weight $\pm 1/2$. This holds for all triples with $2j \le 12$ and is open in general. The second route to large $j_1$, the pencil of theorem 7.1 for triples of unbounded size, would need the next-order structure of A rather than norm bounds.

### 11.1 EQUAL WEIGHTS AT LARGE $j_1$: A SPIN-$j_1$ LADDER

The rest of this section describes what we know about the open case, $j_1 \ge 5/2$, beyond the theorems of section 6. None of it closes the case.

**The reduced obligation.** By lemma 6.4, $q \ge 2\sqrt{\mu}\,\lambda(W_0)$ with $2\sqrt{\mu} \ge 4\sqrt{2s(s+1)/\pi}$, a factor that grows linearly in $s = j_1$. So a much weaker statement about the compressed Kronecker sum than a uniform gap would already suffice:

> (RO) There is $c > 0$ such that every tuple with $j_1 \ge 5/2$ that is not covered by theorem 6.1 has a pairing with $\lambda(W_0) \ge c/(j_1 + 1)$.

Then $q \ge 4c\sqrt{2s/(\pi(s+1))} \ge 2.697\,c$ on those tuples, and conjecture 11.1 would hold with a weaker constant. The evidence for (RO) is strong. A random search over 10 000 uncovered tuples with $5 \le 2j_1 \le 41$ and doubled leg spins up to 200, drawn from six regimes (nearly equal legs, two large legs, two small legs, unbalanced pairs, large gaps, legs near $j_1$), found no counterexample: the smallest $\lambda(W_0)$, in the best pairing, is 0.614 for half-integer $j_1$ and 1.0033 for integer $j_1$, and the smallest $\lambda(W_0)(j_1 + 1)$ is 2.46. Every half-integer minimiser has four nearly equal legs, and along such families $\lambda(W_0)$ decreases slowly with the legs, to 0.583, 0.567 and 0.549 at $2j_1 = 41$, 81 and 161 with doubled legs up to 2048, consistent with a limit of $1/2$. So the factor $1/(j_1 + 1)$ in (RO) is never close to being needed; the data suggest $\lambda(W_0) \ge 1/2$.

**The picture: the low modes are a spin-$j_1$ channel.** One might expect the low singular vectors of $W_0$ to be edge states on the two walls $d = \pm s$, where the cut acts. They are not. Along the extremal family $(s; n - 1/2, n, n, n)$, the weight of the lowest mode on each diagonal matches, to within 0.05, the weights $|\langle m|m_x = 1/2\rangle|^2$ of the eigenvector of $J_x$ in spin $s$ with eigenvalue $1/2$, for $2s = 5$, 9 and 21; along the antidiagonal $u$ the mode is spread evenly over the whole strip. Keeping only one wall gives $\lambda(W_0)$ between 0.90 and 1.03, and the interior strip $|d| \le s - 1$ is only 6–30% better than the full one. The low modes are the $m_x = \pm 1/2$ channel of a spin-$s$ representation of $\mathfrak{su}(2)$ acting across the diagonals, and they need both walls.

The $\mathfrak{su}(2)$ is visible in the operator itself. Between the diagonals $d$ and $d + 1$ the factor $\sqrt{F(d + 1/2)} = 2\ell_s(d)$ of every link is twice the link $\ell_s(d) = 1/2\sqrt{(s - d)(s + d + 1)}$ of $J_x$ in spin $s$, and proposition 4.4 gives at once:

*Proposition 11.1 — $T$ is a $J_x(s)$ ladder with a diagonal defect*

For every tuple and the pairing (23|45), every link of $T$ between the diagonals $d$ and $d+1$ is $\ell_s(d)\,\nu$, with $\nu = {}^1\!/_2 h_1(b)\sqrt{G(u)}$ for an $X$-link and $\nu = {}^1\!/_2 h_2(c)\sqrt{G(u)}$ for a $D$-link, $(b,c)$ its upper end. So $T = \sum_d \ell_s(d)(N_d + N_d^T)$, with $N_d$ the block of $\nu$-links from diagonal $d$ to $d+1$, and on every interior diagonal $N_d^T N_d - N_{d-1}N_{d-1}^T$ is diagonal, with entry ${}^1\!/_4[G(u+1)P(b+1,c+1) - G(u)P(b,c)]$ at $(b,c)$, where $P(b,c) = h_1(b)^2 - h_2(c)^2$.

If that diagonal defect vanished, $T$ would be exactly $J_x(s) \otimes P \oplus 0$ for some $P \succeq 0$, by the argument of theorem 11.2 below with the Gram matrices $N_d^T N_d$ in place of its Casimir (appendix I), and its smallest nonzero eigenvalue would be ${}^1\!/_2$ or 1 times the smallest nonzero eigenvalue of $P$. In the frame of $W_0$ the same structure appears with an absolute constant: the plaquette argument of proposition 6.13 gives, for $W_0$, $M_d^T M_d - M_{d-1}M_{d-1}^T = -d/2 + \Delta_d$ on every interior diagonal, with $M_d$ the block of $W_0$ from diagonal $d$ to $d+1$ and $\Delta_d$ diagonal; and $-d/2$ is exactly the relation $\ell_s(d)^2 - \ell_s(d-1)^2 = -d/2$ of $J_x$. With no defect this structure forces a gap, uniformly in the sizes of the blocks:

*Theorem 11.2 — The su(2) ladder without defect*

Let $V_{-s}, \dots, V_s$ be finite-dimensional real inner-product spaces, $M_d : V_d \to V_{d+1}$ linear, and $W = \sum_d (M_d + M_d^T)$ on $\bigoplus_d V_d$. Suppose $M_d^T M_d - M_{d-1}M_{d-1}^T = -d/2$ on $V_d$ for every $-s < d < s$, and $M_{s-1}M_{s-1}^T \succeq s/2$ on $V_s$, $M_{-s}^T M_{-s} \succeq s/2$ on $V_{-s}$. Then $W$ is an orthogonal sum of copies of $J_x(S)$ with $S \le s-1$, $S \equiv s \pmod 1$, and of the chains $J(\kappa)$ on $-s \le d \le s$ with links $\sqrt{\kappa - d(d+1)/4}$ and $\kappa \ge s(s+1)/4$. Every singular value of the bipartite block of $J(\kappa)$ is nondecreasing in $\kappa$, and $J(s(s+1)/4) = J_x(s)$. Consequently every eigenvalue of $W$ has modulus at least ${}^1\!/_2$ if $s$ is a half-integer, and every nonzero eigenvalue has modulus at least 1 if $s$ is an integer. Both bounds are sharp.

The proof (appendix I) is algebra: $C_d = M_d^T M_d + \frac{d(d+1)}{4}$ is intertwined by the blocks, $M_d C_d = C_{d+1} M_d$, so $W$ commutes with a Casimir and splits into its eigenspaces, on which every block is a scalar multiple of an isometry. The constants ${}^1\!/_2$ and 1 are the bulk limits seen in the data.

**Why the theorem does not apply.** For the real $W_0$ the defect is

$$\Delta = \tfrac{1}{2}\big(\varepsilon_1(b) - \varepsilon_2(c)\big) + [u = s]\big(h_1(b)^2 - h_2(c)^2\big), \qquad \varepsilon(L) = \frac{(4n^2-1)(4m^2-1)}{2(2L-1)(2L+1)(2L+3)},$$

with $\varepsilon_1$, $\varepsilon_2$ formed with the parameters of the two pairs, and with the convention $\varepsilon({}^1\!/_2) = (4n^2-1)/8$ at the bottom label of a pair with $m = {}^1\!/_2$; the last term is present only at the corner sites $u = s$. It vanishes only for two pairs with $m = {}^1\!/_2$ away from their bottom labels and from the corner, which no real tuple achieves everywhere. And

it is not small. On eight tuples near the minimisers its largest entry is 112 to 1 110 in the band $u \le 2s+2$ at the corner, against at most 53 elsewhere, and its size relative to the Gram matrices is between 0.74 and 11. On the tuples with nearly equal legs the lowest mode's Casimir has its mean within about one sector of the bottom value $s(s+1)/4$, but a spread of several sectors that grows with the legs. In the frame of $T$ the defect of proposition 11.1 is at most 9.4% of the Gram diagonal entry by entry on the two tuples measured, but it is still of order one as an operator, because the small singular values of each $N_d$ come from cancellation along $u$. So a perturbation of theorem 11.2 cannot reach the real operator.

**What fails, and what a proof needs.** Two simpler routes are false. The gap is not monotone in $j_1$ with legs 2–5 fixed: $q = 148.70$ at $(4; 1, {}^{11}/_{2}, {}^{11}/_{2}, 6)$ but 70.59 at $(5; 1, {}^{11}/_{2}, {}^{11}/_{2}, 6)$, and 912 of 10 252 such steps without a leg $1/2$ go down. Nor is it monotone along $j_1 - j_5$ fixed, the setting of theorem 7.1: $q = 146.46$ at $(5; 1, 4, 4, 1)$ but 63.30 at $({}^{11}/_{2}; 1, 4, 4, {}^{3}/_{2})$, with 490 of 19 498 steps going down. Reading the strip along the antidiagonals instead of the diagonals gives a transfer-matrix version of lemma 5.1, but its bounds collapse on unbalanced pairs, for example to $1.3 \times 10^{-26}$ where $q = 639$.

A proof of conjecture 11.1 along the lines of this section needs a version of theorem 11.2 that allows the diagonal defect $\Delta$, under a hypothesis that is structural rather than a bound on its size, that the real $W_0$ satisfies in some pairing, and that still yields $\lambda(W_0) \ge c/(j_1+1)$. The defect is explicit and concentrated at the corner of the strip, so the hypothesis presumably says that the corner acts as a trivial boundary condition, the two-dimensional counterpart of the trivial dimerisation of the $j_1 = 1/2$ chain. We have not found it.

**Identities behind this section.** These are proved, and each was checked against an independent computation; the scripts and outputs are in `certificates/su2l`, `w0ob` and `t1all`. *Lemma O*: on the invariant space $T = \frac{1}{2i}[J_2 \cdot J_3 + J_4 \cdot J_5, \Omega]$ with $\Omega = (J_3+J_5)^2 - (J_2+J_4)^2$ (residual $5 \times 10^{-15}$ by brute force on all 512 tuples with doubled spins at most 4). *Lemma U*: along the antidiagonals the Gram matrices of $T$ satisfy the analogue of proposition 6.13 with ratio $G(u+1)/G(u)$ (exact, 97 597 entries). *Lemma S*: for a Stark matrix, $h(L+1)^2 - h(L)^2 = -\frac{2L+1}{4} + {}^{1}/_{2}\varepsilon(L)$, and for $m = 1/2$ the Stark matrix is $J_x$ of spin $n - 1/2$ on its odd subspace (exact, 20 540 labels). *Lemma L*: the Gram identity for $W_0$ with the defect $\Delta$ above (exact, 195 361 sites, 27 993 at the corner). *Lemma W*: for half-integer $j_1$, in every pairing exactly one wall has the corner's colour, and the leak through the other wall obeys the same $J_x$ law (exact, 12 900 tuples). *Lemma Q*: an explicit two-spin Casimir commutes with $K$ up to the $\varepsilon$-terms (float, 16 875 pairs of pairs). Proposition 11.1 was checked exactly on all 55 000 tuples with $2j_1 \le 11$ and doubled legs at most 10, and theorem 11.2 on 3 312 honest two-spin strips and 400 random exact ladders.

**Beyond valence five.** At valence six we have one embedding class (theorem 8.1), chosen because Brunnemann and Rideout's data single it out; its $\sigma$ has only two nonzero entries, and its mechanism is close to the five-valent one. Whether the classes that Brunnemann and Rideout found increasing at valence six, including the cubic vertex, have a gap is open, as is everything non-degenerate at valence seven and higher.

**Other operators.** The general five-valent gap of the Rovelli–Smolin operator is open, and nothing here concerns the polyhedral volume of Bianchi, Donà and Speziale [9], for which Haggard's level-repulsion argument suggests a generic gap [25]. The semi-classical status of the small Ashtekar–Lewandowski eigenvalues is itself debated: the operator does not reproduce the polyhedral volume at generic five-valent vertices [9, 23], and the gap closing found here may be read as a property of its regularisation rather than of quantum geometry. It may still matter for dynamics. Brunnemann and Thiemann noted that the spectrum near zero volume enters the Hamiltonian constraint through the curvature operator, and that valleys of zero volume could make the relevant differences of the spectrum unbounded [16, p. 46]; eigenvalues as small as those of theorem 5.3 and theorem 8.1 sharpen that concern for the embeddings that carry them. The classification says which embeddings carry the effect; whether they should be counted is a question about the dynamics that this paper does not address.

## 12 MOTIVATION AND THE USE OF AI

This paper did not begin in loop quantum gravity. Since late June 2026 I had been building an attention layer, which I call the transporter, in which three tokens interact through one trilinear form, so that its basic cell is a triangle, and whose edges carry operators on the Albert algebra, the exceptional Jordan algebra of $3 \times 3$ octonionic Hermitian matrices. On that algebra the tensor square of a traceless element splits into five irreducible channels, $26 \otimes 26 = 1 \oplus 26 \oplus 52 \oplus 273 \oplus 324$, and the layer is built from them [22]. In October I asked whether the next simplex, the tetrahedron, has any place in this setting. It does. Four primitive idempotents of the Albert algebra with equal pairwise overlaps $1/3$ form a regular tetrahedron of pure states that generates the whole algebra. And a quantum tetrahedron in the sense of Barbieri and of Baez and Barrett [6, 5], with the exceptional group $F_4$ in place of $\mathrm{SU}(2)$ and the 26 on each of its four faces, has a five-dimensional space of shapes, and the five channels label a basis of it. Its orientation operator is the commutator of two pair Casimirs, which for $\mathrm{SU}(2)$ faces is the triple product $J_1 \cdot (J_2 \times J_3)$, the four-valent volume operator of loop quantum gravity.

The volume operator is what took me to the dynamics. In a Claude session on 17 and 18 October I asked why more physicists had not talked about loop quantum gravity, and the answer named the unsolved dynamics, the quantisation of the Hamiltonian

constraint and the anomaly-free closure of its algebra, as the central problem. In the same session I wrote a research prompt for the rank-three Jordan family, and one of its four tasks was to define the analogue of the Baez–Barrett volume operator and compute its spectrum, calibrated against the published volume spectra of Brunnemann and Thiemann and of Bianchi and Haggard [16, 10]. In the dynamics the volume has a role of its own: in Thiemann's construction the extrinsic curvature is obtained from the commutator of the Hamiltonian constraint with the volume [38]. The first lemma proved in the work that followed was that this commutator vanishes on the off-shell space that Varadarajan built for the Euclidean theory [39], and the question became whether a Lorentzian Hamiltonian constraint can be made to close there. I did not settle it. But the same triple product kept reappearing inside the calculation, and the Ashtekar–Lewandowski volume entered it through the sign factors of the edge tangents that are the subject of section 4. The questions I could settle turned out to be about the volume operator itself. Looking for a named open problem on which these tools could act, I found Bianchi and Haggard's question whether the volume has a gap for all spins [10]. At five-valent vertices Brunnemann and Rideout had seen gaps both open and close [13], and this paper explains why. The last paragraph of section 11 returns to dynamics, where the work began. The companion paper [21] comes from the same calculation, continued to $\mathrm{SU}(1,1)$. Appendix J describes how the results were reached.

**Use of AI.** I posed the questions, chose the problems and the order in which to pursue them, and set the standard a result had to meet. I directed the computations and decided what this paper claims. Large language models were used throughout. Anthropic's Claude was used in two ways. In sessions on claude.ai it answered my first questions about tetrahedra in Jordan algebras, carried out the exact calculations on the rank-three Jordan family, and began the calculation on the dynamics. Through Claude Code it wrote and ran nearly all of the code and the certificate scripts for this paper, ran the agent campaigns, drafted and revised the text, and gave the first written proofs of most of the results here, among them the normal form of theorem 1.1, the gap closing of theorem 5.3, the sharp four-valent law of theorem 3.1 with its Stark factorisation, and the interval certificate behind theorem 6.14. Separate Claude agents, working from the written results, refereed them one by one, and an independent Claude referee read the whole paper. OpenAI's GPT models were used through ChatGPT and Codex. ChatGPT produced the first document on the Albert tetrahedron, and GPT ran long exact-arithmetic computations on the Jordan family for two further reports. Codex sessions continued part of the dynamics calculation and audited a note on the equal-weight case. Google's Gemini was used only as a reviewer of this paper, as were further sessions of GPT and Claude. Review comments were adopted only after they had been checked against the source, the rendered pages and our own code. The claims rest on the following, and not on any model's assertion: the written proofs in this paper; the exact rational, outward-rounded interval and rigorously bounded floating-point certificates of section 10, each shipped with its script and checksum; our recomputation

of the published numbers of Brunnemann and Rideout and of Bianchi and Haggard; and cited results. Responsibility for the content, and for any error that remains, is mine.

## A THE FOUR-VALENT MULTIPLIER

This appendix proves lemma 3.5. Notation is that of section 3: sorted spins, $A_2 = j_3 + j_4 + 1 = \varpi - |j_3 - j_4|$ with $\varpi = 2j_4 + 1$, $m_>$ and $m_<$ the larger and smaller of $|j_1 - j_2|$, $|j_3 - j_4|$, $k_1 = m_> + 1$ the label of the first link, $K = j_1 + j_2$ that of the last, and $f(k) = (k^2 - m_<^2)(A_2^2 - k^2)$.

*Proof of (F).* Write $B_2 = |j_3 - j_4|$. Then $k_1 \le K \le 2j_3 \le j_3 + j_4 = \varpi - B_2 - 1$. *Case* $m_> = B_2 \ge m_<$. Here $k_1^2 - m_<^2 \ge 2B_2 + 1$, with equality iff $m_< = B_2$, so $f(k_1) \ge (2B_2 + 1)(\varpi - 2B_2 - 1)(\varpi + 1)$, and $(2B_2 + 1)(\varpi - 2B_2 - 1) - (\varpi - 1) = 2B_2(\varpi - 2B_2 - 2) = 4B_2(j_3 - 1/2) \ge 0$. *Case* $m_> = |j_1 - j_2| > m_< = B_2$. In $v = k_1^2$, $f$ is a concave quadratic on $[(B_2 + 1)^2, (\varpi - B_2 - 1)^2]$; at the left end it is at least $\varpi^2 - 1$ by the first case, at the right end it equals $2j_3(\varpi - 1)(\varpi + 2j_3) \ge \varpi^2 - 1$, so it is at least $\varpi^2 - 1$ throughout and strictly so in the interior. *Equality* forces the first case with $m_< = B_2$ and $B_2(2j_3 - 1) = 0$, and both branches give $|j_1 - j_2| = |j_3 - j_4| = 0$; the branch $j_3 = 1/2$ forces all spins $1/2$. □

*Proof of (C).* $f$ is a product of four linear factors positive on the range, hence log-concave in $k$; it is a concave quadratic in $v = k^2$, symmetric about $v_* = (A_2^2 + m_<^2)/2$, hence unimodal, and $f(v') - f(v) = (v' - v)(A_2^2 + m_<^2 - v - v')$. For (C2), $f_L > f_2$, we need $(m_> + 2)^2 + K^2 < A_2^2 + m_<^2$. Put $R = j_3 + j_4 - K$; since $j_3 \ge j_2$, $R - |j_1 - j_2| - |j_3 - j_4| = 2(j_3 - j_2) \ge 0$, so $R \ge m_>$. With $K \ge m_> + 2$ (dimension at least three), $A_2^2 - K^2 = (R + 1)(2K + R + 1) \ge (m_> + 1)(3m_> + 5) > (m_> + 2)^2$. The inequality (C0), $f_2 > f_1$, is the same with $m_> + 1$ in place of $K$. For (C3), $f_L^2 \ge f_{L-1} f_2$ (dimension at least four), put $\mathsf{T} = f(K)$, $p = f(K - 1) - f(K)$, $\mathsf{q} = f(K) - f(m_> + 2)$ and $E = A_2^2 + m_<^2 - K^2 - (m_> + 2)^2$. Then $p = (2K - 1)((K - 1)^2 - (m_> + 2)^2 - E)$, $\mathsf{q} = (K^2 - (m_> + 2)^2)E$, $\mathsf{T} = (K^2 - m_<^2)(E + (m_> + 2)^2 - m_<^2)$, and the claim is $\mathsf{F} := p\mathsf{q} - \mathsf{T}(p - \mathsf{q}) \ge 0$. The constraints $m_< \le m_>$, $K \ge m_> + 3$ and $R \ge m_>$ give $E \ge E_{\min} = 2K(m_> + 1) - 2m_> - 3 + m_<^2$ (indeed $E - E_{\min} = A_2^2 - (K + m_> + 1)^2$). Substituting $m_< = y$, $m_> = y + z$, $K = m_> + 3 + x$, $E = E_{\min} + w$ with $x, y, z, w \ge 0$ turns $\mathsf{F}$ into a polynomial with 252 terms, all with nonnegative coefficients. □

The polynomial certificates for (F), (C0), (C2) and (C3) were produced in two independent encodings (the shift variables above, and the sorted spins $j_1 = 1/2 + a$, $j_2 = j_1 + b$, $j_3 = j_2 + c$, $j_4 = j_3 + e$ with the dimension condition imposed as a shift). Both have no negative coefficient, and in the second the monomial support of (F) vanishes exactly on the equality locus. The sharp law itself was re-checked by exact rational $LDL^T$ factorisation of $B^T B - \bar{\mu}$ on all 3 784 040 sorted tuples with spins at most 60; the equality set is exactly the 120 tuples $(1/2, 1/2, j, j)$.

## B STARK MATRICES WITH AN END REMOVED

This appendix proves lemma 6.6, the window identity used in corollary 6.8, and theorem 6.7. All three are statements about one Stark matrix $H = H(n,m)$ of dimension $N$, with eigenvectors $v_\lambda$ ($\lambda = N-1-2k$), and both rest on the end components of its eigenvectors. Write $w_0(\lambda) = v_\lambda(0)^2$ and $w_N(\lambda) = v_\lambda(N-1)^2$ for the squared components at the two ends.

**Lemma B.1** (End weights). $w(\lambda) = w(-\lambda)$ at both ends, and with $\lambda_k = N-1-2k$,

$$\frac{w_0(\lambda_{k-1})}{w_0(\lambda_k)} = \frac{k(N+m-k)}{(m+k)(N-k)}, \qquad \frac{w_N(\lambda_{k-1})}{w_N(\lambda_k)} = \frac{k(m+k)}{(N+m-k)(N-k)}.$$

Both ratios are at most 1 for $k \le N/2$, so each end weight decreases away from the middle of the spectrum. Relative to the central weight, the bottom weights decrease and the top weights increase in $m$, the latter towards the binomial ratios $\binom{N-1}{k}/\binom{N-1}{k_c}$.

*Proof.* $H \cong -H$ through $(-1)^k$, which gives the symmetry. For integer $m$, $H$ is $J_z^{(a)} - J_z^{(b)}$ on $(j \otimes j)_{M=m}$ with $2j+1 = n$ (lemma 3.2); its eigenvectors are product states, the bottom site is $|m,m\rangle$ and the top site $|2j,m\rangle$, and the end weights are squared Clebsch–Gordan coefficients with the stated product forms. For fixed $N$ both sides are rational in $m$ (the end weights of a Jacobi matrix are rational functions of its squared links), and they agree at every integer $m \ge 0$, so they agree for all real $m \ge 0$. The monotonicity statements follow factor by factor. □

*Proof of lemma 6.6.* Let $Q = (H-1)^2 - 1 = H(H-2)$. Its eigenvalues are $\lambda(\lambda-2)$, which is $-1$ at $\lambda = 1$ and at least 3 otherwise, so $Q$ is nonsingular with exactly one negative eigenvalue. For such a form, $Q \succeq 0$ on $e^\perp$ if and only if $e^T Q^{-1} e \le 0$: with $y = Q^{-1}e$, the space splits $Q$-orthogonally as $e^\perp \oplus \mathbb{R}y$ and $y^T Q y = e^T y$. For the end vector $e$, with $\mathcal{G}(z) = \sum_\lambda w(\lambda)/(\lambda - z)$, $e^T Q^{-1} e = {}^1\!/_2[\mathcal{G}(2) - \mathcal{G}(0)]$, and $\mathcal{G}(0) = 0$ by symmetry. Pairing $\lambda$ with $-\lambda$,

$$-\mathcal{G}(2) = \tfrac{4}{3} w(1) - \sum_{k \ge 1} \frac{4\, w(2k+1)}{(2k-1)(2k+3)},$$

and the coefficients sum to ${}^4\!/_3 - \frac{1}{2K+1} - \frac{1}{2K+3} < {}^4\!/_3$, so $-\mathcal{G}(2) > 0$ as soon as $w(2k+1) \le w(1)$, which lemma B.1 gives. Hence $\|(H-1)\psi\|^2 - \|\psi\|^2 = \psi^T Q \psi \ge 0$ for $\psi \perp e$. The case $H+1$ follows by the gauge $(-1)^k$. □

**Lemma B.2** (Window identity). Let one pair be $({}^1\!/_2, j)$, with rails $b \in \{\beta, \beta+1\}$, $\beta = j - {}^1\!/_2$, let $V$ be the label range of the other pair, and let $s$ be a half-integer. For either colour class the block of $W_0$ is $P_{J'}(H'+1)P_J$, where $J, J'$ are the positions of the two classes. Either one class is closed, or $J, J'$ are intervals of equal length with $J' = J \pm 1$, so that the block is a shifted window of $H'+1$.

*Proof.* On the rail $b$ the sites are the positions $c \in V \cap [|b-s|, b+s]$. Since $\beta - s \in \mathbb{Z}$, either $\beta \geq s$ or $\beta + 1 \leq s$, so before intersecting with $V$ the two rail intervals differ by exactly one position at each end. Let $I$ be their common part inside $V$; the remaining positions are at most one $p$ below $I$ and one $q$ above it. At each position the two rails carry opposite colours, so $p$ and $q$ each join one class, and each class is $I$ plus its extras. The rung has weight 1, and a site at $c$ and the opposite-colour site at $c \pm 1$ lie on the same rail, because colour alternates along a rail; so the block is $P_{J'}(H' + 1)P_J$. If one class receives both extras or the other receives none, the other class sits on $I$, all its rectangle neighbours lie in $R$, and it is closed. Otherwise $J = I \cup \{p\}$ and $J' = I \cup \{q\}$, intervals of equal length shifted by one. Finally $H' + 1 \cong -(H' - 1)$ by $(-1)^k$, and the gauge preserves windows. □

## PROOF OF THE WINDOW THEOREM

This is the proof of theorem 6.7.

*Reduction to one window.* $(H-1)_{rc} = 0$ for $|r-c| \geq 2$. If $B_{a,b}$ denotes rows $[a+1, b+1]$ by columns $[a, b]$, then $B_{a,N-2} = \left(\begin{smallmatrix} B_{a,b} & * \\ 0 & * \end{smallmatrix}\right)$, so $\sigma_{\min}(B_{a,N-2}) \leq \sigma_{\min}(B_{a,b})$; and $W := B_{0,N-2} = \left(\begin{smallmatrix} * & * \\ 0 & B_{a,N-2} \end{smallmatrix}\right)$, so $\sigma_{\min}(W) = \sigma_{\min}(W^T) \leq \sigma_{\min}(B_{a,N-2})$. It suffices to bound $W$.

*Variational form.* For $\psi$ with $\psi_{N-1} = 0$, $\|W\psi\|^2 = \|(H-1)\psi\|^2 - (u^T\psi)^2$ with $u = (H-1)e_0$. The corner identity of Jacobi matrices gives $v_\lambda(0)v_\lambda(N-1) = \prod h/P'(\lambda)$ with $P(z) = \prod(z - \lambda)$.

*Even $N$.* Here $\sigma_{\min}(W) \geq 1$. With $Q = H(H-2)$ as above, $u^T Q^{-1} u = 1 + {}^1\!/_2 \mathscr{G}_0(2) < 1$, so $Q - uu^T$ has one negative eigenvalue, and $\sigma_{\min}(W) \geq 1$ iff $e_{N-1}^T (Q - uu^T)^{-1} e_{N-1} \leq 0$. By Sherman–Morrison this is $(X(0) + X(2))^2 \leq \mathscr{G}_0(2)\mathscr{G}_N(2)$ with $X(z) = -\prod h/P(z)$. Here $X(0) + X(2) = 2X(0)/(N+1)$ and $X(0)^2 = w_0(1)w_N(1)R_N^2$ with $R_N = |P'(1)/P(0)| = 2^{2K+1}K!(K+1)!/((2K+1)!!)^2$, $K = N/2 - 1$, so the criterion is $\gamma_0\gamma_N \geq 4R_N^2/(N+1)^2$ with $\gamma_e = -\mathscr{G}_e(2)/w_e(1)$. Lemma B.1 gives $\gamma_0 \geq 2N/(N^2-1)$ and $\gamma_N \geq \Gamma_N := \sum_{i=1}^{K} 4(1 - \rho_i)/((2i-1)(2i+3)) + 2N/(N^2-1)$ with $\rho_i = \binom{N-1}{K-i}/\binom{N-1}{K}$, uniformly in $m$, and

$$\Gamma_N \;\geq\; \frac{2R_N^2(N-1)}{N(N+1)} \tag{9}$$

holds for all even $N$. For $N \leq 400$ it is checked exactly in rationals. For $N = 2P$ with $P \geq 100$, let $k = \lceil \sqrt{P} \rceil$. Since $\log(1-x) \leq -x$,

$$\rho_i = \prod_{r=1}^{i} \frac{P-r}{P+r} \leq \exp\Big(-\sum_{r=1}^{i} \frac{2r}{P+r}\Big) \leq e^{-i(i+1)/(P+i)},$$

which is at most $e^{-1} < {}^1\!/_2$ for $i \geq k$, because $i^2 \geq P$. Keeping only the terms $i \geq k$, where $4(1-\rho_i) > 2$, and telescoping $\frac{2}{(2i-1)(2i+3)} = {}^1\!/_2\big(\frac{1}{2i-1} - \frac{1}{2i+3}\big)$, with $2N/(N^2-1) = \frac{1}{2P-1} + \frac{1}{2P+1}$, gives

$$\Gamma_N > \tfrac{1}{2}\Big(\frac{1}{2k-1} + \frac{1}{2k+1}\Big) + \tfrac{1}{2}\Big(\frac{1}{2P-1} + \frac{1}{2P+1}\Big) > \frac{1}{2k+1} \geq \frac{1}{2\sqrt{P}+3} > \frac{4}{P},$$

the last step because $P > 8\sqrt{P} + 12$ for $P \geq 100$. On the other side, $R_2 = 2$ and each step $P \to P+1$ multiplies $R_N$ by $4P(P+1)/(2P+1)^2 < 1$, so $R_N \leq 2$ and the right side of (9) is at most $8(N-1)/(N(N+1)) < 4/P$. The exact range $N \leq 400$ ($P \leq 200$) overlaps the analytic range $P \geq 100$; at $N = 2$ the inequality is an equality.

*Odd N.* For $\theta \in (0,1)$, with $\rho(\lambda) = 1/((\lambda-1)^2 - \theta^2)$, the same argument gives $\sigma_{\min}(W)^2 \geq \theta^2$ iff $\theta^2 S_0 S_N \leq \Xi_N(\theta^2)$, where $S_e = \sum_\lambda (w_e(\lambda)/w_e(0))\rho(\lambda)$ and, by residues,

$$\sqrt{\Xi_N} = \frac{\pi N\, \Gamma(J+1)^2}{4\cos(\pi\theta/2)\,\Gamma(J+{}^3\!/_2+\theta/2)\,\Gamma(J+{}^3\!/_2-\theta/2)}, \qquad J = \tfrac{N-1}{2},$$

independent of $m$. Lemma B.1 gives $S_e \leq \frac{\pi}{2\theta}\tan\frac{\pi\theta}{2}$ and log-convexity of $\Gamma$ reduces the criterion to $\sin(\pi\theta/2) \leq N/(N+1)$, which holds at $\theta = c_*$ for all odd $N \geq 151$ and at $\theta = 2/\pi \arcsin {}^3\!/_4$ for all odd $N \geq 3$ (the analytic constant). For $N = 3$ the window is $\left(\begin{smallmatrix} g_1 & -1 \\ 0 & g_2 \end{smallmatrix}\right)$ with $g_1^2 = 4(m+2)/(2m+3)$, $g_2^2 = 4(m+1)/(2m+3)$, so $\sigma_{\min}^2 = {}^1\!/_2\big(5 - \sqrt{9 + 16/(2m+3)^2}\big)$, increasing in $m$ and equal to $c_*^2$ at $m = 0$. For odd $5 \leq N \leq 149$: $S_0$ decreases and $S_N$ increases in $m$, so on an interval $[m_a, m_b]$ it suffices that $\theta^2 S_0(m_a) S_N(m_b) < \Xi_N$; the certificate bisects $[0, 2^{24}]$ at rational breakpoints, evaluates the weights exactly and $\rho$, $\Xi$ in 40-digit interval arithmetic at $\theta^2 = c_*^2$, and uses the binomial limit of $S_N$ on $[2^{24}, \infty)$. As a check, at $\theta^2 = 0.93$, above the computed value 0.9249 at $N = 5$, it fails, as it must.

## C CHAINS AT $j_1 = {}^1\!/_2$

This appendix proves lemma 6.9 and theorem 6.10. Throughout, the weight ratio is $t = 1$, and we use doubled units, in which the links below are those of $8T$: for the pair $X = (j_2, j_3)$ put $m_X = 2|j_2 - j_3|$ and $n_X = 2(j_2 + j_3 + 1)$, likewise $m_D, n_D$ for $(j_4, j_5)$, and $m = \max(m_X, m_D)$, $n = \min(n_X, n_D)$. A type *owns* $m$ (or $n$) if its $m_P$ (or $n_P$) attains it. By proposition 4.4 with $s = {}^1\!/_2$ and $\alpha = \beta = 2$, a link of type $P$ at doubled label $l$ has squared weight

$$c^2 = \tfrac{1}{4}(n_P^2 - l^2)(l^2 - m_P^2).$$

**Lemma C.1** (Shape of the chain). If $\dim \mathrm{Inv} \geq 2$, the links carry the labels $l = m+1, \ldots, n-1$ in order, so the chain has $N = n - m$ sites; their types alternate; $m_X \not\equiv m_D$ and $n_X \not\equiv n_D$ modulo 2; the first link has the type that does not own $m$, and the last link the type that does not own $n$.

*Proof.* Since $j_1 = {}^1\!/_2$ and the total spin is an integer, $j_2 + j_3$ and $j_4 + j_5$ differ by a half-odd integer, which gives both parity statements. If $X$ owns $m$, so that $B_0 = |j_2 - j_3| > |j_4 - j_5|$, the lowest site is $(b, c) = (B_0, B_0 - {}^1\!/_2)$, and the link leaving it changes $c$; it is of type $D$, at doubled label $m+1$. Each link raises the doubled label by one and the sites form one interval of $u$ (proposition 5.2). The top end is symmetric. □

*Proof of lemma 6.9.* There are $N-1$ links of alternating type. If $N$ is even, the first and last links are both odd-numbered, so they have one type $O$, which by lemma C.1 owns neither $m$ nor $n$; the parities then give $m_O \le m-1$ and $n_O \ge n+1$. For an even link (type $E$) at $L = m+2\tau$ followed by an odd link at $L+1$,

$$\frac{c_{2\tau}^2}{c_{2\tau+1}^2} = \frac{(n^2-L^2)(L^2-m^2)}{(n_O^2-(L+1)^2)((L+1)^2-m_O^2)} \le \frac{(n-L)(n+L)}{(n-L)(n+L+2)} \cdot \frac{(L-m)(L+m)}{(L-m+2)(L+m)} \le \frac{\tau}{\tau+1}.$$

If $N$ is odd, the first and last link types differ, which forces the stated ownership. □

*Proof of theorem 6.10, N even.* Write $K = N/2$. By lemma 5.1 and the ratio bound, the products telescope: $E_{ij}^2 \le c_{2j-1}^{-2} \prod_{k=j}^{i-1} \frac{k}{k+1} = j/i\, c_{2j-1}^{-2}$. The odd link at $l = m+2j-1 \le n-1$ has type $O$ with $n_O \ge n+1$ and $m_O \le m-1$, so

$$c_{2j-1}^2 \ge \tfrac{1}{4}\big((n+1)^2-l^2\big)\big(l^2-(m-1)^2\big) \ge \tfrac{1}{4} \cdot 4n \cdot 4j(m+j-1) \ge 4nj^2,$$

using $(n+1)^2-l^2 \ge 4n$ for $l \le n-1$, $l^2-(m-1)^2 = 2j(2m+2j-2)$, and $m \ge 1$, which holds because $m \ge m_O+1$. Therefore

$$q_{\min}^{-2} \le \|E\|_F^2 \le \sum_{1 \le j \le i \le K} \frac{1}{4n\,ij} = \frac{H_K^2 + H_K^{(2)}}{8n},$$

with $K \le (n-1)/2$ and $n \ge 3$. The right side of $q_{\min} \ge \sqrt{8n/(H_K^2+H_K^{(2)})}$ has minimum 3.3806 (at $n=5$) over $3 \le n \le 2\times 10^4$ in interval arithmetic (`code/b2_constants.py`); for larger $n$, $H_K \le 1+\ln n$ and $H_K^{(2)} \le \pi^2/6$ give an increasing lower bound that is already 36.4 at $n = 2\times 10^4$. □

*Proof of theorem 6.10, N odd.* Write $K = (N-1)/2$; the bipartite block $B$ is $(K+1)\times K$. The odd links have the type $Q$ owning $n$ and the even links the type $P$ owning $m$, with $m_Q \le m-1$ and $n_P \ge n+1$, so $m \ge 1$ again. Put $\rho_k = c_{2k}^2/c_{2k+1}^2$, the squared $P$-link at $L = m+2k$ over the squared $Q$-link at $L+1$, $k = 1,\dots,K-1$. It is strictly increasing in $k$, because both $(n_P^2-L^2)/(n^2-(L+1)^2)$ and $(L^2-m^2)/((L+1)^2-m_Q^2)$ increase in $L$; the derivative of the second has numerator $2[L(L+1)+L(m^2-m_Q^2)+m^2] > 0$. Let $r-1$ be the number of $k$ with $\rho_k \le 1$.

Delete the odd site $2r+1$. The remaining matrix is $B' = B_L \oplus B_R$, the blocks of the even chains on sites $1,\dots,2r$ and $2r+2,\dots,N$, and $B^TB = B'^TB' + vv^T$, so $\sigma_{\min}(B) \ge \min(\sigma_{\min}(B_L), \sigma_{\min}(B_R))$ and $B$ has full column rank.

On the left chain every ratio is at most 1, so each entry of $E_L$ is at most $c_{2j-1}^{-1}$ and the column of $j$ has $r-j+1$ entries: $\|E_L\|_F^2 \le \sum_{j \le r}(r-j+1)/c_{2j-1}^2$. The odd link at $l = m+2j-1$ has type $Q$, and $n-l = 2(K-j+1) \ge 2(r-j+1)$, $l-m+1 = 2j$, $l+m-1 \ge 2j$ (by $m \ge 1$), so $c^2 \ge 2(r-j+1)(n+l)j^2$ and

$$\|E_L\|_F^2 \le \sum_j \frac{1}{2(n+l)j^2} \le \frac{\pi^2}{12(n+m+1)} \le \frac{\pi^2}{72},$$

since $n \ge m+3 \ge 4$. Read the right chain from its right end. Its odd links are then the $P$-links at $l = m+2k$, $k = K,\dots,r+1$, its ratios are $1/\rho_k < 1$, and the column

of the link at $k$ has $k-r$ entries, each at most $1/c$. With $n+1-l \geq 2(K-k+1)$, $c^2 \geq {}^1\!/_4(n+1-l)(n+1+l)(l-m)(l+m) \geq 2(K-k+1)(n+m+3)\,k\,(m+1)$, and $(k-r)/k \leq 1$,

$$\|E_R\|_F^2 \leq \sum_k \frac{1}{2(K-k+1)(n+m+3)(m+1)} \leq \frac{H_K}{2(n+m+3)(m+1)} \leq \frac{H_K}{4(n+4)},$$

the last step by $m \geq 1$. So $q_{\min} \geq \min(6\sqrt{2}/\pi, \sqrt{4(n+4)/H_K})$, and the second term has minimum 4.899 over $3 \leq n \leq 2\times 10^4$ (interval arithmetic) and is increasing beyond. □

The link inequalities of these two proofs were also checked in exact rational arithmetic on all 52 780 chains with spins at most 13 and 341 random chains with spins up to 300; against $q_{\min}$ computed in floating point, the smallest ratio of $q_{\min}$ to the bound is 1.35.

## D THE SLICE $j_1 = 1$

This appendix proves theorem 6.12 from the pencil of proposition 6.11. The pencil is a weighted birth–death problem: each edge contributes $A_e(\Phi_e - \Phi_{e-1})^2$ plus a nonnegative term, where here and below $A_e$ stands for the weight $A_e^+$ of proposition 6.11, the masses are $\mu_\ell$, and a missing end is a Dirichlet condition. We need the smallest nonzero eigenvalue $\lambda = (q/8)^2$ of the pencil to satisfy $\lambda \geq \lambda_* := (4.558/8)^2 = 0.32461506\ldots$.

**Lemma D.1** (Discrete Hardy inequality). On a chain with $\Phi = 0$ at a ghost site at one end, let $\tau_k$ be the sum of $1/A_e$ over the edges from the ghost to site $k$ and $\mathbb{M}_k$ the $\mu$-mass of the sites from $k$ onwards, $k$ included. Then $\sum \mu\Phi^2 \leq 4\mathrm{B}\sum A(\Delta\Phi)^2$ with $\mathrm{B} = \max_k \tau_k \mathbb{M}_k$, so the smallest eigenvalue is at least $1/(4\mathrm{B})$.

*Proof.* By Cauchy–Schwarz with weights $\sqrt{\tau}$,

$$\Phi_k^2 \leq 2\sqrt{\tau_k}\sum_{e\leq k} A_e\Delta_e^2\sqrt{\tau_e}, \qquad \text{because} \qquad \sum_{e\leq k}\Delta\tau_e/\sqrt{\tau_e} \leq 2\sqrt{\tau_k}.$$

Since $\tau_k \leq \mathrm{B}/\mathbb{M}_k$,

$$\sum_{k\geq e}\mu_k\sqrt{\tau_k} \leq \sqrt{\mathrm{B}}\sum_{k\geq e}\mu_k/\sqrt{\mathbb{M}_k} \leq 2\sqrt{\mathrm{B}\,\mathbb{M}_e} \leq 2\mathrm{B}/\sqrt{\tau_e}.$$

Exchange the sums. □

This is Muckenhoupt's weighted Hardy inequality with constant 4 [33].

**Lemma D.2** (Identical pairs). If the two pairs are identical with $m > 0$, then $A^- \equiv 0$, both ends are free, the kernel is the constants, and the second eigenvalue satisfies $\lambda_2 \geq 1/(2\tau_{int}\mathbb{M})$, with $\tau_{int}$ the sum of $1/A$ over the interior edges and $\mathbb{M}$ the total mass.

*Proof.* The eigenvector of $\lambda_2$ of the irreducible Jacobi pencil changes sign once, say after site $k$. Then $\lambda_2 \sum \mu (v^+)^2 = \langle v^+, Lv \rangle \geq \mathscr{E}_{left}(v^+)$, since the cross term $-A_{k+1} v_k v_{k+1}$ is nonnegative, and $\mathscr{E}_{left}$ is the left piece with a ghost at $k+1$; likewise on the right. Lemma D.1 on each piece gives $\lambda_2 \geq \max(1/(4\tau_L \mathbb{M}_L), 1/(4\tau_R \mathbb{M}_R))$, and $\tau_L + \tau_R \leq 2\tau_{int}$, $\mathbb{M}_L + \mathbb{M}_R = \mathbb{M}$ finish the proof. □

**The estimates.** Order the pairs so that $n := n_1 \leq \nu := n_2$, and put $m = \max(m_1, m_2)$, $N = n - m$. Since $j_1 = 1$, $n$ and $\nu$ are both integers or both half-integers, so $\nu > n$ means $\nu \geq n+1$. For an interior edge, $P_i(e) \geq (n_i^2 - e^2)(e^2 - m^2)$ and $(a+b)^2 \geq 3a^2 + b^2$ for $b \geq a$ give

$$(\sqrt{P_1} + \sqrt{P_2})^2 \geq (e^2 - m^2)\big[(\nu^2 - n^2) + 4(n^2 - e^2)\big],$$

hence $1/A_e \leq 2/((n+m)(e-m)(n-e))$, using $e(n+m) \leq (e+m)(n+e)$, and also $1/A_e \leq 8/((e-m)(\nu^2 - n^2))$. At a Dirichlet top edge ($\nu > n$), $1/A_n \leq 8n/((\nu^2 - n^2)(n^2 - m^2)) \leq 4/((n-m)(n+m))$; at a Dirichlet bottom edge $1/A \leq 8/(n^2 - m^2)$, or $2/(n^2 - 1)$ at the corner. For the masses, $\mu_\ell \leq 2/\ell$, so $\mathbb{M} \leq 4 + 2\ln 2n$ (counting $\mu_{1/2} = 8/3$ in the half-integer case), and $\mathbb{M} \leq 4N/n$ when $m > n/2$. Summing $\sum_e 1/((e-m)(n-e)) \leq 2H_{N-1}/N$ gives two forms of the Hardy product with the reference at the top:

$$\mathsf{B} \leq \mathsf{F} := \mathbb{M} \cdot \min\Big( \frac{4H_{N-1} + 4}{(n-m)(n+m)},\ \frac{8H_N}{\nu^2 - n^2} \Big).$$

*Proof of theorem 6.12.* Since $(4/4.558)^2 > 0.7701$, it suffices that $\mathsf{B} \leq 0.7701$, or $\tau_{int}\mathbb{M} \leq 2 \cdot 0.7701$ for identical pairs.

*Analytic part, $\nu \geq 33$.* If $\nu > n \geq 25$: $\mathsf{F} \leq 8(H_n + 1)(4 + 2\ln 2n)/n^2$, which is 0.729 at $n = 25$ and decreases (each factor's ratio between $n$ and $n+1$ is below $(n+1)^2/n^2$); for $m > n/2$ the bound $(32/3)(H_n + 2)/n^2$ is smaller still. If $\nu > n$ and $n \leq 24\frac{1}{2}$: take the minimum of the two forms. The second gives $\mathsf{F} \leq 8H_{24}(4 + 2\ln 48)/(33^2 - 24^2) = 0.691$ for $n \leq 24$; at $n = 24\frac{1}{2}$ the second form gives 0.7806 but the first gives 0.750. If $\nu = n$ and the pairs are not identical (or the corner is present), the bottom end is Dirichlet and the same estimate with $H + 2$ gives 0.729 at $n = 28$, decreasing. If the pairs are identical, lemma D.2 with $\tau_{int}\mathbb{M} \leq 8H_n(4 + 2\ln 2n)/n^2$ gives 0.483 at $n = 28$.

*Finite part, $\nu \leq 32\frac{1}{2}$.* For every unordered pair of pairs with $n_1, n_2 \leq 32$ (164 176 cases, half-integers included) the matrix $\mathscr{E} - \lambda_* \mu$, with $\lambda_*$ the exact rational $(4.558/8)^2$, was factorised exactly: 163 246 are positive definite and the other 930 are identical pairs with exactly one negative pivot, the kernel, so $\lambda_2 \geq \lambda_*$. The 10 416 pairs of pairs with $\max(n_1, n_2) = 32\frac{1}{2}$ were certified the same way (10 385 positive definite, 31 with exactly the kernel pivot); for them the analytic bound is also at most 0.4949. The pencil depends only on the two pairs, and the spectrum of $T$ does not depend on the pairing (proposition 4.5), so this covers every tuple with $j_1 = 1$. □

For large spins the bound grows like $n/\ln n$, and on the computed range it tracks the computed value within a factor of about 2.6. The smallest value on this slice over spins at most 19 is 7.3534, at $(1; 1/2, 1, 1, 3/2)$.

# E THE SLICE $j_1 = 3/2$

This appendix proves theorem 6.14. The proof has four parts: the reduction, the chain lemmas with their 30 polynomial certificates, a finite box of $6.4 \times 10^7$ parameter sets evaluated in outward-rounded interval arithmetic (section E.5), and the tail lemmas with their 50 certificates.

## E.1 ZIGZAGS AND LINKS

Fix an ordered pairing with pairs $(n_1, m_1)$, $(n_2, m_2)$ and write $P_i(L) = (n_i^2 - L^2)(L^2 - m_i^2)$. For $s = 3/2$, $F(x) = 4 - x^2$ and $G(u) = (u + 5/2)(u - 3/2)$. Zigzag 0 has rows $R_b = (b, b + 3/2)$ and columns $C_b = (b, b + 1/2)$; zigzag 1 has rows $R'_b = (b, b - 1/2)$ and columns $C'_b = (b, b - 3/2)$. In the order of $u$ each zigzag is a path $C_b, R_b, C_{b+1}, \dots$, and the block is $B = \left(\begin{smallmatrix} D_0 & 0 \\ N & D_1 \end{smallmatrix}\right)$. By proposition 4.4 the squared links of $T$ are

| LINK | BETWEEN | SQUARED WEIGHT |
|---|---|---|
| $q_b$ | $C_b$–$R_b$ | $3/16 P_2(b + 3/2)\, b/(b + 1)$ |
| $p_b$ | $R_b$–$C_{b+1}$ | $3/16 P_1(b + 1)\,(2b + 5)/(2b + 3)$ |
| $q'_b$ | $C'_b$–$R'_b$ | $3/16 P_2(b - 1/2)\,(b + 1)/b$ |
| $p'_b$ | $R'_b$–$C'_{b+1}$ | $3/16 P_1(b + 1)\,(2b - 1)/(2b + 1)$ |
| $n_b^+$ | $R'_b$–$C_b$ | $P_2(b + 1/2)(2b + 3)(2b - 1)/(16b(b + 1))$ |
| $n_b^-$ | $R'_b$–$C_{b-1}$ | $P_1(b)(b^2 - 1)/(4b^2 - 1)$ |

(checked exactly against the grid on 4 171 links). The site ranges follow from the region: rows of zigzag 0 have $b \in [\max(m_1, m_2 - 3/2, 0), \min(n_1 - 1, n_2 - 5/2)]$, its columns $b \in [\max(m_1, m_2 - 1/2, 1/2), \min(n_1 - 1, n_2 - 3/2)]$, rows of zigzag 1 $b \in [\max(m_1, m_2 + 1/2, 1), \min(n_1 - 1, n_2 - 1/2)]$, and its columns $b \in [\max(m_1, m_2 + 3/2, 3/2), \min(n_1 - 1, n_2 + 1/2)]$. A zigzag is of type CR, RC, RR or CC according to whether its path starts with a column or a row and ends with a column or a row; its imbalance is 0, 0, +1 or −1.

## E.2 REDUCTION

**Lemma E.1** (All-tall pairings). If in some ordered pairing both zigzags have imbalance 0 or +1, then $B$ is injective and $\lambda(T) = \sigma_{\min}(B)$.

*Proof.* Each $D_k$ is a path with nonzero links and at least as many rows as columns, so it is injective (delete an end row if it is tall). If $Bx = 0$ then $D_0 x_0 = 0$, so $x_0 = 0$, and then $D_1 x_1 = 0$. □

The proof needs a pairing in which neither zigzag has more columns than rows, and which pairing works depends on the gaps between the sorted spins of legs 2–5. The regimes below sort the tuples by those gaps. In R1 and R4 one outer gap exceeds the other by at least $3/2$ ($g_1$ in R1, $g_3$ in R4); in R2$^{\mp}$ they differ by exactly $1/2$; R1c and R4c are the boundary cases in which the smaller outer gap vanishes; and Tm, Tp are the two one-parameter families in which only one leg differs from the others, by $-1/2$ (Tm, the smallest leg) or $+1/2$ (Tp, the largest).

**Lemma E.2** (Regimes). Sort $j_2 \le j_3 \le j_4 \le j_5$ and put $g_1 = j_3 - j_2$, $g_2 = j_4 - j_3$, $g_3 = j_5 - j_4$; then $g_1 + g_3 \equiv 1/2 \pmod 1$. Every tuple falls in exactly one of the following regimes, and in the pairing listed both zigzags have imbalance in $\{0, 1\}$.

| REGIME | GAPS | PAIRING | ZIGZAG 0 / ZIGZAG 1 |
|---|---|---|---|
| R1 | $g_1 - g_3 \ge 3/2$ | $(j_2 j_4 \mid j_3 j_5)$ | CR / CR |
| R2$^-$ | $g_1 - g_3 = 1/2$, $g_3 \ge 1/2$ | $(j_2 j_4 \mid j_3 j_5)$ | CR / RR |
| R1c | $g_1 = 1/2$, $g_3 = 0$, $g_2 \ge 1/2$ | $(j_2 j_3 \mid j_4 j_5)$ | CR / CR |
| Tm | $g_1 = 1/2$, $g_2 = g_3 = 0$ | $(j_3 j_4 \mid j_2 j_5)$ | RC / RR |
| R4 | $g_3 - g_1 \ge 3/2$ | $(j_2 j_4 \mid j_3 j_5)$ | RR / RR |
| R2$^+$ | $g_3 - g_1 = 1/2$, $g_1 \ge 1/2$ | $(j_2 j_4 \mid j_3 j_5)$ | CR / RR |
| R4c | $g_1 = 0$, $g_3 = 1/2$, $g_2 \ge 1/2$ | $(j_2 j_3 \mid j_4 j_5)$ | RR / RR |
| Tp | $g_1 = 0$, $g_3 = 1/2$, $g_2 = 0$ | $(j_2 j_3 \mid j_4 j_5)$ | RC / RR |

The one-parameter families are Tm $= (3/2; j, j+1/2, j+1/2, j+1/2)$ and Tp $= (3/2; j, j, j, j+1/2)$.

*Proof.* In $(j_2 j_4 | j_3 j_5)$, $n_2 - n_1 = g_1 + g_3$ and $m_2 - m_1 = g_3 - g_1$, with $m_1 = g_1 + g_2$, $m_2 = g_2 + g_3$; in $(j_2 j_3 | j_4 j_5)$, $n_2 - n_1 = g_1 + 2g_2 + g_3$ and $(m_1, m_2) = (g_1, g_3)$. The conditions are disjoint and exhaustive because $g_1 - g_3$ is a nonzero half-odd integer. Substituting into the four range formulas gives the types; for example in R1, $m_2 + 3/2 \le m_1$ and $m_1 \ge 3/2$ make every lower end $m_1$, and $n_2 \ge n_1 + 3/2$ makes every upper end $n_1 - 1$. □

The census of all 54 019 tuples with spins at most 20 (legs 2–5 sorted, $\dim \mathrm{Inv} \ge 2$) confirms the types, the imbalances and the ratio hypotheses below in exact arithmetic.

*Proposition E.3 — The coupling identity*

On the columns of zigzag 0, $N^TN = {}^4\!/_3 D_0^T D_0 + \operatorname{diag}(\Lambda)$ with $\Lambda_c = {n_c^+}^2 + {n_{c+1}^-}^2 - {}^4\!/_3(q_c^2 + p_{c-1}^2) = \Lambda_c^1 + \Lambda_c^2$,

$$\Lambda_c^1 = \frac{P_1(c+1)\,c(c+2)}{(2c+1)(2c+3)} - \frac{P_1(c)(2c+3)}{4(2c+1)}, \qquad \Lambda_c^2 = \frac{P_2(c+{}^1\!/_2)(2c+3)(2c-1)}{16c(c+1)} - \frac{P_2(c+{}^3\!/_2)\,c}{4(c+1)}.$$

This is proposition 6.13 with $\mathrm{r} = F(0)/F(-1) = {}^4\!/_3$; the formula holds at every column, the ends included, because the link of a missing row vanishes.

**Corollary E.4.** Let $F_0 \geq 1/\lambda_{\min}(D_0^T D_0)$, $F_1 \geq 1/\lambda_{\min}(D_1^T D_1)$ and $C_\Lambda \geq \sup_x x^T\Lambda x/|D_0 x|^2$ ($C_\Lambda = 0$ if $\Lambda \leq 0$). Then for every $\tau > 0$,

$$\sigma_{\min}(B)^2 \geq \min\Big(\frac{1-\tau({}^4\!/_3 + C_\Lambda)}{F_0},\ \frac{\tau}{(1+\tau)F_1}\Big).$$

When $D_0$ is square one may take $C_\Lambda = \sum_c \Lambda_c^+ R_c$ with $R_c \geq \sum_k (D_0^{-1})_{ck}^2$.

*Proof.* If $1 - \tau({}^4\!/_3 + C_\Lambda) \leq 0$, the first term of the minimum is at most 0 and there is nothing to prove. Otherwise, by (8) and $D_0^T D_0 - \tau N^T N = (1 - {}^4\!/_3\tau)D_0^T D_0 - \tau\Lambda \geq (1 - \tau({}^4\!/_3 + C_\Lambda))D_0^T D_0$; for the last statement write $y = D_0 x$ and bound the operator norm of $\Lambda^{+1/2} D_0^{-1}$ by its Frobenius norm. □

## E.3 CHAINS

Each lemma uses only the explicit inverse of a bidiagonal path. For a row $R_b$ write $r_b = p_b^2/q_b^2$ (same-row ratio) and $\bar r_b = p_b^2/q_{b+1}^2$ (same-column ratio).

**Lemma E.5** (CR chains). If $D$ is square with $(Dx)_{R_b} = q_b x_b + p_b x_{b+1}$, then for $k \geq c$

$$|(D^{-1})_{ck}| = q_c^{-1} \prod_{t=c}^{k-1} \sqrt{\bar r_t}.$$

If every $\bar r_t \leq 1$, then $1/\lambda_{\min}(D^T D) \leq \sum_c (\mathrm{hi} - c + 1)/q_c^2$ and $R_c \leq (\mathrm{hi} - c + 1)/q_c^2$ (count form). With $\ell_c \leq q_c^2$ the entries obey $E_{ck}^2 \leq \ell_c^{-1} \prod_{t=c}^{k-1} p_t^2/\ell_{t+1}$, which gives the forward recursion $F = \sum_k \Phi_k$, $\Phi_k = (p_{k-1}^2 \Phi_{k-1} + 1)/\ell_k$, and the backward recursion $R_c = (1 + p_c^2 R_{c+1})/\ell_c$ (exact-product form).

**Lemma E.6** (RC chains). If the rows $R_{lo}, \dots, R_{hi}$ are matched to the columns above them by $p$, then $|E_{c,i}| = p_i^{-1} \prod_{t=i+1}^{c-1} q_t/p_t$ for $i < c$; if every $r_t \geq 1$, then $1/\lambda_{\min} \leq \sum_i (\mathrm{hi} - i + 1)/p_i^2$ and $R_c \leq \sum_{i<c} 1/p_i^2$.

**Lemma E.7** (Tall chains). If the rows are $R_{lo}, \dots, R_{hi}$, the columns $C_{lo+1}, \dots, C_{hi}$, and $r_t$ is strictly decreasing on the interior rows, then $1/\lambda_{\min}(D^T D) \leq \sum_{i<\mathrm{hi}} (\mathrm{hi} - i)/p_i^2 + \sum_{i>\mathrm{lo}} (i - \mathrm{lo})/q_i^2$.

*Proof.* Delete the first row $j$ with $r_j \le 1$ (or the top row). The lower piece is matched by $p$ with ratios $q/p \le 1$, the upper by $q$ with ratios $p/q \le 1$; add their Frobenius norms. □

In every bound the length factors $(\mathrm{hi} - c + 1,\ \mathrm{hi} - i,\ i - \mathrm{lo})$ are cancelled by a factor $n - L$ or $L - m$ of the links, which is what makes the bounds uniform. The sums that remain are handled by the identity $\sum_{1\le c\le k\le N} \frac{1}{k(N+1-c)} = H_N^{(2)} < \frac{\pi^2}{6}$, which follows from grouping $\sum_{j+k\le N} 1/(jk)$ by $j + k$.

### E.4 WORST CASES AND CERTIFICATES

Pair 1 enters exactly. Pair 2 enters only through monotone dependences ($P_2(L)$ increases in $n_2$ and decreases in $m_2$; $\Lambda^2$ is nonincreasing in $n_2^2$ and $m_2^2$ for $c \ge 1$), so each regime is evaluated at the boundary of pair 2: $n_2 = n_1 + 3/2$, and $m_2$ at its extreme value; in R4 the pair-1 parameter is worst-cased instead. Every bound is then a function of two parameters, $(n_1, m_1)$ or $(n_1, m_2)$, or of $n_1$ alone in R1c, R4c, Tp and Tm. The hypotheses are 30 exact polynomial certificates, each showing that a polynomial in nonnegative shift variables has coefficients of one sign (or, once, that a univariate polynomial has no root on $(1, \infty)$):

| LABEL | CLAIM | USED FOR |
|---|---|---|
| S1 | $\Lambda_c \le 0$ on zigzag 0 in R4 and R4c | $C_\Lambda = 0$ there |
| S2 | $r$ strictly decreasing on interior rows of every tall chain | lemma E.7 |
| S3 | $\bar r \le 1$ for the CR chains of R1, R1c and R2$^\pm$ (zigzag 0) | lemma E.5 |
| S4 | $r \ge 1$ for the RC chains of Tp and Tm | lemma E.6 |
| S5 | $\partial\Lambda^2/\partial(n_2^2) \le 0$, $\partial\Lambda^2/\partial(m_2^2) \le 0$ for $c \ge 1$ | worst cases |
| S6 | $\Lambda \le m_1^2[(4c+9)n_1^2 + c(4c^2+11c+8)]/(4(2c+1)(2c+3))$ | tails |
| S7 | telescoping bounds $\bar r \le \frac{\Theta_c}{\Theta_c+1}\cdot(\dots)$, $\Theta_c = c - \mathrm{lo} + 1$ | tails |
| S8 | two elementary rational inequalities | tails |

Negative controls (false variants of S1–S3) fail, as they must. The two columns $c = 1/2$ that occur (in R1c and in R2$^+$ with $m_1 = 1/2$) are treated exactly: $\Lambda_{1/2} = 5/16(n_1^2 - 9/4) - 1/3(n_2^2 - 4) \le 0$ in R1c and $\Lambda_{1/2} \le n_1^2/16$ in R2$^+$.

## E.5 THE BOX

For each parameter set the bound of corollary E.4 is evaluated with $F_0, F_1, C_\Lambda$ the smaller of the count and exact-product forms (lemma E.7 for tall chains) and $\tau$ chosen to balance the two terms. The parameter sets are all those with $n_1 \le 4000$ in the four two-parameter regimes and $n_1 \le 10^5$ in the four one-parameter ones, $6.4 \times 10^7$ in all. The minima of the resulting lower bound for $q = 8\sigma_{\min}(B)$ are:

| REGIME | PARAMETER SETS | MIN BOUND FOR $q$ | AT |
|---|---|---|---|
| R1 | 15 980 006 | 10.873 | $n_1 = 7/2$, $m_1 = 3/2$ |
| R2$^+$ | 15 988 002 | 7.125 | $n_1 = 9/2$, $m_1 = 1/2$ |
| R2$^-$ | 15 984 004 | 4.723 | $n_1 = 3$, $m_1 = 1$ |
| R4 | 15 999 998 | 6.507 | $n_1 = 2$, $m_2 = 3/2$ |
| R1c | 99 998 | 8.219 | $n_1 = 5/2$ |
| R4c | 99 999 | 4.621 | $n_1 = 3$ |
| Tp | 99 999 | 4.386 | $n_1 = 4$ |
| Tm | 99 998 | 3.777 | $n_1 = 4$, i.e. $(3/2; 1, 3/2, 3/2, 3/2)$ |

These values come from a binary64 evaluation, truncated to three decimals, and were reproduced to six decimals by an independent exact-rational evaluation at the minimising parameters. The certified version re-runs the whole box with every operation rounded outward by one ulp in the safe direction and bounds the cancellation $1-\tau(4/3+C_\Lambda)$ from below directly. Every one of the 64 352 004 parameter sets is certified positive, with no NaN, and the certified lower bounds for the minima, truncated to six decimals, are 10.873596 (R1), 7.125868 (R2$^+$), 4.723102 (R2$^-$), 6.507913 (R4), 8.219252 (R1c), 4.621864 (R4c), 4.386733 (Tp) and 3.777422 (Tm), each at the same parameters as above. The smallest certified cancellation margin $1 - \tau(4/3 + C_\Lambda)$ is $3.02 \times 10^{-5}$ (truncated), at Tm. The binding value 3.7774 exceeds the constant 3.7 of theorem 6.14.

## E.6 TAILS

Beyond the box we use a sufficient condition that needs no choice of $\tau$.

**Lemma E.8.** Let $\lambda_* = (3.7/8)^2$ and $\mathscr{C} = \lambda_*[F_0 + 7/3 F_1 + C_\Lambda F_1]$. If $\mathscr{C} \le 1$, then $\sigma_{\min}(B)^2 \ge \lambda_*$.

*Proof.* Take $\tau = \theta/(4/3 + C_\Lambda)$ with $\theta = 1 - \lambda_* F_0$. The first term of corollary E.4 is then $\lambda_*$, and the second is at least $\lambda_*$ iff $\theta \ge \lambda_*(4/3 + C_\Lambda + \theta)F_1$, which follows from $(1 - \lambda_* F_0)(1 - \lambda_* F_1) \ge 1 - \lambda_* F_0 - \lambda_* F_1 \ge \lambda_*(4/3 + C_\Lambda)F_1$, a consequence of $\mathscr{C} \le 1$. □

In each regime the three box quantities are bounded by closed forms in $n = n_1$ and $\mathsf{L} = 1 + \ln n$, uniformly in the second parameter $a$. The factorisations behind them are exact identities: for instance in R2$^-$ ($m_2 = a - 1/2$, $\Theta_c = c - a + 1$) the squared links of zigzag 0 satisfy

$$\frac{16}{3}\,\frac{c+1}{c}\,q_c^2 = (n-c)(n+c+3)(\Theta_c+1)(c+1+a),$$

and the same-column ratio is

$$\bar r_t = \frac{\Theta_t}{\Theta_t+2}\cdot\frac{t+1+a}{t+2+a}\cdot\frac{n+t+1}{n+t+4}\cdot\frac{(2t+5)(t+2)}{(2t+3)(t+1)},$$

whose products telescope; $\Lambda$ is a quadratic in $n$ whose three coefficients are bounded by certified polynomial inequalities. The resulting bounds are:

| REGIME | $F_0$ | $F_1$ | $C_\Lambda$ |
|---|---|---|---|
| R2$^-$ | $\frac{32}{3n}$ | $12/n + \frac{32\mathsf{L}}{3n(n+2)}$ | $8(2.5+\ln n)\mathsf{L} + \frac{400(2.5+\ln n)}{n-10}$ |
| R2$^+$ | $16/n$ | $\frac{32}{3n} + \frac{32\mathsf{L}}{3n(n+2)}$ | $16\mathsf{L}^2 + 2/3\mathsf{L}$ |
| R1 | $\frac{4.22}{n}$ | $\frac{16}{3n}S(2a-3)$ | $a[2.67\mathsf{L}^2 + 12.6\mathsf{L} + 9.3]$ |
| R4 | $\frac{16\pi^2}{18n} + \frac{64\mathsf{L}}{3n(n+3)}$ | $\frac{3.52}{n} + \frac{32\mathsf{L}}{3n(n+2)}$ | 0 |
| R4c | $\frac{16\pi^2}{18n} + \frac{32\mathsf{L}}{3n(n+2)}$ | $\frac{16(\pi^2/6-1)}{n} + \frac{32\mathsf{L}}{3n(n+3)}$ | 0 |
| R1c | $\frac{16(\mathsf{L}+\ln 2)(1+2\mathsf{L})}{3n^2}$ | $\frac{32\mathsf{L}^2}{n(n+1)}$ | $16/225(\mathsf{L}+\ln 2)(1+2\mathsf{L})$ |
| Tm | $\frac{32\mathsf{L}^2}{3n^2}$ | $\frac{16(\pi^2/6-1)}{n} + \frac{32\mathsf{L}}{3n(n+1)}$ | $6\mathsf{L}^2$ |
| Tp | $\frac{64\mathsf{L}^2}{3n^2}$ | $\frac{16(\pi^2/6-1)}{n} + \frac{32\mathsf{L}}{3n(n+2)}$ | $6\mathsf{L}^2$ |

Here $S(x) = \sum_{i\ge1} 1/(i(i+x)) \le 1/(1+x) + \ln(1+x)/x$, and in R1 the product $C_\Lambda F_1$ is made uniform in $a$ by $aS(2a-3) \le 3 + 1/2\ln 2n$. Writing $\mathscr{C}(n)$ as a sum of monomials $K\mathsf{L}^p/\prod(n+\gamma_i)$, each is nonincreasing for $n \ge n_0$ once its logarithmic derivative is negative at $n_0$, and $\mathscr{C}(n_0)$ is evaluated in 120-bit interval arithmetic, with every decimal constant entered as an exact rational and the derivative test also done in intervals: $\mathscr{C} \le 0.7644$ (R1), $0.7960$ (R2$^+$), $0.5189$ (R2$^-$), $0.000915$ (R4) at $n_0 = 4000$, and $3\times10^{-6}$ (R1c), $7.03\times10^{-5}$ (R4c), $0.0208$ (Tp, Tm) at $n_0 = 10^5$ (upper bounds, rounded up). All are below 1. (An earlier version of the script entered the decimal constants as binary64 numbers, so that its intervals enclosed the nearest binary64 points rather than the decimals, and did the derivative test in ordinary precision. The margins are wide enough that its verdicts were the same; the certificate of record is the repaired script, `certificates/z32rigor/scripts/tail_certify_exact.py`.) The 50 polynomial facts used are exact certificates, and every closed form was also checked to dominate the corresponding box quantity at $n_1 \in \{60, 60\,1/2, 200, 200\,1/2, 500, 999\,1/2\}$ for all $a$.

### E.7 SMALL SPINS EXACTLY

Independently of box and tails, the two matrices of (8) are tridiagonal in the column index with rational diagonals and off-diagonals whose squares are rational, so their positivity is decided by exact $LDL^T$ pivots at rational $\tau$. All 54 019 tuples with $j_1 = 3/2$ and spins at most 20, counted as above, are certified this way at $q \geq 7.18$; the smallest certified value is 7.190 at $(3/2; 1/2, 1, 1, 1)$, where the computed value is 7.476, so the two-chain inequality loses 4% there.

## F THE SLICE $j_1 = 2$

This appendix proves theorem 6.15. Tuples whose legs 2–5 are not all integers or all half-integers at least $3/2$ are covered by theorem 6.5 with $q \geq 4\sqrt{12/\pi} > 7.8$, and tuples with spins at most 19 by the finite certificate. It remains to treat the *even-sum class* (legs all integers, or all half-integers at least $3/2$) with $2j_{\max} \geq 39$. Throughout, $\lambda_0 = (4.5579607/8)^2$.

### F.1 THE TRIANGULAR REDUCTION

For any $j_1$ and a colour class $C$ of columns, let $Z(\delta, \rho)$ be the block of $B$ from column diagonal $\delta$ to an adjacent row diagonal $\rho = \delta \pm 1$. A row meets at most two columns of $\delta$, consecutive in $u$, so $Z(\delta, \rho)$ is a path matrix and $Z^T Z$ is tridiagonal.

**Lemma F.1.** $\|Bx\|^2 = \sum_\rho \|Z(\rho-1,\rho)x_{\rho-1} + Z(\rho+1,\rho)x_{\rho+1}\|^2$, and for a column diagonal $\delta$, with $D = Z(\delta, \delta-1)$, $N = Z(\delta, \delta+1)$ and $\mathrm{r}_\delta = F(\delta+1/2)/F(\delta-1/2)$, $N^T N - \mathrm{r}_\delta D^T D = \mathrm{diag}(\Lambda)$, where at the column $(b, c)$

$$\Lambda_{(b,c)} = \tfrac{1}{16} F(\delta + \tfrac{1}{2}) \big[ G(u+1)\big(h_1(b+1)^2 - h_2(c+1)^2\big) - G(u)\big(h_1(b)^2 - h_2(c)^2\big) \big].$$

*Proof.* $T$ links only adjacent diagonals. The off-diagonal entries are the plaquette argument of proposition 6.13; the diagonal is the difference of the two sums of squared links at $(b, c)$, read off proposition 4.4. □

*Theorem F.2 — Ownership*

Let each row diagonal $\rho$ that touches two column diagonals be dropped, or owned by one of its neighbours with a parameter $\theta_\rho \in (0,1)$; row diagonals touching one column diagonal are owned by it with coefficient 1. Put

$$\mathrm{Stage}_\delta = \sum_{\rho \text{ owned by } \delta} c_\rho Z(\delta,\rho)^T Z(\delta,\rho) - \sum_{\rho \text{ owned by the other neighbour}} \big(\theta_\rho^{-1} - 1\big) Z(\delta,\rho)^T Z(\delta,\rho),$$

with $c_\rho = 1$ for walls and $1-\theta_\rho$ otherwise. Then $\|Bx\|^2 \geq \sum_\delta x_\delta^T \mathrm{Stage}_\delta x_\delta$; if every $\mathrm{Stage}_\delta \succeq \lambda$ with $\lambda > 0$, then $B$ is injective and $\sigma_{\min}(B)^2 \geq \lambda$.

*Proof.* Apply $\|a+b\|^2 \geq (1-\theta)\|a\|^2 - (\theta^{-1}-1)\|b\|^2$ to each owned row diagonal, with $a$ the owner's part, and discard dropped rows. □

Each $\mathrm{Stage}_\delta - \lambda$ is tridiagonal with rational diagonal and off-diagonal entries that are rational multiples of one square root (by the proportionality of lemma F.1), so its $LDL^T$ pivots are rational: *stage certificates* are exact.

**Count forms.** Let $D$ be a square path matrix with a perfect matching of rows to columns that follows the path from its Dirichlet end, matched link $\ell_k$ at row $k$. Then $|(D^{-1})_{c,k}| = \ell_k^{-1} \prod(\text{unmatched/matched ratios from } k \text{ to } c)$, and if every ratio is at most 1, $\sigma_{\min}(D)^2 \geq 1/\sum_k (\#\text{columns reached by } k)/\ell_k^2$; grouping by columns gives the same with column ratios. The row norms obey the same bounds, and $\sup_x x^T \Lambda^+ x / \|Dx\|^2 \leq \sum_c \Lambda_c^+ R_c$. A tall path is split by deleting one row. Chains are typed by their ends as before: RC, CR (square), RR (tall), CC (wide, never used).

## F.2 THE PROOF FOR $j_1 = 2$

In the even-sum class all $n_i, m_i$ are integers and $N_i \geq 3$. Take $C = \{d = \pm 1\}$; the row diagonals are the walls $d = \pm 2$ and the middle $d = 0$. Write $W_- = Z(-1,-2)$, $W_+ = Z(1,2)$, $M_- = Z(-1,0)$, $M_+ = Z(1,0)$, so that

$$\|Bx\|^2 = \|W_- x_-\|^2 + \|M_- x_- + M_+ x_+\|^2 + \|W_+ x_+\|^2. \tag{10}$$

With $s = 2$: $F(\pm 1/2) = 6$, $F(\pm 3/2) = 4$, $G(u) = (u+3)(u-2)$; write $P(L) = (n_1^2 - L^2)(L^2 - m_1^2)$ and $Q(L) = (n_2^2 - L^2)(L^2 - m_2^2)$.

**Lemma F.3** (Chains). $W_-$ has rows $(b, b+2)$ and columns $(b, b+1)$; the row $(b, b+2)$ has squared links $q_b^2 = Q(b+2)\, b/(2(2b+3))$ to $(b, b+1)$ and $p_b^2 = P(b+1)(b+3)/(2(2b+3))$ to $(b+1, b+2)$. $M_+$ has rows $(e, e)$ and columns $(c+1, c)$; the row $(e, e)$ has $\alpha_e^2 = {}^3/_4 Q(e)(e-1)(2e+3)/((2e-1)(2e+1))$ to $(e, e-1)$ and $\beta_e^2 = {}^3/_4 P(e+1)(e+2)(2e-1)/((2e+1)(2e+3))$

to $(e+1, e)$. $W_+$, $M_-$ are the mirror images. On diagonal $-1$, $M_-^T M_- = {}^3\!/_2 W_-^T W_- + \mathrm{diag}(\Lambda)$ with $\Lambda_c = 3N_c/(4(2c+1)(2c+3))$ and

$$N_c = (n_1^2 + m_1^2 - n_2^2 - m_2^2)K(c) + 2n_1^2 m_1^2(c+3) - 2n_2^2 m_2^2(c-1) - 2(n_2^2 + m_2^2) + 2(2c^2 + 4c + 1)^2,$$

$K(c) = c(2c^2 + 6c + 5)$. $W_-$ is of type RC, CR or RR, or wide exactly when $m_1 \geq 1$, $m_2 \leq m_1 + 1$ and $n_2 \leq n_1 + 1$.

The proof needs a pairing in which the wall chain $W_-$ is not *wide* (more columns than rows), so that it is injective. The routing lemma finds one in three ways: in case NN neither wall chain is wide; in cases C1 and C3 only $W_-$ is used, and the pair $(n_2, m_2)$ is far from $(n_1, m_1)$ in its upper end (C1) or close to it (C3), which decides how the middle row is handled.

**Lemma F.4** (Routing). Sort $j_2 \leq j_3 \leq j_4 \leq j_5$ with integer gaps $g_1, g_2, g_3$. Some ordered pairing gives one of: (NN) neither wall chain is wide; (C1) $W_-$ is not wide, $n_2 \geq n_1 + 2$, and $m_2 \geq m_1 + 2$ or $(m_1, m_2) = (0, 1)$; (C3) $W_-$ is not wide, $|n_2 - n_1| \leq 1$, and $m_2 \geq m_1 + 2$ or $(m_1, m_2) = (0, 1)$. In C1 and C3, $W_+$ is wide or empty, and this is not used. If $2j_{\max} \geq 39$, then $n_2 \geq 22$ in C1 and $\min(n_1, n_2) \geq 21$ in C3.

*Proof.* For $(23|45)$, $(n_2 - n_1, m_1, m_2) = (g_1 + 2g_2 + g_3, g_1, g_3)$; for $(45|23)$ the negatives and swap; for $(34|25)$, $(g_3 - g_1, g_2, g_1 + g_2 + g_3)$. If $|g_1 - g_3| \leq 1$ and $g_1 + g_3 \geq 2$, then $(34|25)$ is C3; if $(g_1, g_3) = (0, 0)$, $(23|45)$ is NN; if $(g_1, g_3) = (1, 0)$, $(23|45)$ is NN when $g_2 \geq 1$ and $(45|23)$ is C3 otherwise; if $(g_1, g_3) = (0, 1)$, $(23|45)$ is C3 when $g_2 = 0$ and C1 otherwise; if $g_3 \geq g_1 + 2$, $(23|45)$ is C1; if $g_1 \geq g_3 + 2$, $(45|23)$ is NN. The pair containing $j_5$ has $n + m = 2j_5 + 1$ and $m \leq n - 3$. □

The ratio conditions that the count forms need (same-row ratio of $W_-$ at least 1 in type RC, same-column ratio at most 1 in type CR, monotone same-row ratio in type RR, and the same for $M_+$), the bounds on $N_c$, and the telescoping of the coupling are 24 exact polynomial certificates (S1–S10). The same script also runs three negative controls, false variants of S1, S2 and S7 that must fail and do; they are checks, not part of the proof. With $S(m) = \sum_{i \geq 1} 1/(i(i+2m))$ ($= \pi^2/6$ for $m = 0$, $H_{2m}/(2m)$ otherwise) and $\Theta_0 = \sum_{k \geq 1} \varphi(k)/(k+1)^2 \leq 2.21190$, $\varphi(k) = (2k+1)(2k+3)/((k+2)(2k-1))$, they give:

**Lemma F.5** (Chain bounds). (i) $W_-$ of type RC: $\sigma_{\min}^2 \geq (n_1 + m_1 + 1)/(4S(m_1))$. (ii) Type CR: $\sigma_{\min}^2 \geq n_2/(4\pi^2/6)$. (iii) Type RR: $\sigma_{\min}^2 \geq 1/\max\big(4S(m_1)/(n_1 + m_1 + 1),\ {}^5\!/_4 H_{n_2-3}/n_2\big)$, where for the lower piece the column count is at most $n_1 - L$ because the rows end at $n_1 - 1$. (iv) If $m_2 \geq 1$ and ($m_2 \geq m_1 + 2$ or $(m_1, m_2) = (0, 1)$), then $\sigma_{\min}(M_+)^2 \geq 1/\max\big({}^4\!/_3 \Theta_0/(n_1 + m_2 + 1),\ {}^{40}\!/_7 H_{n_2}/(n_2(n_2+1))\big)$, where the second term is present only when $n_2 \geq n_1$.

**Lemma F.6** (Coupling and two stages). Let $\mathrm{C} = \sup \|M_- x\|^2/\|W_- x\|^2$ on diagonal $-1$, so that $\mathrm{C} \leq {}^3\!/_2 + \sum_c \Lambda_c^+ R_c$. In C1, $\Lambda_c \leq 0$ for every column, so $\mathrm{C} \leq {}^3\!/_2$. In C3, $W_-$ is RC and its same-row ratios satisfy $\rho_t \leq \nu_t\, t/(t+1)$ with an exactly telescoping $n$-ratio $\nu_t$, which

gives $C \le 3/2 + W$ with an explicit $W$. If $\sigma_{\min}(W_-)^2 \ge \lambda_W > \lambda_0$, $\sigma_{\min}(M_+)^2 \ge \lambda_M$, $C \le \bar{C}$ and $\lambda_M \ge \lambda_0\big(1 + \bar{C}\,\lambda_W/(\lambda_W - \lambda_0)\big)$, then $\sigma_{\min}(B)^2 \ge \lambda_0$.

*Proof.* For the last statement apply theorem F.2 to (10): the walls are owned by their own diagonals, the middle row by $x_+$ with $1-\theta = \lambda_0/\lambda_M$; then $x_-$ receives $(1-(\theta^{-1}-1)\bar{C})\lambda_W \ge \lambda_0$ and $x_+$ receives $(1-\theta)\lambda_M = \lambda_0$, and $x \mapsto x/(x-\lambda_0)$ is decreasing, so lower bounds may be substituted. □

*Proof of theorem 6.15.* Take the pairing of lemma F.4. *NN.* Drop the middle row. Each wall chain has $\sum \le \max(4\cdot\frac{\pi^2}{6}/4,\ 4\cdot\frac{\pi^2}{6}/5,\ \max_{n\ge5} 5/4H_{n-3}/n) = 1.6449$, so $\sigma_{\min}(B)^2 \ge 0.6079$ and $q \ge 6.237$. *C1.* Take $\lambda_W$ from (iii), $\lambda_M$ from (iv), $\bar{C} = 3/2$. Both bounds are nondecreasing in all four parameters and the requirement of lemma F.6 decreases in $\lambda_W$, so the worst case is the corner $(n_1, m_1, n_2, m_2) = (3, 0, 22, 1)$, where $\lambda_W = 0.6079$, $\lambda_M = 1.6954$ and the requirement is 1.3694. *C3.* For fixed $(n_1, n_2-n_1, m_1)$ the coupling bound $W$ is nonincreasing in $m_2$ and $\lambda_M$ nondecreasing, so it suffices to check the slices $m_2 = m_1 + 2$ and $(m_1, m_2) = (0, 1)$. For $21 \le n_1 \le 999$ all 1 489 059 slice configurations satisfy lemma F.6 in outward-rounded interval arithmetic, with smallest margin $\lambda_M$/requirement $> 3.2356$ at $(21, 0, 22, 1)$. For $n_1 \ge 1000$: $(k+1)/p_k^2 \le 4/((n_1^2 - L^2)(L - m_1))$ and partial fractions give $R_c \le 8\nu H_{n_1}/(cn_1(n_1 - m_1))$ and $N_c^+ \le 8an_1c(c+2)^2 + 8(c+1)^4$, where $\nu$ and $a$ are fixed together by the case: $(\nu, a) = (1, 1)$ for $n_2 = n_1 - 1$, $(2, 0)$ for $n_2 = n_1$ and $(4, 0)$ for $n_2 = n_1 + 1$. So $W \le \bar{W}(n_1) = 12\nu H(5a + 1 + 3/n_1 + (2H + 2.6)/n_1)$ with $H = H_{n_1}$, which at $n_1 = 1000$ is 540.8, 183.3 and 366.7 in the three cases, while the requirement allows $\bar{C} \le 1043.4$ there. This comparison persists for every $n_1 \ge 1000$. With $\lambda_M \ge 3(n_1 + 2)/(4\Theta_0)$ (from lemma F.5(iv), whose second term is the smaller one for $n_1 \ge 1000$, and $m_2 \ge 1$) and $\lambda_W \ge 3(n_1 + 1)/(2\pi^2)$ (from (i), as $S(m_1) \le \pi^2/6$) the allowance is at least $f(n_1) = (A(n_1+2)-1)(1-B/(n_1+1))$ with $A = 3/(4\Theta_0\lambda_0) > 1.04$ and $B = 20\lambda_0/3$, and $f(n+1)-f(n) = A+B(A-1)/((n+1)(n+2)) \ge A$. The case $(\nu, a) = (1, 1)$ dominates the other two, since its excess over $(4, 0)$ is $24/5H(5n_1 - 15H - 42)/n_1 > 0$; and its bound $\bar{W} = 72H_{n_1} + (24H_{n_1}^2 + 67.2H_{n_1})/n_1$ grows by less than $72/(n_1 + 1) < 0.072$ per unit of $n_1$, because the second term decreases for $n_1 \ge 1000$: with $H' = H + 1/(n_1 + 1)$, $n_1(n_1 + 1)$ times its increment is at most $-24H^2 - 19.2H + 67.2 + 24/(n_1 + 1) < 0$, as $H > 2$. □

As independent checks, the closed forms were compared with the actual chain matrices on random configurations (never exceeded; the coupling bound is within 0.1% in places), and every even-sum tuple with $39 \le 2j_{\max} \le 44$ was given an exact stage certificate directly.

## G THE THREE-SPIN PENCIL AND THREE BOUNDED LEGS

### PROOF OF THE THREE-SPIN PENCIL

This is the proof of theorem 7.1. Write $\mathscr{H}_{2345} = j_2 \otimes j_3 \otimes j_4 \otimes j_5$, $L = J_2 + J_3 + J_4 + J_5 = \mathrm{G} + J_5$, and $H_s = \{h \in \mathscr{H}_{2345} \,:\, L_z h = sh,\ L_+ h = 0\}$ for the highest-weight vectors of weight $s = j_1$. All adjoints are taken in the natural inner products.

*Proof. Step 1: removing the distinguished leg.* By lemma 4.3, $J_1 \cdot (J_4 \times J_5) = J_1 \cdot (J_2 \times J_4 + J_3 \times J_4)$ on invariant vectors, so $T = J_1 \cdot U$ there. $U$ acts on legs 2–4 and commutes with $J_1$, and on an invariant $\psi$ we have $J_1^a \psi = -L^a \psi$; hence $T\psi = -\sum_a U^a L^a \psi$. The two orders of the product agree, $\sum_a U^a L^a = \sum_a L^a U^a =: L \cdot U$, because $U$ is a vector operator for G and commutes with $J_5$, so $\sum_a [L^a, U^a] = \sum_a [\mathrm{G}^a, U^a] = i \sum_a \epsilon_{aab} U^b = 0$; for the same reason $L \cdot U$ is Hermitian and commutes with $L$. Write an invariant as $\psi = \sum_m |s, m\rangle \otimes w_m$. The component $w_{-s}$ has weight $s$ and is annihilated by $L_+$ (the $|s, -s\rangle$ component of $(J_1 + L)_+ \psi = 0$), and $\psi \mapsto w_{-s}$ is $(2s+1)^{-1/2}$ times a unitary map from $\mathrm{Inv}(j_1 \otimes \mathscr{H}_{2345})$ onto $H_s$. Since $L \cdot U$ acts on $\mathscr{H}_{2345}$ alone and commutes with all rotations of $\mathscr{H}_{2345}$, this map carries $T$ to $-(L \cdot U)|_{H_s}$.

*Step 2: the extremal projector.* On the weight space $\mathscr{H}_{2345}[s]$ put $P = \sum_k \chi_k L_-^k L_+^k$. On a vector $|l, s\rangle$ of total spin $l \geq s$, $L_-^k L_+^k |l, s\rangle = \prod_{i<k} (l - s - i)(l + s + 1 + i)\, |l, s\rangle$, so $P|l, s\rangle$ is $|l, s\rangle$ times the terminating series ${}_2F_1(-(l-s), l+s+1; 2s+2; 1) = (s+1-l)_{l-s}/(2s+2)_{l-s}$ (Chu–Vandermonde), which is 1 for $l = s$ and 0 for $l > s$. So $P$ is the orthogonal projection of $\mathscr{H}_{2345}[s]$ onto $H_s$ [3]. It is a polynomial in $L_\pm$, so it commutes with $L \cdot U$.

*Step 3: three spins suffice.* Let $t_5 = |j_5, j_5\rangle$ and $\iota v = v \otimes t_5$ for $v \in V[M_0]$; $\iota$ is an isometry into $\mathscr{H}_{2345}[s]$. Put $R = P\iota : V[M_0] \to H_s$. $R$ is onto: if $h \in H_s$ is orthogonal to $R(V[M_0])$, then $\langle h, \iota v\rangle = \langle Ph, \iota v\rangle = 0$ for all $v$, so $h$ has no component with $m_5 = j_5$; its top nonzero component in $m_5$ then sits below $j_5$, and $L_+ = \mathrm{G}_+ + J_{5+}$ maps it to a nonzero component one step higher in $m_5$, which nothing else in $L_+ h$ can cancel, contradicting $L_+ h = 0$ unless $h = 0$.

*Step 4: the Gram matrix.* Since $J_{5+} t_5 = 0$ and $J_{5+}$ commutes with $\mathrm{G}_+$, $L_+^k \iota v = \iota(\mathrm{G}_+^k v)$. Hence

$$R^* R = \iota^* P \iota = \sum_k \chi_k (L_+^k \iota)^* (L_+^k \iota) = \sum_k \chi_k \mathrm{G}_-^k \mathrm{G}_+^k = \mathrm{B},$$

so $\mathrm{B} \succeq 0$. On the part with $\mathrm{G}^2 = g(g+1)$, $\mathrm{G}_-^k \mathrm{G}_+^k = \prod_{i<k} (g - M_0 - i)(g + M_0 + 1 + i)$ and the same Chu–Vandermonde evaluation gives $\beta_g = (2s + 1 - g - M_0)_{g-M_0} / (2s+2)_{g-M_0}$, with $2s + 1 - M_0 = s + j_5 + 1$. Every factor $(s + j_5 + 1 - g + i)/(2s + 2 + i)$ lies in $[0, 1]$ because $g \geq |M_0| \geq j_5 - s$, and the product vanishes exactly when one numerator factor is 0, that is when $g > s + j_5$.

*Step 5: the operator, short form.* Put $\mathsf{A} := R^*(L\cdot U)R = \iota^*(L\cdot U)P\iota$, using $P(L\cdot U)P = (L\cdot U)P$; it is Hermitian. For $v \in V[M_0]$, $P\iota v \in H_s$, so $L_zP\iota v = sP\iota v$ and $L_+P\iota v = 0$. Writing $L\cdot U = U_zL_z + {}^1\!/_2(U_+L_- + U_-L_+)$,

$$\langle \iota w, (L\cdot U)P\iota v\rangle = s\langle U_z\iota w, P\iota v\rangle + \tfrac{1}{2}\langle L_+U_-\iota w, P\iota v\rangle = \big\langle sU_zw + \tfrac{1}{2}\mathsf{G}_+U_-w,\ \mathsf{B}v\big\rangle,$$

because $U$ acts on legs 2–4, so $U_z\iota w = \iota(U_zw)$ and $L_+U_-\iota w = \iota(\mathsf{G}_+U_-w)$, and $\iota^*P\iota = \mathsf{B}$. So $\mathsf{A} = (sU_z + {}^1\!/_2U_+\mathsf{G}_-)\mathsf{B}$.

*Step 6: the printed form.* Expand instead $(L\cdot U)\iota w = \iota(\mathsf{X}w) + {}^1\!/_2(U_+w)\otimes J_{5-}t_5$, using $L_-\iota w = \iota(\mathsf{G}_-w) + w\otimes J_{5-}t_5$. The first term contributes $\langle w, \mathsf{X}^*\mathsf{B}v\rangle$ to $\langle (L\cdot U)\iota w, P\iota v\rangle$. For the second, $L_+^k(y\otimes J_{5-}t_5) = \mathsf{G}_+^k y\otimes J_{5-}t_5 + k\,\mathsf{G}_+^{k-1}y\otimes J_{5+}J_{5-}t_5$ and $J_{5+}J_{5-}t_5 = 2j_5t_5$, so only the second piece survives the pairing with $L_+^k\iota v = \iota(\mathsf{G}_+^kv)$, and with $k\chi_k = -\chi'_{k-1}/(2s+2)$ the contribution is $j_5\sum_k k\chi_k\langle U_+w, \mathsf{G}_-^{k-1}\mathsf{G}_+^kv\rangle = -\frac{j_5}{2s+2}\langle w, U_-\mathsf{B}'\mathsf{G}_+v\rangle$. This is the form $\mathsf{A} = \mathsf{X}^*\mathsf{B} - \frac{j_5}{2s+2}U_-\mathsf{B}'\mathsf{G}_+$ of the theorem; the two forms are equal because both equal $R^*(L\cdot U)R$.

*Step 7: the spectrum.* $R$ is onto $H_s$ and $R^*R = \mathsf{B}$, so its polar decomposition $R = Y\mathsf{B}^{1/2}$ has $Y$ a unitary from the range of $\mathsf{B}$ onto $H_s$. Then $Y^*(L\cdot U)Y = \mathsf{B}^{-1/2}\mathsf{A}\mathsf{B}^{-1/2}$ on the range of $\mathsf{B}$, and with Step 1 the spectrum of $T$, with multiplicities, is that of $-\mathsf{B}^{-1/2}\mathsf{A}\mathsf{B}^{-1/2}$ there. □

The theorem was also checked numerically, against $T$ built on the full tensor product with no recoupling theory, on 15 tuples with doubled spins up to 8, including one ($j_1 = 1$, all spins 1) where $\mathsf{B}$ has a kernel: the two forms of $\mathsf{A}$ agree to $1.1\times10^{-15}$ and the spectra, with multiplicities, to $1.8\times10^{-15}$ (`certificates/pencil71`).

### THREE BOUNDED LEGS

The rest of this appendix proves theorem 7.3. By proposition 4.5 the bounded legs may be taken to be legs 2, 3 and 4. Call the fourth leg $j_5$, put $M_0 = j_1 - j_5$ (so $|M_0| \le j_2+j_3+j_4$ when the invariant space is nonzero), use the pairing $(j_2j_3|j_4j_5)$ and the coordinates $(b,\gamma)$ with $c = s+\gamma$.

**Lemma G.1** (Structural stability). For $s \ge s_{\min} := \max\big(M_0+j_4,\ {}^1\!/_2(j_2+j_3+M_0+j_4),\ \max|\gamma|+1,\ 1\big)$ the sites are exactly $b\in[|j_2-j_3|, j_2+j_3]$, $\gamma\in[-M_0-j_4,\ j_4-M_0]$ with $|\gamma|\le b$, independent of $s$, and the links are

$$X\ :\ \tfrac{1}{4}h_1(b)\sqrt{b^2-\gamma^2}\sqrt{(2s+1+\gamma)^2-b^2}, \qquad D\ :\ \tfrac{1}{4}\sqrt{k_D}\sqrt{\frac{\mathcal{N}(s)}{4(s+\gamma)^2-1}},$$

with $k_D = (j_4+1-M_0-\gamma)(\gamma+M_0+j_4)(b-\gamma+1)(b+\gamma)$ and $\mathcal{N}(s) = (2s+j_4+1-M_0+\gamma)(2s+\gamma-M_0-j_4)(2s+\gamma-b)(2s+b+\gamma+1)$.

*Proof.* The region conditions are linear in $s$. The condition $c \in [|j_4 - j_5|, j_4 + j_5]$ becomes $\gamma \in [-M_0 - j_4, j_4 - M_0]$ once $j_5 \geq j_4$, that is once $s \geq M_0 + j_4$; $c - b \leq s$ becomes $\gamma \leq b$; $s \leq b + c$ becomes $\gamma \geq -b$; and $b - c \leq s$ becomes $b \leq 2s + \gamma$, which holds once $2s \geq j_2 + j_3 + M_0 + j_4$. Substituting $c = s + \gamma$ into proposition 4.4 gives the links, and every factor is positive for $s \geq s_{\min}$; the condition $s > |\gamma|$ keeps $4(s+\gamma)^2 - 1 > 0$. □

So for $s \geq s_{\min}$ the bipartite block $Y(s)$ (from the smaller colour class) has a fixed shape, and $Y(s)/s \to Y_\infty$ entrywise.

**Error bounds.** For $s \geq s_0$ each link satisfies $|w(s) - s\,w_\infty| \leq$ an exact rational multiple of $\sqrt{k}$, where $k$ is the rational squared limit link ($k = h_1(b)^2(b^2 - \gamma^2)$ for an $X$-link and $k = k_D$ for a $D$-link, so that $w_\infty = {}^1\!/_2\sqrt{k}$): for $X$-links, $|\sqrt{(2s+1+\gamma)^2 - b^2} - 2s| \leq 2|1+\gamma| + |(1+\gamma)^2 - b^2|/(2s_0)$; for $D$-links, $\mathcal{N}(s) - 4s^2(4(s+\gamma)^2 - 1)$ is a cubic in $s$ whose coefficients are bounded termwise, and $2s(4(s+\gamma)^2 - 1) \geq 8s^3[(1 - |\gamma|/s_0)^2 - 1/(4s_0^2)]$. Summing gives an exact rational $E(s_0)^2 \geq \|Y(s) - sY_\infty\|_F^2$ for all real $s \geq s_0$.

**The limit.** $Y_\infty$ is in general rectangular, and its nonzero singular values are the positive eigenvalues of the three-spin chirality $U_z$ on $V[M_0]$, with multiplicity (theorem 7.1 as $s \to \infty$). For every triple with $2j \leq 6$ and every $M_0$ with nonzero invariant space, $Y_\infty$ is injective in at least one of the three choices of which bounded leg pairs with $j_5$ (the only exceptions are $|M_0| = j_2 + j_3 + j_4$, where $T \equiv 0$), and a lower bound $\sigma_\infty \leq \sigma_{\min}(Y_\infty)$ is certified by an interval Cholesky factorisation of $Y_\infty^T Y_\infty - \sigma_\infty^2$ in 40-digit interval arithmetic. Weyl's inequality then gives $q = 8\sigma_{\min}(Y(s)) \geq 8(s\sigma_\infty - E(s_0))$ for all $s \geq s_0$; in particular $Y(s)$ is injective, so this is the smallest nonzero eigenvalue. We take $s_0$ to be the least half-integer at least $s_{\min}$ with $s_0\sigma_\infty - E(s_0) \geq 4.5579607/8$. There are 532 cases $(j_2, j_3, j_4; M_0)$ for $\mathscr{C}_6$, with $s_0 \leq 113$ ($s_0 \leq 9.5$ for $\mathscr{C}_2$ and $s_0 \leq 38.5$ for $\mathscr{C}_4$).

For all but two of the 532 cases the certified $\sigma_\infty$ exceeds $\nu_0 = \sqrt{(5 - 3\sqrt{2})/2} = 0.6153695\ldots$, and is in fact at least 0.7087 (at least 0.8660 on $\mathscr{C}_2$); we compared each certified value with $\nu_0$ in exact arithmetic. The two exceptions are the triple $(1, 1, {}^1\!/_2)$ on legs $(2, 3, 4)$ with $M_0 = \pm{}^1\!/_2$, the extremal triple of theorem 7.2, where the interval certificate gives only $\sigma_\infty = 0.6153689 < \nu_0$. There the limit matrix has five sites, for $M_0 = {}^1\!/_2$ the sites $(b, \gamma) = (0,0), (1,-1), (1,0), (2,-1), (2,0)$, with colour classes of sizes 3 and 2, and its characteristic polynomial is exactly $x(4x^4 - 20x^2 + 7)/4$, as for $U_z$ in theorem 7.2. So $Y_\infty$ is $3 \times 2$ with singular values $\nu_0$ and $\sqrt{(5 + 3\sqrt{2})/2} = 2.1497$, and $\sigma_{\min}(Y_\infty) = \nu_0$ exactly; the same holds for $M_0 = -{}^1\!/_2$. Hence $\sigma_{\min}(Y_\infty) \geq \nu_0$ in every case, and Weyl's inequality gives

$$q \;\geq\; 8\nu_0\, s - 8E(s_0) \qquad (s \geq s_0)$$

in every case; for the extremal case $M_0 = {}^1\!/_2$ this reads $q \geq 8\nu_0 s - 15.64$ for $s \geq 4.5$.

**The finite part.** For $s < s_0$ there are finitely many tuples. Those with spins at most 19 (17 376 tuples) are certified exactly by rational $LDL^T$ with exact rank; the remaining 6 137 are certified individually: their site and link structure is first checked to equal the formulas of lemma G.1 at that $s$, then $Y(s)^T Y(s) - (c/8)^2$ with $c = \max(4.5579607,\ 8\sigma_K s - C_K)$ is shown positive definite by interval Cholesky.

**Assembly.** $C_K = \max\bigl(8 \max E(s_0),\ 8\sigma_K \max s_0 - 4.558\bigr)$ rounded up, with $\sigma_K$ the smallest certified $\sigma_\infty$, is 97, 453 and 1370 for $K = 2, 4, 6$; in each class the first term decides ($8 \max E(s_0) = 96.05$, 452.08 and 1369.52). For $s \geq s_0$ the display above gives $q \geq 8\nu_0 s - C_K$. For $s < s_0$ the finite part gives $q \geq 4.5579607$, while $8\nu_0 s - C_K < 8\nu_0 \max s_0 - C_K < 0$ ($8\nu_0 \max s_0 < 46.8$, 189.6 and 556.3); so $q \geq 8\nu_0 s - C_K$ for every tuple of $\mathscr{C}_K$, which is theorem 7.3. (The finite certificates were run against the slightly smaller target $\max(4.5579607,\ 8\sigma_K s - C_K)$, which is never below 4.5579607; only that is used here.) The class constant is crude (a Frobenius-norm Weyl bound); the individual bounds in the shipped outputs are much better. The exact comparison with $\nu_0$ and the characteristic polynomials of the two exceptional limits are checked by `certificates/lj/scripts/lj_exact_slope_check.py`. For $K = 8$ only the part $s \geq s_0$ was run (1 500 cases, $s_0 \leq 253$); the finite part, about $10^5$ interval certificates, was not, and $\mathscr{C}_8$ is not claimed.

## H THE SIX-VALENT LADDER

This appendix proves theorem 8.1. Recall $A = -8i(X - Y)$, so $q = 8\times$ the singular values of the real antisymmetric matrix $\mathsf{K} = X_r - Y_r$, where $X = iX_r$ and $Y = iY_r$ with $X_r$, $Y_r$ real antisymmetric in a real recoupling basis, and with spins $j_1 = j_2 = 1/2$, $j_3 = j_4 = n/2$, $j_5 = j_6 = \frac{n+1}{2}$, $\dim \mathrm{Inv} = 4n + 2$.

### H.1 THE LADDER

Use $c = |J_1 + J_3| \in \{c_-, c_+\}$, $e = |J_4 + J_5|$ with $i = e - 1/2 \in \{0, \ldots, n\}$, and either $g = |J_1 + J_2 + J_3|$ (basis $\mathfrak{B}_1$) or $h = |J_2 + J_4 + J_5|$ (basis $\mathfrak{B}_2$). In $\mathfrak{B}_2$ write $L_c(i)$ for $h = e - 1/2$ and $R_c(i)$ for $h = e + 1/2$; row $c_-$ has $h \in [1, n]$ ($2n$ states) and row $c_+$ has $h \in [0, n + 1]$ ($2n + 2$ states).

*Proposition H.1 — The ladder*

Up to the signs of its entries: (Y) for each row $c$ and each $h = 1, \dots, n$, $Y$ has the single link $R_c(h-1)$–$L_c(h)$ of modulus $y_h = \tfrac{1}{4}\sqrt{(n+1-h)(n+2+h)h(h+1)}$, and its only zero rows are $L_+(0)$ and $R_+(n)$; (X) in each block $i$, $X = x(u_+u_-^T - u_-u_+^T)$ with $x = \tfrac{1}{4}\sqrt{n(n+2)}$ and $u_\pm(i) = \pm|\alpha_\pm|L_\pm \pm |\beta_\pm|R_\pm$, where

$$\alpha_-^2 = \frac{(n+2+i)\,i}{n(2i+2)}, \quad \beta_-^2 = \frac{(n-i)(i+2)}{n(2i+2)}, \quad \alpha_+^2 = \frac{n+1-i}{2n+4}, \quad \beta_+^2 = \frac{n+3+i}{2n+4}.$$

*Proof.* $X$ commutes with $g$, $e$ and legs 4–6; in $\mathfrak{B}_1$ at fixed $(g,e)$ it is the four-valent operator on legs $(1,3,2,g)$ with $k=c$. Both values of $c$ occur only for $g = n/2$, where (4) gives $x^2 = n(n+2)\cdot n(n+2)/(16\,n(n+2))$. $Y$ commutes with $c$ and $h$; at fixed $(c,h)$ it is the four-valent operator on $(4,5,2,h)$ with $k = e = h \pm \tfrac{1}{2}$, and the spin-$\tfrac{1}{2}$ cancellation of proposition 5.2 gives $y_h^2$; for $h=0$ and $h=n+1$ only one $e$ is allowed. The overlap $|\langle c,e,g=n/2|c,e,h\rangle|^2 = (2g+1)(2h+1)\{c\ \tfrac{1}{2}\ g;\ e\ j_6\ h\}^2$ follows from Edmonds' table for $6j$ symbols with a spin-$\tfrac{1}{2}$ entry [20, Table 5], using

$$\begin{Bmatrix} a & b & c \\ \tfrac{1}{2} & c-\tfrac{1}{2} & b+\tfrac{1}{2} \end{Bmatrix}^2 = \frac{(\Sigma-2b)(\Sigma-2c+1)}{(2b+1)(2b+2)\,2c(2c+1)},$$
$$\begin{Bmatrix} a & b & c \\ \tfrac{1}{2} & c-\tfrac{1}{2} & b-\tfrac{1}{2} \end{Bmatrix}^2 = \frac{(\Sigma+1)(\Sigma-2a)}{2b(2b+1)\,2c(2c+1)},$$

with $\Sigma = a+b+c$, after bringing each symbol to this form by its symmetries; this gives the four stated values for all $n$ and $i$. □

The magnitudes were also read off the full operator exactly for $n \le 20$, and the $6j$ forms checked against exact symbols for $n \le 30$.

## H.2 THE ZERO-ENERGY RECURSION AND THE KERNEL

Write the components in block $i$ as $a_i = \psi(L_-(i))$, $b_i = \psi(R_-(i))$, $p_i = \psi(L_+(i))$, $r_i = \psi(R_+(i))$, and $m_i = \langle u_-(i), \psi\rangle$, $P_i = \langle u_+(i), \psi\rangle$, $W_i = (P_i, r_i)$, $U_i = (m_i, b_i)$. By proposition H.1, $(X\psi)$ is $x\,m_i$ times $u_+(i)$ minus $x\,P_i$ times $u_-(i)$ on block $i$, and $Y$ couples $R_c(h-1)$ to $L_c(h)$ with weight $y_h$. So, up to signs, the equations of $\mathrm{K}\psi = 0$ read

$$L_+(i):\ x\alpha_+(i)m_i \pm y_i r_{i-1} = 0, \qquad R_+(i):\ x\beta_+(i)m_i \pm y_{i+1}p_{i+1} = 0,$$
$$L_-(i):\ x\alpha_-(i)P_i \pm y_i b_{i-1} = 0, \qquad R_-(i):\ x\beta_-(i)P_i \pm y_{i+1}a_{i+1} = 0,$$

with the terms involving $y_0$ or $y_{n+1}$ absent. For $i+1<n$ they solve successively, every coefficient nonzero, as two kinds of step (at the right end $\beta_-(n)=0$, and the equations there are treated separately below):

$$W_i \mapsto U_{i+1}: \quad m_{i+1}=\pm\frac{y_{i+1}}{x|\alpha_+(i+1)|}r_i, \quad a_{i+1}=\pm\frac{x\beta_-(i)}{y_{i+1}}P_i, \quad b_{i+1}=\frac{m_{i+1}\mp|\alpha_-(i+1)|a_{i+1}}{\pm\beta_-(i+1)},$$
$$U_i \mapsto W_{i+1}: \quad P_{i+1}=\pm\frac{y_{i+1}}{x|\alpha_-(i+1)|}b_i, \quad p_{i+1}=\pm\frac{x\beta_+(i)}{y_{i+1}}m_i, \quad r_{i+1}=\frac{P_{i+1}\mp|\alpha_+(i+1)|p_{i+1}}{\pm\beta_+(i+1)}.$$

The first uses the equations at $L_+(i+1)$ and $R_-(i)$ and the definition of $m_{i+1}$; the second those at $L_-(i+1)$ and $R_+(i)$ and the definition of $P_{i+1}$. In the coordinates $W=(P,r)$ and $U=(m,b)$ the steps are $U_{i+1}=M_1(i)W_i$ and $W_{i+1}=M_2(i)U_i$, with $2\times 2$ step matrices of the pattern

$$M_1(i)=\begin{pmatrix}0 & \mathsf{a}_1\\ \mathsf{b}_1 & \mathsf{c}_1\end{pmatrix}, \qquad M_2(i)=\begin{pmatrix}0 & \mathsf{a}_2\\ \mathsf{b}_2 & \mathsf{c}_2\end{pmatrix},$$

whose entries have the moduli

$$|\mathsf{a}_1|=\frac{y_{i+1}}{x\alpha_+(i+1)}, \qquad |\mathsf{b}_1|=\frac{x\alpha_-(i+1)\beta_-(i)}{y_{i+1}\beta_-(i+1)}, \qquad |\mathsf{c}_1|=\frac{|\mathsf{a}_1|}{\beta_-(i+1)},$$
$$|\mathsf{a}_2|=\frac{y_{i+1}}{x\alpha_-(i+1)}, \qquad |\mathsf{b}_2|=\frac{x\alpha_+(i+1)\beta_+(i)}{y_{i+1}\beta_+(i+1)}, \qquad |\mathsf{c}_2|=\frac{|\mathsf{a}_2|}{\beta_+(i+1)},$$

all at the index $i$ of the step. So the recursion splits into the chains $(U_0,W_1,U_2,\dots)$ and $(W_0,U_1,W_2,\dots)$. At the left end, $L_+(0)$ is a zero row of $Y$, which forces $m_0=0$, and $\alpha_-(0)=0$ then gives $b_0=0$: $U_0=0$ and the first chain vanishes. At the right end, $R_+(n)$ is a zero row, which forces $m_n=0$ and then, since $\alpha_-(n)=1$ and $\beta_-(n)=0$, $a_n=0$; the equations at $L_+(n)$ and $R_-(n-1)$ then force $W_{n-1}=0$.

*Proof of theorem 8.1(i).* For odd $n$, $W_{n-1}$ lies in the second chain, $W_{n-1}=\mathsf{T}W_0$ with $\mathsf{T}=M_2(n-2)M_1(n-3)\cdots M_2(1)M_1(0)$, a product of steps each of nonzero determinant; so $W_0=0$ and then every component vanishes. For even $n$, $W_{n-1}$ lies in the first chain and vanishes automatically, and the second chain carries a two-dimensional kernel. □

Exact characteristic polynomials of the operator built from (3) confirm a trivial kernel for every odd $n\le 51$ and a kernel of dimension exactly two for every even $n\le 50$.

## H.3 THE BOUND

For odd $n$ group the steps in pairs, $N(i) = M_2(i+1)M_1(i)$ for even $i$, so that $\mathsf{T} = \mathsf{T}_{rest}\mathsf{T}_{left}$ with $\mathsf{T}_{left} = N(0)$ and $\mathsf{T}_{rest} = N(n-3)\cdots N(4)N(2)$. Two ratios measure how far each two-step factor is from its dominant entry:

$$\epsilon_1(i) = \frac{|\mathsf{b}_1|}{|\mathsf{c}_1|} = \frac{x^2\alpha_-(i+1)\alpha_+(i+1)\beta_-(i)}{y_{i+1}^2},$$
$$\epsilon_2(i) = \frac{|\mathsf{b}_2\mathsf{a}_1|}{|\mathsf{c}_2\mathsf{c}_1|} = \frac{x^2\alpha_+(i+2)\beta_+(i+1)\alpha_-(i+2)\beta_-(i+1)}{y_{i+2}^2},$$

with $\mathsf{a}_1, \mathsf{b}_1, \mathsf{c}_1$ the entries of $M_1(i)$ and $\mathsf{a}_2, \mathsf{b}_2, \mathsf{c}_2$ those of $M_2(i+1)$. Every $\alpha_\pm$, $\beta_\pm$ is at most 1, so $\epsilon_1(i) \le x^2/y_{i+1}^2$ and $\epsilon_2(i) \le x^2/y_{i+2}^2$.

**Lemma H.2** (Quasimode). For odd $n$, let $v$ be a unit singular vector of $\mathsf{T}$ for its smaller singular value $|\det \mathsf{T}|/\|\mathsf{T}\|$ and build $\psi$ from $W_0 = v$. All equations hold except two at block $n$, and $\sigma_{\min}(\mathsf{K}) \le \sqrt{3}\,(y_n + {}^1\!/_4\sqrt{n})\,|\det \mathsf{T}|/\|\mathsf{T}\|$.

*Proof.* Choose $p_0, r_0$ with $(P_0, r_0) = v$; this is possible because $\alpha_+(0) \neq 0$. Run the second chain from $W_0$ to $W_{n-1} = \mathsf{T}v$ and set the first chain to zero. Every equation used by a step then holds, and so do the equations at $L_+(0)$ and $R_+(0)$, which involve only the first chain. At block $n$ define $a_n$ by the equation at $R_-(n-1)$, so $|a_n| = x\beta_-(n-1)|P_{n-1}|/y_n$, and put $p_n = r_n = 0$, which is what the first chain gives. The equation at $L_-(n)$ holds because $P_n = 0 = b_{n-1}$. The two equations left, at $L_+(n)$ and $R_+(n)$, have residuals $x\alpha_+(n)m_n \pm y_n r_{n-1}$ and $x\beta_+(n)m_n$, where $m_n = \pm a_n$ because $\alpha_-(n) = 1$ and $R_-(n)$ does not exist. Since $\alpha_+(n)^2 + \beta_+(n)^2 = 1$,

$$\|\mathsf{K}\psi\| \le x|m_n| + y_n|r_{n-1}| \le \Big(y_n + \frac{\beta_-(n-1)x^2}{y_n}\Big)\|W_{n-1}\|, \qquad \frac{\beta_-(n-1)x^2}{y_n} = \frac{n+2}{8\sqrt{n(n+1)}} \le \frac{\sqrt{n}}{4}.$$

On the other side $\|\psi\|^2 \ge p_0^2 + r_0^2$, and $P_0^2 \le (\alpha_+(0)^2 + \beta_+(0)^2)(p_0^2 + r_0^2) = p_0^2 + r_0^2$, so $1 = \|v\|^2 = P_0^2 + r_0^2 \le 2\|\psi\|^2$. With $\|W_{n-1}\| = \|\mathsf{T}v\| = |\det \mathsf{T}|/\|\mathsf{T}\|$ this gives $\sigma_{\min}(\mathsf{K}) \le \|\mathsf{K}\psi\|/\|\psi\| \le \sqrt{2}\,(y_n + {}^1\!/_4\sqrt{n})|\det \mathsf{T}|/\|\mathsf{T}\|$, which is stronger than stated. □

**Lemma H.3** (Determinant). Each two-step factor has $|\det N(i)|^2 = (n+3+i)/(n+5+i)$, so $|\det \mathsf{T}_{rest}| = \sqrt{(n+5)/(2n+2)} \le 1$.

*Proof.* For the pattern $\left(\begin{smallmatrix} 0 & \mathsf{a} \\ \mathsf{b} & \mathsf{c} \end{smallmatrix}\right)$, $|\det| = |\mathsf{ab}|$, so

$$|\det N(i)| = \frac{\alpha_-(i+1)\beta_-(i)}{\alpha_+(i+1)\beta_-(i+1)} \cdot \frac{\alpha_+(i+2)\beta_+(i+1)}{\alpha_-(i+2)\beta_+(i+2)}.$$

From the four formulas of proposition H.1, $\alpha_-(i+1)^2/\alpha_-(i+2)^2 = (n+3+i)(i+1)(i+3)/((i+2)^2(n+4+i))$, $\beta_-(i)^2/\beta_-(i+1)^2 = (n-i)(i+2)^2/((i+1)(i+3)(n-i-1))$,

$\alpha_+(i+2)^2/\alpha_+(i+1)^2 = (n-i-1)/(n-i)$ and $\beta_+(i+1)^2/\beta_+(i+2)^2 = (n+4+i)/(n+5+i)$, whose product is $(n+3+i)/(n+5+i)$. Over $i = 2, 4, \dots, n-3$ the product telescopes to $(n+5)/(2n+2)$. □

**Lemma H.4** (Cone). For $n \geq 5$ and $3 \leq j \leq n-1$, $y_j \geq 3x$. On each two-step factor $N(i)$ of $\mathsf{T}_{rest}$ the cone $|P| \leq 2|r|$ is invariant, and

$$\|\mathsf{T}_{rest}\| \;\geq\; e^{-4} \prod_{j=3}^{n-1} \frac{y_j}{x}.$$

*Proof. The hypothesis.* $y_j^2/x^2 = j(j+1)(n+1-j)(n+2+j)/(n(n+2))$ is a product of positive linear factors, hence log-concave in $j$, so its minimum over $[3, n-1]$ is at an end; there $12(n-2)(n+5) - 9n(n+2) = 3(n-4)(n+10) \geq 0$ and $2(n-1)(2n+1) - 9(n+2) = (n-4)(4n+5) \geq 0$. So $\epsilon_1, \epsilon_2 \leq 1/9$.

*The cone.* Apply $N(i)$ to $(P, r)$. Up to signs, $P'' = \mathsf{a}_2\mathsf{c}_1(r \pm \epsilon_1 P)$ and $r'' = \mathsf{c}_2\mathsf{c}_1\big((1 \pm \epsilon_2)r \pm \epsilon_1 P\big)$. If $|P| \leq 2|r|$, then $|P''| \leq |\mathsf{a}_2\mathsf{c}_1|(1+2\epsilon_1)|r|$ and $|r''| \geq |\mathsf{c}_2\mathsf{c}_1|(1-\epsilon_2-2\epsilon_1)|r|$. Since $|\mathsf{a}_2|/|\mathsf{c}_2| = \beta_+(i+2) \leq 1$, the cone is invariant as soon as $\beta_+(1+2\epsilon_1) \leq 2(1-\epsilon_2-2\epsilon_1)$, and with $\epsilon_1, \epsilon_2 \leq 1/9$ the left side is at most $11/9$ and the right side at least $4/3$.

*The growth.* Start from $(0, 1)$, which is in the cone. Each factor multiplies $|r|$ by at least $|\mathsf{c}_1\mathsf{c}_2|(1-\epsilon_2-2\epsilon_1)$, and $|\mathsf{c}_1\mathsf{c}_2| = y_{i+1}y_{i+2}/(x^2\alpha_+(i+1)\beta_-(i+1)\alpha_-(i+2)\beta_+(i+2)) \geq y_{i+1}y_{i+2}/x^2$. The brackets satisfy $1-\epsilon_2-2\epsilon_1 \geq 1 - x^2/y_{i+2}^2 - 2x^2/y_{i+1}^2 \geq 2/3$, and $\ln(1-t) \geq -3/2t$ on $[0, 1/3]$, so their product is at least $e^{-3S}$ with $S = \sum_{j=3}^{n-1} x^2/y_j^2$. Finally $x^2/y_j^2 \leq n/(j(j+1)(n+1-j))$, and the partial fractions $\frac{1}{j(n+1-j)} = \frac{1}{n+1}(1/j + \frac{1}{n+1-j})$ give $S \leq \sum_{j\geq 3} \frac{1}{j(j+1)} + \frac{1}{n+2}\sum_{j=3}^{n-1}\Big(\frac{1}{j+1} + \frac{1}{n+1-j}\Big) \leq 1/3 + \frac{2\ln n}{n+2} \leq 4/3$. □

*Proof of theorem 8.1(ii).* We have $\|\mathsf{T}\| \geq \|\mathsf{T}_{rest}\|\,\sigma_{\min}(\mathsf{T}_{left})$ and $\sigma_{\min}(\mathsf{T}_{left}) \geq |\det \mathsf{T}_{left}|/\mathrm{Y}_0$, where $\mathrm{Y}_0 = \|M_1(0)\|_F\,\|M_2(1)\|_F$ bounds $\|\mathsf{T}_{left}\|$. The step entries give

$$\mathrm{Y}_0^2 = \frac{(57n^2+146n-72)(649n^2+1298n-1944)}{72\,n(n-1)(n+2)(n+5)};$$

substituting $n = 7+k$ into $900 - \mathrm{Y}_0^2$ gives a fraction with all coefficients positive, so $\mathrm{Y}_0 \leq 30$ for $n \geq 7$ (its limit is 22.67). Since $|\det \mathsf{T}| = |\det \mathsf{T}_{rest}|\,|\det \mathsf{T}_{left}|$, lemmas H.2 to H.4 give $q = 8\sigma_{\min}(\mathsf{K}) \leq 8\sqrt{3}\cdot 30\cdot e^4\,(y_n + 1/4\sqrt{n})/\prod_{j=3}^{n-1}(y_j/x)$, where $y_n = 1/4(n+1)\sqrt{2n}$ and $8\sqrt{3}\cdot 30\,e^4 = 22696.0 < 22700$. Finally

$$\Big(\prod_{j=3}^{n-1}\frac{y_j}{x}\Big)^2 = \frac{(n-1)!\,n!\,(n-2)!}{12}\cdot\frac{\prod_{k=n+5}^{2n+1}k}{(n(n+2))^{n-3}} \;\geq\; \frac{((n-2)!)^3}{12\,(n+2)^{n-3}},$$

because the $n-3$ factors of $\prod_{k=n+5}^{2n+1}k$ each exceed $n$ and $(n-1)!\,n! = (n-1)^2 n\,((n-2)!)^2$. This gives the stated bound. By Stirling the logarithm of the product $\prod_{j=3}^{n-1}(y_j/x)$ is $n\ln n + O(n)$, and the logarithm of the lower bound on the right is $n\ln n + O(n)$ as well. □

The proof uses the moduli of the entries and the sign structure of proposition H.1: independent signs for the components of $u_\pm$ and for the dimers. For arbitrary independent signs of the entries the rank-two form of the rungs can fail, and the proof does not apply. Numerically, 1 000 random sign patterns of the entries for each $n \le 9$ leave the kernel dimension unchanged and move $\log_{10} q_{\min}$ by less than 0.18, never near the bound. The identities in these proofs were checked in exact arithmetic (`certificates/v6proofs`). The bound is far from sharp. Before the last display, that is with the exact product, it exceeds $q_{\min}$ by a factor between $10^{4.8}$ and $10^{6.4}$ for odd $7 \le n \le 81$; the closed form of theorem 8.1(ii), which drops $\prod_{k=n+5}^{2n+1} k$ and replaces $n$ by $n+2$, exceeds $q_{\min}$ by a factor $8.5 \times 10^9$ at $n = 7$, $1.9 \times 10^{11}$ at $n = 9$ and about $10^{35}$ at $n = 51$, a factor of order $e^{O(n)}$ that does not affect the leading term $n \ln n$.

## I THE $\mathfrak{su}(2)$ LADDER

This appendix proves proposition 11.1 and theorem 11.2. Write $G_d^+ = M_d^T M_d$ on $V_d$ for $d \le s-1$ and $G_d^- = M_{d-1} M_{d-1}^T$ on $V_d$ for $d \ge -s+1$; $J(\kappa)$ is the zero-diagonal Jacobi matrix on $d = -s, \dots, s$ with links $\ell_d(\kappa) = \sqrt{\kappa - c_d}$, $c_d = d(d+1)/4$.

*Proof of proposition 11.1.* By proposition 4.4 the link from $(b, c-1)$ or $(b-1, c)$ to $(b, c)$ is $\tfrac{1}{4}h\sqrt{F(d' \pm \tfrac{1}{2})G(u)}$, where $F$ is evaluated at the midpoint between the two diagonals it joins, and $F(d+\tfrac{1}{2}) = (s-d)(s+d+1) = 4\ell_s(d)^2$. Dividing the plaquette identity of lemma F.1 by $\ell_s(d)^2$ gives the defect, since the ratio $F(d+\tfrac{1}{2})/F(d-\tfrac{1}{2})$ there equals $\ell_s(d)^2/\ell_s(d-1)^2$. □

*Proof of theorem 11.2. A conserved Casimir.* Put $C_d = G_d^+ + c_d$ for $d \le s-1$ and $C_s = G_s^- + s(s-1)/4$. By hypothesis $C_d = G_d^- + c_{d-1}$ also for $-s < d < s$, since $c_d - c_{d-1} = d/2$. Then

$$M_d C_d = M_d M_d^T M_d + c_d M_d = (G_{d+1}^- + c_d) M_d = C_{d+1} M_d$$

for $d \le s-1$, using the second form of $C_{d+1}$. So $C = \bigoplus_d C_d$ commutes with $M = \bigoplus_d M_d$, hence with $M^T$ and with $W = M + M^T$, and it preserves each $V_d$.

*Splitting.* On the $\kappa$-eigenspace $E_\kappa$ of $C$, with $E_{\kappa,d} = E_\kappa \cap V_d$, $G_d^+ = a_d$ and $G_{d+1}^- = a_d$ with $a_d = \kappa - c_d$, scalars, nonnegative wherever the space they act on is nonzero. If $a_d > 0$, then $a_d^{-1/2} M_d$ is an isometry of $E_{\kappa,d}$ onto $E_{\kappa,d+1}$; if $a_d = 0$, $M_d$ vanishes on $E_{\kappa,d}$ and $M_d^T$ on $E_{\kappa,d+1}$. Split $\{d : E_{\kappa,d} \ne 0\}$ into maximal runs joined by links with $a_d > 0$. A run cannot end at an interior $d_1 > -s$ unless $a_{d_1-1} = 0$, because $G_{d_1}^- = a_{d_1-1} > 0$ on $E_{\kappa,d_1} \ne 0$ forces $E_{\kappa,d_1-1} \ne 0$; likewise at the top. Transporting an orthonormal basis along a run shows that $W$ on the run is the chain with links $\sqrt{a_d}$ tensored with an identity.

*The two kinds of run.* If a run contains $d = s$, the wall hypothesis gives $a_{s-1} \ge s/2$, that is $\kappa \ge s(s+1)/4$; the same follows at $d = -s$, where $a_{-s} = \kappa - s(s-1)/4$. An interior end $d_2 < s$ forces $\kappa = c_{d_2} \le s(s-1)/4$, so a run with an interior end has both ends

interior, $c_{d_2} = c_{d_1-1}$, that is $(d_1+d_2)(d_2-d_1+1) = 0$, so $d_1 = -d_2 =: -S$ and $\kappa = S(S+1)/4$: the links are $\frac{1}{2}\sqrt{(S-d)(S+d+1)}$, the chain is $J_x(S)$, and $S \le s-1$, $S \equiv s \pmod 1$. Every other run is the full range with $\kappa \ge s(s+1)/4$, the chain $J(\kappa)$.

*Monotonicity in $\kappa$.* Let $s(s-1)/4 < \kappa < \kappa'$, so all links are positive. The ratios $\rho_d = \ell_d(\kappa')/\ell_d(\kappa) = \sqrt{1+(\kappa'-\kappa)/(\kappa-c_d)}$ depend on $d$ only through $|d+\frac{1}{2}|$ and increase with it. Define $\vartheta$ on the sites $j = d+s \in \{0,\dots,2s\}$ by $\vartheta_0 = 1$, $\vartheta_{j+1} = \rho_j/\vartheta_j$; then $J(\kappa') = \Theta J(\kappa)\Theta$ with $\Theta = \mathrm{diag}(\vartheta)$, and the bipartite blocks satisfy $B(\kappa') = \Theta_E B(\kappa)\Theta_O$. Since $\vartheta_{j+2}/\vartheta_j = \rho_{j+1}/\rho_j$, each colour's $\vartheta$ decreases up to the first site $j > s-1$ and increases after it, so the two colour minima sit at the neighbouring sites $j_*$, $j_*+1$, where $j_*$ is the least integer above $s-1$, and their product is $\vartheta_{j_*}\vartheta_{j_*+1} = \rho_{j_*} \ge 1$, the ratio at a middle link. By Ostrowski's bound $\sigma_k(XBY) \ge \sigma_{\min}(X)\sigma_{\min}(Y)\sigma_k(B)$ for invertible diagonal $X, Y$, every singular value of $B(\kappa')$ is at least the corresponding one of $B(\kappa)$.

*The constants.* At $\kappa = s(s+1)/4$ the links are those of $J_x(s)$, whose nonzero eigenvalues are $\pm\frac{1}{2}, \pm\frac{3}{2}, \dots$ (half-integer $s$) or $\pm1, \pm2, \dots$ (integer $s$). For integer $s$ the bipartite block of $J(\kappa)$ is $(s+1)\times s$ of full column rank, so there is exactly one zero eigenvalue. The pieces $J_x(S)$ with $S \equiv s$ have the same nonzero eigenvalues. Equality holds for $W = J_x(s)$. □

The *flat* version used after proposition 11.1 has the same proof with $C_d = N_d^T N_d$: if $N_d^T N_d = N_{d-1} N_{d-1}^T$ on every interior diagonal, then $N_d C_d = C_{d+1} N_d$, on the $g$-eigenspace every $g^{-1/2} N_d$ ($g > 0$) is an isometry across the whole strip, and $T = \sum_d \ell_s(d)(N_d + N_d^T)$ is $J_x(s) \otimes \sqrt{g}$ there; the walls impose nothing, because $\ell_s$ vanishes beyond them.

## J HOW THE RESULTS WERE REACHED

The work was kept in a running findings ledger and a task log, with corrections recorded as they happened. The question about tetrahedra was put to a Claude session on 12 October 2026: does the tetrahedron connect to Jordan algebras? The answer was that it marks where the exceptional algebra stops. Four orthogonal idempotents force associative coordinates, so there is no octonionic algebra of rank four, and the session knew of no natural tetrahedron inside the Albert algebra. Two days later a ChatGPT session exhibited one. Four primitive idempotents with pairwise overlaps $\frac{1}{3}$, the overlaps of the qubit tetrahedron of quantum measurement theory, generate all 27 dimensions, as an integer determinant of $-256$ certifies.

Between 17 and 19 October the quantum tetrahedron was worked out over all four rank-three Jordan algebras, with SO(3), SU(3), Sp(6) and $F_4$ on the faces, and with the SU(2) case of Barbieri and of Baez and Barrett reproduced as a calibration [6, 5]. Several tools shared the work. In claude.ai sessions Claude computed the four rows, their channel decompositions, recoupling matrices and volume-type spectra, the sixteen-dimensional spin-closure quotient of the exceptional case, and the bridge between the

tetrahedron and the attention layer, in exact rational Python, python-flint, PARI/GP and LiE built from source. A parallel session checked the exceptional case again in GAP with a different octonion convention and wrote a thirteen-page draft. GPT ran long exact-arithmetic computations for two reports, on measurement and spin closure and on quantum-shape observables in the rank-three Jordan family, with an archive of certificates. An agent on my own machine extended the $F_4$ sweep to level 140. The four-face invariant spaces have dimensions 5, 8, 5 and 5. For $F_4$, opening one leg of a four-face invariant gives exactly one of the attention layer's equivariant value maps, so the shapes of the exceptional tetrahedron are the five channels the layer already used. One correction from that stage set the habit of keeping a cumulative record. An assembled "complete" package turned out to have dropped two of the four earlier write-ups. They had to be restored from the original archives, and the build scripts showed how the omission had happened.

The turn to dynamics, described in section 12, led to a second Claude chat, which opened with a request to prove loop quantum gravity. Claude's first answer was that a physical theory cannot be proved, and it listed what is open. My next message attached a statement of a closure target and a criticism, and with reviews I supplied the chat narrowed to one checkable claim: on Varadarajan's off-shell space [39], can a correction $T$ make $H - 2T$ close with the Lorentzian sign? The chat ended with a no-go theorem for Lorentzian corrections that preserve the vertices, which an AI referee checked and which went into my research ledger on 20 October. A Codex session continued the calculation on the evening of 19 October, and it then moved to Claude Code, with a second Claude Code session working alongside. The tetrahedron came back twice. On the intertwiner space of the parent vertex, one candidate correction acted as a real multiple of $i\, J_1 \cdot (J_2 \times J_3)$. And a tangent-sensitive four-valent volume carries the sign factor $\epsilon_{123} - \epsilon_{124} + \epsilon_{134} - \epsilon_{234}$, which equals $\pm 4$ exactly at positively spanning vertices. Two days of campaigns found no closure. Several apparent successes took their sign from conventions the test states cannot see, or diverged in the ordered limits. An internal referee overturned a headline result, "memoryless corrections close with the Euclidean sign", with a counterexample, and it was retracted. Early in this work I had set myself the goal of a large new result. A general law whose constant could be predicted in advance did not count.

The work turned to quantum geometry through the volume. In self-dual variables, with $\mathrm{SU}(1,1)$ labels, the four-valent volume becomes an infinite zero-diagonal Jacobi matrix that is not essentially self-adjoint. That observation, the Bohr–Sommerfeld rule it led to, and their generalisation are the subject of [21]. A search of the literature for named open problems on which these tools could act then found Bianchi and Haggard's question [10]. On 21 October, within a day, came a uniform four-valent gap; the sharp four-valent law, through the observation that the four-valent matrix is a hydrogen Stark matrix times a smooth multiplier (theorem 3.1); an explicit five-valent family on which the gap closes, through a dimerised chain; the six-valent ladder; the Rovelli–Smolin contrast; and the normal form of theorem 1.1. A referee of the six-valent result found

its rate wrong: the exponent is $n \ln n$, not $\tfrac{3}{2}\, n \ln n$. Another check found that a briefing had confused the normalisations $4T$ and $8T$. When I asked what structural property separates the families that keep a gap from those that lose it, the answer was already in the normal form: the ratio $t$ of the two weights.

The equal-weight case took most of the remaining time. The slice $j_1 = \tfrac{3}{2}$ was first closed with a floating-point sweep that allowed for rounding. An independent referee found that this was not rigorous as written and that a tail estimate was missing. The tail lemmas were then written out, and the finite box was re-run on a GPU in outward-rounded interval arithmetic over $64\,352\,004$ parameter sets, with smallest certified bound $3.777422$ against the constant $3.7$ of theorem 6.14 (appendix E). The slices $j_1 = 1$ and $2$ were proved, and every tuple with spins at most $19$ was certified. One strategy for large spins was refuted by its own test vector, which became theorem 7.4. Three further rounds aimed at $j_1 \geq \tfrac{5}{2}$ (a layer decomposition, wall modes and an $\mathfrak{su}(2)$ ladder) did not succeed, and a fourth was stopped at its start. I ended the campaign on 22 October. That case is conjecture 11.1, and the ladder picture is in section 11.1.

The paper was written from the primary sources from 21 October. It was then read by an independent Claude referee, by Gemini, by a Claude session on claude.ai and by GPT, and each report was triaged against the source and checked with our own code before anything was adopted. The Claude referee asked for written proofs of two lemmas of appendix H that had only been stated, for a full proof of the three-spin pencil (theorem 7.1), which had only been sketched, and for a fairer attribution of the four-valent law to the numerics of Brunnemann and Thiemann [16]. It also asked us to remove an unverified assumption about how the runtime library rounds square roots, and the rigorous-float certificates were re-run with every square root checked in exact arithmetic, with identical minima on all $2\,274\,052$ tuples. Writing out one of the missing proofs exposed an inequality slip in the argument it came from, which was repaired with no change to the theorem. Gemini had read the text layer of the PDF, and nine of its typographical findings were artefacts of the extraction; its points of wording were adopted. The claude.ai referee showed that theorem 1.1, as then worded for non-coplanar tangents, was false, because $1\,680$ sign vectors have a coplanar triple. The theorem is now stated for tangents in general position, with the remaining case described in remark 4.1. The same referee showed that a heuristic limit stated in section 7 was contradicted by larger computations, and it asked for a statement of how AI was used, which is section 12. GPT found that the reciprocal clause of theorem 5.3 holds only after the two pairs of spins are exchanged. It found that the tail certificate for $j_1 = \tfrac{3}{2}$ had entered its decimal constants as binary64 numbers before enclosing them; re-run with the constants as exact rationals, it passes in all eight regimes. And it proposed the exact slope $8\nu_0 = 4\sqrt{10 - 6\sqrt{2}}$ for the linear law on three bounded legs, which theorem 7.3 now states and theorem 7.2 shows cannot be improved.

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